Volume I.
Introduction.

We presuppose, in our considerations, the natural numbers $1, 2, 3, \ldots$ and the rules according to which calculations are made with these numbers as known and given. The fundamental arithmetical operations, the so-called four species, are addition, multiplication, to which, as repetition, exponentiation belongs, subtraction, and division. The first two are called the direct operations of calculation; they are distinguished by the fact that in the realm of the natural numbers they can be carried out without restriction. The first of the indirect or inverse operations, subtraction, can be carried out only when the minuend is greater than the subtrahend.

The problem of division can be understood in two ways. In the first elementary interpretation one asks how often the divisor is contained in the dividend. A number is not contained in a smaller one. But if the dividend is equal to or greater than the divisor, then the answer to the question gives a quotient and, in most cases, a remainder which is smaller than the divisor. If no remainder is left, one says that the division comes out exactly, or that the dividend is divisible by the divisor, or that the divisor is a factor or divisor of the number which forms the dividend.

This problem, which already presents itself at the first stages of the art of calculation, leads to a more deeply lying distinction among numbers, which is the foundation of all number theory.

Since a factor can never be greater than the number of which it is a factor, every number has only a finite number of factors. Every number is divisible by 1 and by itself, and a number which has no other divisor is called a prime number. The number 1 itself is customarily not designated a prime number, for reasons of expediency. If two numbers are divisible by a third, then the sum and the difference of the first two are also divisible by the third; and if one number is divisible by a second and this second by a third, then the first is also divisible by the third.

Two numbers always have the common divisor 1. If they have no other common divisor, as for example the pairs of numbers 5 and 7, or 21 and 38, then the two numbers are called relative prime numbers, or also numbers prime to one another.

Among the common divisors of any two given numbers one will be the greatest, and it is a very important problem to find this greatest common divisor. This is accomplished by a procedure known under the name algorithm of the greatest common divisor, which already occurs in Euclid.[1]

Let a, a_1 be the two numbers; take the larger of them, say a, as dividend and the smaller a_1 as divisor, and determine the quotient q_1 and, if the division does not come out exactly, the remainder a_2, so that

$$a = q_1 a_1 + a_2,$$

and $a_2 < a_1$. Every common divisor of a and a_1 is then also a common divisor of a_1 and a_2, and conversely. If one proceeds with a_1, a_2 just as with a, a_1, and puts, when the division does not come out exactly,

$$a_1 = q_2 a_2 + a_3,$$

then again $a_3 < a_2$, and every common divisor of a_1 and a_2 is also a common divisor of a_2 and a_3, and conversely. If one continues dividing in this way, then, since the numbers $a, a_1, a_2, a_3, \ldots$ continually decrease, the division must necessarily come out exactly after a finite number of steps; that is, at last a pair of equations

$$a_{\nu-2} = q_{\nu-1} a_{\nu-1} + a_\nu, \qquad a_{\nu-1} = q_\nu a_\nu$$

must occur. Then a_ν is a common divisor of all preceding a's, hence also of a and a_1, and every common divisor of a and a_1 is a divisor of a_ν. Thus a_ν is the greatest common divisor of a and a_1, and at the same time we have arrived at the theorem that every common divisor of two numbers must divide their greatest common divisor.

[1]Elements, Book VII, 11, vol. II of Heiberg's edition.

If a and a_1 are relatively prime numbers, then the last divisor is $a_\nu = 1$. Multiplying, under this assumption, the preceding equations by an arbitrary number b, it follows that b is the greatest common divisor of ab and $a_1 b$, and hence that every common divisor of ab and a_1, or of a and $a_1 b$, must be a divisor of b. Thus if a_1 is relatively prime to a and to b, it is also relatively prime to ab; and from the special assumption that a_1 is a prime number it follows that a product can be divisible by a prime number only when at least one of the factors is divisible by it. Thus, if the product ab is divisible by a_1 and a_1 is relatively prime to a, then b must be divisible by a_1.

Let a, b be any two numbers with greatest common divisor d, and let $a = da'$, $b = db'$. Then a' and b' are relatively prime to one another. Every number m which is at once a multiple of a and of b has the form $m = am' = da'm'$, and in this $a'm'$ must be a multiple of b'; therefore m' too must be a multiple of b'. That is, every number which is at once a multiple of a and of b is divisible by $a'b'd$. This number $a'b'd$, which itself is a common multiple of a and b, is therefore called the least common multiple of a and b.

If a number is divisible by two relatively prime numbers, then it is also divisible by their product; and if, therefore, a number m is divisible by several numbers of which each two are relatively prime to one another, then it is also divisible by the product of all these numbers.

On this rests the proof of the important theorem that a number m can always be represented, and in only one way, as a product of prime numbers. For since a divisor cannot be greater than the dividend, m can certainly be divisible by only a finite number of prime numbers. If a is one of these prime numbers and a^α the highest power of a which divides m, then α too is a definite number. Let likewise $b^\beta, c^\gamma, \dots$ be the highest powers of the remaining prime numbers $b, c, \dots$ which divide m. Then, since any two of the numbers $a^\alpha, b^\beta, c^\gamma, \dots$ are relatively prime, m must also be divisible by the product $a^\alpha b^\beta c^\gamma \cdots$; and it must be equal to this product, since otherwise another prime number, or a higher power of one of the prime numbers $a, b, c, \dots$, would have to divide m.

We now give an overview of the extensions of the concept of number which are necessary in mathematics and which were gradually introduced.

By a manifold or set, or by the abbreviated sign M, we understand a system of objects or elements of any kind, which is so delimited and complete in itself that, for every arbitrary object, it is completely determined whether it belongs to the system or not, regardless of whether we are able, in each particular case, actually to make the decision or not.

A set is called ordered when, of any two different elements, one that is fully determined in itself always counts as the greater, and in such a way that from $a > b$, $b > c$ one always has $a > c$. If $a > b$ and $b > c$, or $a < b$ and $b < c$, we say that b lies between a and c.

The natural numbers form an ordered set; between two successive elements of it there lies no further element. Such a manifold is called discrete. An ordered set with the property that between any two elements further elements can always be found is called dense. One can form a dense set by combining the natural numbers in pairs and regarding these pairs as elements of a set. These pairs shall be called fractions and denoted by $m : n$ or m/n, and two such fractions $m : n$ and $m' : n'$ are set equal when $mn' = nm'$. If all fractions equal to one another are combined into one element, one obtains a manifold which is ordered once one further stipulates that $m : n$ is greater than $m' : n'$ when $mn' > nm'$. That this manifold is dense is seen as follows: if $\mu = m : n$, $\mu' = m' : n'$ are two fractions and $\mu > \mu'$, then, with h arbitrary, one can put

$$\frac{hmn'}{hnn'}, \qquad \frac{hm'n}{hnn'},$$

and therein $hmn' > hm'n$. One can always take h so large that further numbers lie between hmn' and $hm'n$, and if p is such a number, then $p : hnn'$ lies between μ and μ'.

The points of a straight line can also be regarded as an ordered set if by greater and smaller one understands some indication of position, for instance farther right and farther left, or higher and lower.

A division of an ordered set M into two parts A, B of such a kind that every element a of A is smaller than every element b of B is called a cut in M and is suitably denoted by (A, B). Such a cut arises when one selects an element μ of M, assigns all smaller elements to A, all larger elements to B, and assigns μ itself at will to A or to B. Strictly speaking, according as one does the one or the other, two cuts arise, but we shall always regard them as equal. If, in a cut (A, B), either A contains a greatest element or B a least element μ, then we say that μ produces the cut (A, B). But it may also occur that neither A has a greatest nor B a least element.

If every cut in a dense set is produced by a definite element μ, then the set is called continuous.

Continuity as well as density are properties which, by the nature of the matter, are inaccessible to our sense perception; consequently they also cannot be demonstrated with rigor in things of the external world, in spatial magnitudes, intervals of time, or masses, however much they may seem to lie in the essence of our intuition. But pure conceptual systems can very well be constructed to which density without continuity, or also density and continuity, belong.[2]

This very abstract way of looking at the matter gives us the assurance that the assumption of a continuous set contains no contradiction, and that such sets at least exist in the realm of thought. Geometry, and also analysis, which has always willingly attached itself to geometrical intuition, long tacitly assumed the existence of continuous sets, for example in the points of a straight line or of any other connected line-segment, as a kind of axiom. The difference between a dense and a continuous set, which lies at the basis of the distinction between commensurable and incommensurable segments, was also not missed by the ancients.[3]

We too shall in what follows not renounce the aid of geometrical intuition, and shall, for example, without hesitation regard the points of a straight line as a continuous set.

An ordered set M is called measurable under the following assumptions. Addition and multiplication are generally executable in M, and likewise the subtraction of a smaller element from a greater one; that is, from any two elements a, b (which may also be identical) a new element $a + b$ of M can be derived according to a definite rule, in such a way that $a + b$ is greater than a and greater than b, and the familiar laws of addition expressed in the formulae

$$a + b = b + a, \qquad (a + b) + c = a + (b + c)$$

hold. Moreover, for two elements a, c, of which the second is the greater, a third element b can be found such that $a + b = c$, which is also expressed by the sign of subtraction $b = c - a$. Thus two unequal elements of M always have a definite difference. From these assumptions it follows that a sum becomes greater when one of its summands becomes greater. The repeated, say m-fold, addition of the same element a is called multiplication, and its result is denoted by ma.

Finally there is added one more assumption, namely that, for every given a, a sufficiently high multiple ma is greater than any other element b given in advance. Among the elements of a measurable set there is therefore no greatest element.

In a dense measurable set there is also no least element. For if a were the least and b an arbitrary element, then between b and $b + a$ no elements could lie; for if $b < c < b + a$, then it would follow from the definition of measurability that $a' = c - b$ would be smaller than a. Conversely it also follows that a measurable set in which no least element occurs is dense. For if a and $a + b$ are any two elements, one need only add to a an element smaller than b in order to obtain an element between a and $a + b$.

If a set is continuous, then the assumptions for measurability can be simplified still further, because then subtraction is a consequence of addition. For if a and c are two elements of a continuous ordered set M in which addition exists, and if $c > a$, then one obtains a cut (A, B) in M by assigning to A all elements x for which $a + x < c$, and to B those for which $a + x > c$. This cut is produced by an element b, for which $a + b = c$ must hold.

[2]This was proved by Dedekind, to whom we owe in general the definition of continuity given above. Compare Dedekind's writings, *Stetigkeit und irrationale Zahlen*, Braunschweig 1872, 1892, and *Was sind und was sollen die Zahlen?*, Braunschweig 1888, 1893. Other manifolds to which continuity belongs were formed by Weierstrass and G. Cantor.

[3]Euclid, Elements, Book X.

The natural numbers form, according to our definition, a measurable set in which a least element, namely 1, occurs. The manifold of rational fractions is likewise measurable when addition and subtraction are explained according to the known rules of fractional calculation. Especially important, and as it were typical for the measurable sets, is the manifold of rectilinear segments, or lengths of a line, which are added simply by laying them end to end. Quantities of material, compared by means of the balance, and intervals of time, measured by the clock, also furnish examples of measurable sets. The manner of measuring does not lie in the nature of the manifold itself, but is put into it by the thinking observer; thus, for example, it would be just as admissible to understand by the sum $a+b$ of two segments a and b the hypotenuse of a right triangle with legs a, b, instead of, as is usually assumed, the segment composed from a and b by laying them end to end.

In order to make the continuous set B of cuts in the manifold R of rational fractions into a measurable one, observe first the following. If $\alpha = (A, B)$ is an element in B and μ is an arbitrarily given rational fraction, then one can always choose an element a' in A so that $a' + \mu = b'$ is contained in B. For if one chooses two arbitrary elements a, b, then one can determine the natural number m so that $m\mu > b - a$, hence $a + m\mu$ is contained in B. If then h is the smallest integer for which $a + h\mu$ is contained in B, then $a + (h-1)\mu = a'$ is in A and hence $a' + \mu = b'$ is in B.

We now understand by the sum $(A, B) + (A', B') = (A'', B'')$, or $\alpha + \alpha' = \alpha''$, the cut in R which is obtained by assigning a rational fraction a'' to A'' only when an a in A and an a' in A' exist such that $a'' \leq a + a'$. In fact (A'', B'') is a cut; for if a'' is contained in A'', the same holds for every smaller fraction, and there are fractions which are in A'' and fractions which are not in A'', namely fractions of the forms $a + a'$ and $b + b'$. Further, α'' is greater than α and than α'. For A'' first contains all of A, and if a' is an arbitrary fraction in A', one can choose a in A so that the element $a + a'$ of A' is contained in B; hence A'' is more comprehensive than A. The cuts produced by rational fractions μ, μ' give, by addition, the cut produced by $\mu + \mu'$.

We now pass to the definition of ratios, which since antiquity have been regarded as the foundation of the theory of numbers, and here follow Euclid first of all.[4]

If one combines the elements of a measurable set M into pairs, and regards these pairs themselves as elements, then a new manifold arises. We denote such a pair by $a : b$, or also by a/b; but, when a and b are different elements, we distinguish $a : b$ from $b : a$, and call a the numerator and b the denominator of $a : b$. We shall call these pairs ratios, and shall now order and make measurable this new set.

Assume first that two whole numbers m, n exist such that $na = mb$, as is always the case, for example, when M is the system of natural numbers, or when a, b are two commensurable segments. Then, if p, q are two other whole numbers, $qa = pb$ holds if and only if $mq = np$. The pair of numbers p, q is completely determined by this requirement if the condition is added that p, q are to be relatively prime. Then, if h is an arbitrary whole number, $m = hp$, $n = hq$. In this case we call the ratio $a : b$ rational and set it equal to the rational fraction $m : n$ or $p : q$. Accordingly these rational fractions may be regarded as ratios of whole numbers.

All rational ratios equal to one another form a rational number, and the rational numbers form, like the rational fractions, an ordered, dense, and measurable manifold. The natural numbers themselves are inserted into the manifold of rational numbers if by the natural number m one understands the ratio $m : 1$.

We now return to any measurable manifold M and take from it any two elements a and b. If one chooses, which is always possible, two natural numbers m, n so that $na > mb$, then the ratio $a : b$ is called greater than the rational ratio $m : n$, or

$$\frac{a}{b} > \frac{m}{n};$$

[4]Elements, Book V.

and if $m : n = p : q$, then also $a : b > p : q$. Similarly, if $n'a < m'b$, then

$$\frac{a}{b} < \frac{m'}{n'}.$$

If $a : b > m : n$, then one can insert one, and consequently also arbitrarily many, rational numbers $m_1 : n_1$ such that

$$\frac{a}{b} > \frac{m_1}{n_1} > \frac{m}{n}.$$

For this choose an arbitrary whole number k and determine the whole number h so that $h(na-mb) > kb$, which is always possible. Then

$$\frac{a}{b} > \frac{hm+k}{hn} > \frac{m}{n}.$$

And in exactly the same way, if $n'a < m'b$ and $h'(m'b - n'a) > k'a$, one obtains

$$\frac{a}{b} < \frac{h'm'}{h'n'+k'} < \frac{m'}{n'}.$$

If now $a : b$ and $\alpha : \beta$ are any two ratios, which for brevity we shall also denote by ϵ, ϵ', and whose elements belong to the same or to different manifolds, then two cases are possible: either no rational ratio μ lies between ϵ and ϵ', or a rational ratio lies between them.

In the first case the two ratios ϵ and ϵ' are called equal. One sees that two ratios which are equal to a third are also equal to one another; for if $\epsilon < \mu < \epsilon'$, then every other ratio ϵ'' is either smaller than, equal to, or greater than μ. If ϵ'' is equal to μ, then rational ratios lie both between ϵ and ϵ'' and between ϵ'' and ϵ'. If, however, ϵ'' is smaller than μ, then μ lies between ϵ'' and ϵ', and if ϵ'' is greater than μ, then μ lies between ϵ and ϵ''; hence ϵ'' cannot at the same time be equal to ϵ and equal to ϵ'.

In the second case the ratios ϵ, ϵ' are called unequal. Then either $\epsilon < \mu < \epsilon'$ or $\epsilon > \mu' > \epsilon'$ must hold, and these two cases exclude one another; for from $\epsilon < \mu < \epsilon'$ and $\epsilon > \mu' > \epsilon'$ it would follow that $\epsilon < \mu'$ and consequently $\mu < \mu'$. The relation of magnitude remains unchanged when ϵ or ϵ' is replaced by an equal element. In the case $\epsilon < \mu < \epsilon'$ the ratio ϵ is called smaller than ϵ', and in the opposite case greater. Between two unequal ratios one can insert an arbitrary number of rational ratios.

If now we collect all ratios equal to one another, we obtain a generic concept which we designate as a number in the general sense of the word. A number is therefore a name or sign for a certain manifold whose elements are precisely the ratios equal to one of them.[5] Under this concept of number are included the rational ratios, and hence also the natural numbers as the ratios $m : 1$; they form the rational numbers. Numbers which do not arise from rational ratios are called irrational numbers. According to what has been developed so far, the numbers form an ordered set, and their order can be fixed if, for each number, any one of the ratios contained under it is chosen as representative.

Of two ratios with the same denominator and unequal numerators, that one is greater whose numerator is greater; and of two ratios with the same numerator and unequal denominators, that one is smaller whose denominator is greater. For if a, a', b are arbitrary elements of a measurable set and $a' > a$, choose first a whole number n such that $na > b$ and $n(a' - a) > b$. Then take m to be the least whole number satisfying $mb > na$; then $na < mb$, but $mb < na'$. Thus

$$\frac{a}{b} < \frac{m}{n} < \frac{a'}{b},$$

and hence $a : b < a' : b$. In exactly the same way one proves that if $b' > b$, then $a : b > a : b'$. This theorem is also expressed by saying that a ratio increases with the numerator and decreases with increasing denominator.

[5]The natural numbers too can be traced back in a simple and consistent way to the generic concept.

From this one easily obtains the following theorem. If a, b, c, d are elements of the same measurable set and $a : b = c : d$, then also $a : c = b : d$. For suppose that $a : c < b : d$; then there would have to be two whole numbers m, n such that

$$na < mc, \qquad nb > md.$$

Then, however, by the theorem just proved, $na : nb < mc : md$, and hence $a : b < c : d$, contrary to the supposition.

To this is now attached the following principal theorem. If of four quantities a, b, c, d any three are given arbitrarily from a continuous measurable set, then the fourth can be determined in the same set so that $a : b = c : d$.

The theorem is an immediate consequence of the assumed continuity. For if, for example, an element x of M is put into A or into B according as $x : b$ is smaller or greater than $c : d$, then a cut is obtained, which is produced by an element a satisfying $a : b = c : d$. But this theorem also holds in certain non-continuous sets, for example for the rational fractions.

It follows that as representatives of two numbers one can always choose two ratios whose elements belong to the same manifold and have the same arbitrarily chosen denominator. The addition is then defined by

$$\frac{a}{c} + \frac{b}{c} = \frac{a+b}{c}.$$

This rule contains as a special case the addition of rational fractions; and to justify it generally it remains only to show that if $a : c = a' : c'$ and $b : c = b' : c'$, then also $(a + b) : c = (a' + b') : c'$. We prove this by showing that if $a : c = a' : c'$ and $(a + b) : c > (a' + b') : c'$, then also $b : c > b' : c'$ must hold. Let therefore, with m, n two whole numbers,

$$\frac{a+b}{c} > \frac{m}{n} > \frac{a'+b'}{c'}.$$

Then $n(a + b) > mc$ and hence

$$\frac{b}{c} > \frac{mc - na}{nc}.$$

On the other hand, from the supposition $a : c = a' : c'$ it follows easily that

$$\frac{mc - na}{nc} = \frac{mc' - na'}{nc'},$$

and therefore $b : c > b' : c'$, as was required to be proved.

It is thus shown that the numbers, as we have defined them, also form a measurable set. If a and c are taken from a continuous manifold, then, with fixed c, the ratios $a : c$ also form a continuous set; hence, since continuous sets exist at all, it follows that the numbers too form a continuous set.

If $\alpha, \beta, \gamma, \delta$ are now numbers, then from the proportion $\alpha : \beta = \gamma : \delta$ any one of the four numbers can be determined by the three others given. If one puts $\delta = 1$ and seeks α, one obtains the multiplication $\alpha = \beta\gamma$; the commutativity of the factors is a consequence of the theorem that from $\alpha : \beta = \gamma : \delta$ there follows $\alpha : \gamma = \beta : \delta$. If one seeks γ, one obtains division; and from the definition of addition given above there follows the fundamental formula $\alpha(\beta + \gamma) = \alpha\beta + \alpha\gamma$. The four fundamental operations are therefore executable in the domain of numbers, with the sole restriction that in subtraction the subtrahend must be smaller than the minuend.

The construction of a cut in the sequence of numbers always supplies the proof for the existence of a number satisfying prescribed requirements. Thus one obtains a cut (A, B) by putting into A all and only those numbers whose square is smaller than a prescribed number α; to this cut there corresponds a definite number whose square is equal to α and which is denoted by $\sqrt{\alpha}$. In this way the existence of square roots is proved.

The numerical sequences introduced by G. Cantor for the definition of irrational numbers can also be referred back to cuts.[6]

According to Cantor, by a numerical sequence one is to understand any unbounded system of numbers ordered in a definite way

$$S = x_1, x_2, x_3, x_4, \ldots$$

having the property that there is a definite number g below which none of the numbers of S sinks, and that, if δ is any chosen number and x_n, x_m are different elements of S, then $x_n - x_m$ or $x_m - x_n$ always remains smaller than δ when m and n have exceeded a sufficiently large number.

There are always numbers which are not exceeded by the numbers of such a sequence; for, once δ has been chosen and n suitably determined, x_m, whatever the value of m, never exceeds the greatest of the numbers $x_1, x_2, \ldots, x_n, x_n + \delta$. One now obtains a cut (A, B) by throwing into B those numbers which, when n has a sufficiently high value, are no longer exceeded by any x_n, and all other numbers, which are therefore exceeded by infinitely many x_n, into A. If this cut is produced by the number α, then, however small ε may be, there are always infinitely many numbers x_n between $\alpha - \varepsilon$ and α; and one may say that this number α, completely determined by S, is produced by the numerical sequence S. According to Cantor the numerical sequence S is directly the definition of the number α. Of course one and the same number α can be produced by very different numerical sequences. These numerical sequences, however, are all to be regarded as equal to one another. Among other things, one may take all numbers of S to be rational fractions. Conversely, it is also easily shown that for every given number α one can always give numerical sequences S by which α is produced, so that the totality of the numerical sequences S likewise forms a continuous set.

In the explanation of the fundamental operations an inconvenient restriction has appeared in subtraction, from which we can free ourselves by introducing negative numbers.

Let x denote any element of the number system defined so far, which we shall now call the system of positive numbers. We take this number system a second time and, to distinguish it, denote in this second system, which is to be called the system of negative numbers, every element by $-x$. This second system we now order in exactly the reverse way from the first, so that wherever in the system x "greater" stands, in the system $-x$ "smaller" is put, and conversely. Addition and subtraction in $-x$ are explained just as in x, so that

$$(-x) + (-y) = -(x + y), \qquad (-x) - (-y) = -(x - y)$$

shall hold.

But we wish to order these two number systems together in such a way that every $-x$ is to be smaller than every x. We thus obtain an ordered set in which there is no greatest and no least element. This system is also in general continuous; only the single cut $(-x, x)$ is produced by no element, and here, where the two systems meet, a violation of continuity is still present. To restore continuity we must therefore add, corresponding to the cut $(-x, x)$, a number zero, or 0, which is precisely defined by this cut. Then we have an ordered continuous set unbounded on both sides, the complete series of real numbers.

In this enlarged number domain we now explain addition generally by putting, by definition,

$$x + (-x) = 0, \qquad x + 0 = x,$$

$$x + (-y) = x - y \quad \text{if } x > y, \qquad x + (-y) = -(y - x) \quad \text{if } y > x.$$

With this explanation of addition, if z_1, z_2, z_3 are any three numbers of the whole number domain, the laws

$$z_1 + z_2 = z_2 + z_1, \qquad (z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$$

[6] Cantor, *Ueber die Ausdehnung eines Satzes aus der Theorie der trigonometrischen Reihen.* Mathematische Annalen, vol. 5 (1872); compare also Heine, *Elemente der Functionenlehre.* Journal fuer Mathematik, vol. 74 (1872).

hold, which are called the commutative and the associative laws. One forms the sum of an arbitrary number of summands by uniting, at will and successively, two summands at a time into a sum. Subtraction no longer needs special consideration if one explains $z_1 - z_2$ by $z_1 + (-z_2)$ and sets $-(-z) = z$.

The number line is represented intuitively by points, by laying off from a fixed point of a straight line, denoted by 0, the positive numbers as segments on one side, say the right, and the negative numbers on the other, the left, side. The image of the sum of two segments $z_1 + z_2$ is obtained by laying off from the point z_1 the segment of length $+z_2$ to the right or to the left, according as z_2 is positive or negative.

Multiplication and division are introduced into the enlarged number domain by the equations

$$x(-y) = (-x)y = -xy, \qquad (-x)(-y) = xy, \qquad 0x = 0.$$

Division, as the inverse of multiplication, is always possible except when the divisor is zero.

A further extension of the concept of number consists in the introduction of complex quantities. We combine every two numbers of the complete number line into pairs (x, y), and regard two such pairs (x, y) and (a, b) as equal only when $x = a$, $y = b$. These pairs of numbers form a manifold whose elements cannot indeed be ordered, but with which the arithmetical operations of addition, subtraction, multiplication, and division are to be performed according to the following rules:

$$(x, y) + (a, b) = (x + a, y + b),$$

$$(x, y)(a, b) = (xa - yb, xb + ya).$$

We also stipulate that $(x, 0) = x$, which does not contradict these equations. The pair (x, y) is equal to 0 only when x and y are both zero. For abbreviation we denote $(0, 1)$ by i. The equations above then give the consequences

$$(x, 0)(0, 1) = (0, x) = ix, \qquad (x, 0) + (0, y) = (x, y) = x + yi,$$

$$x + yi + a + bi = (x + a) + (y + b)i,$$

and the inverse of addition

$$x + yi - (a + bi) = (x - a) + i(y - b),$$

and further multiplication

$$(x + yi)(a + bi) = xa - yb + i(xb + ya), \qquad i^2 = -1,$$

$$(x + yi)(x - yi) = x^2 + y^2,$$

$$\frac{x + yi}{a + bi} = \frac{ax + by + i(ay - bx)}{a^2 + b^2},$$

by which division is defined, except when $a + bi = 0$.

Numbers of the form ix are called purely imaginary numbers, and i the imaginary unit. The numbers $a + bi$ are called imaginary or complex. The systems of real and of purely imaginary numbers are contained among them as special cases. Thus, in the domain of complex numbers $x + yi$, the fundamental operations can be carried out without restriction, except for division by zero; and calculation with real numbers is a special case.

Following Gauss, one represents the complex numbers $z = x + yi$ geometrically by the points of a plane, laying down a rectangular system of coordinates and regarding the point with coordinates x, y as the image of the numerical value z. The points of the x-axis represent the real numbers x in the way discussed above. The points of the y-axis are the images of the purely imaginary numbers yi. The origin of coordinates is the image of the number 0. The radius vector from the zero point to the point z has the numerical value $\rho = \sqrt{x^2 + y^2}$, which is called the absolute value, or the magnitude, or, according to the older terminology, the modulus of the complex number z.

The only number having absolute value 0 is 0. Every positive number occurs as the absolute value of infinitely many complex numbers, and the image-points of all numbers with the same absolute value lie on a circle whose center is at the origin of coordinates.

Two imaginary numbers which differ only by the sign of i, thus $x + yi$ and $x - yi$, are called conjugate imaginary. Their product is the square of the absolute value of either of them. If one of two conjugate imaginary numbers is zero, then the other is also zero; hence in every correct numerical equation one may replace i by $-i$ without disturbing its correctness.

We shall also record the often used theorem that the absolute value of the sum of two non-zero complex numbers is never greater than the sum of the absolute values of the summands, and is equal to it only when the ratio, or quotient, of the two summands is real and positive. Namely, let

$$z = x + yi, \qquad c = a + bi, \qquad Z = z + c = (x + a) + (y + b)i,$$

$$\rho^2 = x^2 + y^2, \qquad r^2 = a^2 + b^2, \qquad R^2 = (x + a)^2 + (y + b)^2.$$

Then

$$(\rho + r - R)(\rho + r + R) = (\rho + r)^2 - R^2 = 2(\rho r - ax - by),$$

which is surely positive when $ax + by < \rho r$. If however $ax + by > \rho r$, then it would follow that

$$\rho^2 r^2 - (ax + by)^2 = (ay - bx)^2 < 0,$$

which is impossible; and equality occurs only when $ay - bx = 0$. Thus $\rho + r \geq R$, and only in the special case $\rho + r = R$ when $ay - bx = 0$ and $ax + by > 0$. This is the assertion.

In the geometric representation this theorem says that in a triangle one side is smaller than the sum of the two others. The theorem has the further consequence that the absolute value of a sum cannot be smaller than the difference of the absolute values of the summands. For applying the preceding theorem to the sum $c = Z - z$, one obtains $r \leq R + \rho$, or

$$R \geq r - \rho.$$

Equality here occurs only when the quotient $z : c$ is real and negative.

It remains to say a word about the most important aid of algebra, literal calculation. The use of this aid is so general that one sometimes uses the word literal calculation outright as synonymous with algebra. We presuppose as known the rules according to which calculations are made with such literal expressions. Equations between literal expressions can be of two kinds. Either they are so-called identities, that is, the two expressions set equal to one another can, by applying the rules of calculation, be transformed so that both expressions agree exactly. From such literal equations one then obtains correct numerical equations when the letters are replaced by arbitrary real or complex numbers, provided that the demand to divide by zero does not arise. The letters in such equations are often also designated as variables, because one may imagine, without ever coming to a contradiction, that other and again other numerical values are successively put for the letters.

Another kind of equations between literal expressions does not have this character of identity. They contain rather a requirement imposed on the numbers which may be substituted for the letters without committing an error. Algebra has the task of determining numerical values which satisfy such a requirement, that is, of solving the equation. In these equations the letters are also designated as "unknowns." Very often in one and the same equation letters of two kinds occur: some for which arbitrary numerical values are to be put, and others whose numerical value is first to be determined.

First Section. Rational Functions.

§ 1. Whole functions.

The next object of consideration is whole rational functions, or, briefly, whole functions of one variable. By these we understand expressions of the following form:

$$f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_{n-1}x + a_n, \tag{1}$$

where n is an integral exponent, the degree of the function $f(x)$. The degree is a natural number. Sometimes, however, it is also useful to speak of whole rational functions of degree 0, by which a constant is understood. x is called the variable, $a_0, a_1, a_2 \ldots a_{n-1}, a_n$ the coefficients. Both x and $a_0, a_1, \ldots, a_n$ are symbols for indeterminate quantities, with which one proceeds according to the rules of literal calculation, and for which definite numerical values may also be substituted under some circumstances (cf. the Introduction).

If the function $f(x)$ is written in the way shown in (1), we call it ordered according to descending powers of x. The summands may be arranged in any other order; for example, the function may also be ordered according to ascending powers of x:

$$f(x) = a_n + a_{n-1}x + \cdots + a_1x^{n-1} + a_0x^n.$$

By addition, subtraction, and multiplication of whole rational functions there again arise whole rational functions. The rules of literal calculation directly give the laws of formation of these new functions.

In addition, the coefficient of any power x^ν in the sum is equal to the sum of the coefficients of x^ν in the individual summands. The degree of the function that arises is equal to the highest of the degrees of the summands, and can decrease only in the special case in which the highest degree occurs in several summands and the sum of the coefficients of the highest powers is zero.

In the multiplication of two or more whole rational functions there arises a function whose degree is equal to the sum of the degrees of the factors.

To survey the law of formation of the coefficients for the product, put

$$\begin{aligned} A(x) &= a_0x^m + a_1x^{m-1} + a_2x^{m-2} + \cdots, \\ B(x) &= b_0x^n + b_1x^{n-1} + b_2x^{n-2} + \cdots, \end{aligned} \tag{2}$$

$$A(x)B(x) = C(x) = c_0x^{m+n} + c_1x^{m+n-1} + c_2x^{m+n-2} + \cdots \tag{3}$$

and obtain

$$\begin{gathered} c_0 = a_0b_0, \\ c_1 = a_0b_1 + a_1b_0, \\ c_2 = a_0b_2 + a_1b_1 + a_2b_0, \\ \cdots, \end{gathered}$$

or, in general notation, if ν denotes one of the numbers $0, 1, 2, \ldots, m+n$,

$$c_\nu = a_0b_\nu + a_1b_{\nu-1} + a_2b_{\nu-2} + \cdots + a_{\nu-1}b_1 + a_\nu b_0, \tag{4}$$

where all a whose index is greater than m, and all b whose index is greater than n, are to be set equal to zero. For c_ν is the coefficient of $x^{m+n-\nu}$, and hence $c_\nu x^{m+n-\nu}$ arises by multiplying all terms of the form $a_\mu x^{m-\mu}$ by all terms of the form $b_{\nu-\mu}x^{n-\nu+\mu}$ and then adding. If a_0 and $b_0 = 1$, then also $c_0 = 1$.

If in all factors of such a product the highest power of x has coefficient 1, then in the product too the coefficient of the highest power is 1.

From these rules for calculating with whole functions it follows that the rules of calculation expressed in formulae such as $ab = ba$, $(ab)c = a(bc)$, $(a + b)c = ac + bc$ and the like also hold when a, b, c are whole functions of x, and in the sense that whole functions are to be regarded as equal only when equal powers have equal coefficients.

By whole functions of several variables we understand sums of products of whole, positive powers of the variables $x, y, z \dots$ with arbitrary coefficients that count as constant. One can think of these functions as arising from functions of one variable x when the coefficients $a_0, a_1 \dots a_n$ occurring in them are themselves regarded as whole rational functions of other variables $y, z \dots$. Thus functions of m variables arise from functions of $m - 1$ variables. All terms of such a function are of the form $x^r y^s z^t \cdots$, multiplied by a coefficient; and if the function is properly collected, then each combination of the exponents $r, s, t \dots$ occurs only once. Two whole functions so ordered are regarded as equal only when they contain the same products $x^r y^s z^t \cdots$ with the same coefficients.

§ 2. A theorem of Gauss.

We shall at once make an application of the multiplication rule for two whole rational functions, in order to prove a theorem of Gauss which will be useful to us later, but which here is to serve as an introduction to the mode of calculation and as an example.[7]

We consider here the case in which the coefficients in the functions $A(x), B(x)$ are whole numbers, so that, by (4), the coefficients of $C(x) = A(x)B(x)$ are also whole numbers.

If all integral coefficients $a_0, a_1, \dots, a_m$ of a function $A(x)$ have no common divisor, then the function $A(x)$ is called an original or primitive function, and the theorem we wish to prove is:

If $A(x)$ and $B(x)$ are primitive functions, then their product $C(x)$ is also a primitive function.

The proof follows almost immediately from looking at formula (4).

For if all coefficients $c_0, c_1, c_2 \dots c_{m+n}$, as given in (4), had a common divisor greater than 1, then there would also have to be at least one prime number which divides all these coefficients. Let p be such a prime number. By the assumption that $A(x), B(x)$ are primitive, this prime can divide neither all $a_0, a_1 \dots a_m$ nor all $b_0, b_1 \dots b_n$.

Suppose now that p divides

$$a_0, a_1 \dots a_{r-1}, \quad \text{but not } a_r,$$

and divides

$$b_0, b_1 \dots b_{s-1}, \quad \text{but not } b_s.$$

Then, by assumption, $r \le m$, and r may also be zero if p already does not divide a_0. Likewise $s \le n$.

If now, by (4), we form c_{r+s} and arrange it in the following way,

$$\begin{aligned} c_{r+s} = a_r b_s &+ a_{r-1} b_{s+1} + a_{r-2} b_{s+2} + \cdots \\ &+ a_{r+1} b_{s-1} + a_{r+2} b_{s-2} + \cdots , \end{aligned} \tag{1}$$

one sees at once that c_{r+s} cannot be divisible by p, as had been assumed; for the first term $a_r b_s$ is not divisible by p, while all other terms, since they are multiplied by one of the coefficients $a_0, a_1 \dots a_{r-1}, b_0, b_1 \dots b_{s-1}$, are divisible by p. Thus the assumption that $C(x)$ is not primitive while $A(x)$ and $B(x)$ are primitive leads to a contradiction.

This theorem can be transferred to functions of several variables. We call a whole rational function of m variables with integral coefficients original or primitive if the coefficients have no common divisor, and we state the theorem that the product of two primitive functions is again a primitive function.

[7] Gauss, *Disquisitiones arithmeticae*, Art. 42.

To see its truth, we need only take, in the argument carried out above, the coefficients $a_0, a_1, a_2 \ldots$, $b_0, b_1, b_2 \ldots$ not as whole numbers, but as whole rational functions of $m-1$ variables $y, z \ldots$, and call such a function divisible by a prime number p when all its coefficients are divisible by p. If we then suppose that the theorem to be proved has already been proved for functions of $m-1$ variables, formula (1) gives its correctness for functions of m variables, hence its general validity by complete induction, or by the conclusion from $m-1$ to m.

An imprimitive function is one whose integral coefficients all have a common divisor. We call this divisor the divisor of the function. Then we can also express the theorem just proved as follows:

The divisor of a product of two whole functions is equal to the product of the divisors of the two functions.

For if PA and QB are two whole functions with divisors P and Q, then A and B are primitive functions. Hence $AB = C$ is also a primitive function, and PQ is the divisor of the function $PA \cdot QB = PQC$.

Insofar as it refers to functions of one variable, we can give the theorem the following form, without changing its content essentially; in this form it is especially useful. Let

$$\varphi(x) = x^m + \alpha_1 x^{m-1} + \alpha_2 x^{m-2} + \cdots + \alpha_m,$$

$$\psi(x) = x^n + \beta_1 x^{n-1} + \beta_2 x^{n-2} + \cdots + \beta_n$$

be two whole rational functions in which the highest powers of x have coefficient 1, while the remaining coefficients are rational numbers. Then, in the product

$$\varphi(x)\psi(x) = x^{m+n} + \gamma_1 x^{m+n-1} + \gamma_2 x^{m+n-2} + \cdots + \gamma_{m+n},$$

the coefficients γ cannot all be whole numbers if the coefficients α, β in $\varphi(x)$ and $\psi(x)$ are not all whole numbers.

For let the smallest common denominator of the coefficients α of φ be denoted by a_0, and that of the coefficients β of ψ by b_0. Then $a_0\varphi(x) = A(x)$ and $b_0\psi(x) = B(x)$ are primitive functions of x. If the coefficients γ were whole numbers, the product $a_0 b_0 \varphi(x)\psi(x)$ would have divisor $a_0 b_0$ and, if a_0 and b_0 were not both equal to 1, would therefore not be primitive; but this would contradict the theorem proved above.

§ 3. Division.

Let, as before,

$$\begin{aligned} A = A(x) &= a_0 x^m + a_1 x^{m-1} + \cdots, \\ B = B(x) &= b_0 x^n + b_1 x^{n-1} + \cdots \end{aligned} \tag{1}$$

be two whole rational functions of x; but now suppose that $m \geq n$ and that a_0 and b_0 are different from zero. Then the difference

$$A - \frac{a_0}{b_0} x^{m-n} B \tag{2}$$

is also a whole rational function of x, but its degree is smaller than m, since the highest power in the two terms of the difference has the same coefficient and therefore cancels. We put this difference equal to

$$A' = A'(x) = a'_0 x^{m'} + a'_1 x^{m'-1} + \cdots, \qquad m' < m. \tag{3}$$

If now m' is still $\geq n$, then in (2) we can put A' in place of A and repeat the same reasoning.

This gives a chain of equations:

$$
\begin{aligned}
A - \frac{a_0}{b_0} x^{m-n} B &= A', \\
A' - \frac{a'_0}{b_0} x^{m'-n} B &= A'', \\
A'' - \frac{a''_0}{b_0} x^{m''-n} B &= A''', \\
&\cdots,
\end{aligned} \tag{4}
$$

and this chain can be continued until the degree of the function that has arisen is smaller than n. Since in the sequence of functions $A, A', A'' \ldots$ the degree of each following function is lowered by at least one unit, the chain of equations (4) consists of at most $m - n + 1$ members; but it may also contain fewer members when several powers cancel simultaneously. If we add all equations (4), denote the last of the functions $A, A' \ldots$ by C, and, for abbreviation, set

$$Q = \frac{a_0}{b_0} x^{m-n} + \frac{a'_0}{b_0} x^{m'-n} + \cdots, \tag{5}$$

so that Q also is a whole rational function of x, then there follows

$$A = QB + C. \tag{6}$$

The operation just described, by which the functions Q, C are found from A, B, is called division. A is the dividend, B the divisor, C the remainder, and Q the quotient. The degree of the remainder is always lower than the degree of the divisor.

The coefficients of the functions Q and C are composed from the coefficients a and b by addition, subtraction, multiplication, and division. In the denominator, however, only powers of b_0 occur; and hence if $b_0 = 1$, the coefficients of Q and C are whole functions of the a and b. The highest power of b_0 which can occur in the denominator is the $(m - n + 1)$st, since in the chain (4) the factor b_0 is added once in the denominator in each following equation. In special cases, however, the highest power of b_0 in all denominators may be a lower one.

Take, for example, for A a function of the third degree,

$$f(x) = a_0 x^3 + a_1 x^2 + a_2 x + a_3, \tag{7}$$

and for B the so-called first derivative of $f(x)$, which is of the second degree,

$$f'(x) = 3a_0 x^2 + 2a_1 x + a_2. \tag{8}$$

Then one obtains

$$Q = \frac{1}{3} x + \frac{a_1}{9a_0}, \tag{9}$$

$$C = \frac{6a_0 a_2 - 2a_1^2}{9a_0} x + \frac{9a_0 a_3 - a_1 a_2}{9a_0}. \tag{10}$$

How the calculation prescribed in the equations (4) is expediently arranged may here be assumed as known from the elements. We draw attention to the analogy between this calculation and division in the decimal number system. A function $f(x)$ represents a number written decimally if the coefficients $a_0, a_1 \ldots$ are whole numbers between zero, inclusive, and 10, exclusive, and $x = 10$ is set. If one also permits numbers greater than 10 among the coefficients, then one can represent a number by $f(x)$ in various ways. This is used in the division procedure in order to avoid fractional and negative coefficients in the results in the simplest way.

§ 4. Division by a linear function.

We shall apply the general rule for division to one special case. Let the dividend be

$$f(x) = a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \cdots + a_n, \tag{1}$$

arbitrary, while the divisor is of the first degree, or, as one also says, linear. We shall also suppose the coefficient of the first power of x in the divisor to be 1, and therefore take the divisor in the simple form $(x - \alpha)$, where α remains arbitrary. In this case the quotient Q is of degree $n - 1$, and the remainder C of degree 0, that is, independent of x. Thus set

$$f(x) = (x - \alpha)Q + C. \tag{2}$$

Then C no longer contains x, and if we set

$$Q = q_0 x^{n-1} + q_1 x^{n-2} + \cdots + q_{n-2} x + q_{n-1}, \tag{3}$$

there follows from (2)

$$\begin{aligned} f(x) = q_0 x^n + q_1 x^{n-1} + \cdots + q_{n-2} x^2 + q_{n-1} x + C \\ - \alpha q_0 x^{n-1} - \cdots - \alpha q_{n-3} x^2 - \alpha q_{n-2} x - \alpha q_{n-1}, \end{aligned} \tag{4}$$

and by comparison with (1):

$$\begin{aligned} q_0 &= a_0, \\ q_1 - \alpha q_0 &= a_1, \\ q_2 - \alpha q_1 &= a_2, \\ &\cdots, \\ q_{n-1} - \alpha q_{n-2} &= a_{n-1}, \\ C - \alpha q_{n-1} &= a_n. \end{aligned} \tag{5}$$

From this one obtains

$$\begin{aligned} q_0 &= a_0, \\ q_1 &= a_0 \alpha + a_1, \\ q_2 &= a_0 \alpha^2 + a_1 \alpha + a_2, \\ &\cdots, \\ q_{n-1} &= a_0 \alpha^{n-1} + a_1 \alpha^{n-2} + \cdots + a_{n-1}, \\ C &= a_0 \alpha^n + a_1 \alpha^{n-1} + \cdots + a_{n-1} \alpha + a_n = f(\alpha). \end{aligned} \tag{6}$$

C arises from $f(x)$ by setting $x = \alpha$, and therefore can also be denoted by $f(\alpha)$.

Accordingly we also have the formula

$$\frac{f(x) - f(\alpha)}{x - \alpha} = Q(x), \tag{7}$$

where $Q(x)$ is a whole function of degree $n - 1$.

If in the expressions (6) we put the sign x in place of the indeterminate quantity α, there arises from this a sequence of whole rational functions of x, which, if for simplicity we put $a_0 = 1$, we write as follows:

$$\begin{aligned} f_0 &= 1, \\ f_1 &= x + a_1, \\ f_2 &= x^2 + a_1 x + a_2, \\ &\cdots, \\ f_{n-1} &= x^{n-1} + a_1 x^{n-2} + a_2 x^{n-3} + \cdots + a_{n-1}. \end{aligned} \tag{8}$$

These functions $f_0, f_1 \dots f_{n-1}$ will later render us good service. For the present we add the following remarks.

According to (8), one can express the powers $1, x, x^2 \dots x^{n-1}$ of x linearly in terms of the functions $f_0, f_1 \dots f_{n-1}$, and indeed in such a way that only integral rational combinations of the a occur in the coefficients, for instance

$$1 = f_0,$$

$$x = f_1 - a_1 f_0,$$

$$x^2 = f_2 - a_1 f_1 + (a_1^2 - a_2) f_0,$$

$$\cdots .$$

From this it follows that every whole rational function of x whose degree is not greater than $n-1$ can likewise be expressed linearly by $f_0, f_1, \dots, f_{n-1}$ in the form

$$y_0 f_0 + y_1 f_1 + \cdots + y_{n-1} f_{n-1}, \tag{9}$$

where the coefficients $y_0, y_1 \dots y_{n-1}$ are independent of x.

Thus, if $F(x)$ is any whole rational function of x, then by § 3, regarding $f(x)$ as divisor, one can set

$$F(x) = Qf(x) + y_0 f_0 + y_1 f_1 + \cdots + y_{n-1} f_{n-1}, \tag{10}$$

where Q is also a whole rational function of x.

For the recurrent calculation of the functions $f_r(x)$, relation (8) gives

$$f_r(x) - x f_{r-1}(x) = a_r. \tag{11}$$

§ 5. Rational fractional functions; divisibility.

If $F(x)$ and $f(x)$ are two integral rational functions of x, then the fraction

$$\frac{F(x)}{f(x)} \tag{1}$$

is called a *fractional rational*, or also briefly a *fractional* or *rational function* of x.

If the degree of the numerator is lower than the degree of the denominator, then the function is called a *proper fraction*; in the opposite case it is called an *improper fraction*.

According to § 3 the integral rational functions Q and $\varphi(x)$ can be so determined that

$$F(x) = Qf(x) + \varphi(x) \tag{2}$$

and the degree of $\varphi(x)$ is smaller than the degree of $f(x)$; consequently

$$\frac{F(x)}{f(x)} = Q + \frac{\varphi(x)}{f(x)}, \tag{3}$$

and hence the theorem:

Every fractional function can be decomposed into the sum of an integral function and a proper fractional function.

If n is the degree of $f(x)$, then the degree of $\varphi(x)$ is at most $n-1$; it may also be lower. In particular, the case may occur in which $\varphi(x)$ vanishes identically.

In this case the function $F(x)$ is said to be divisible by $f(x)$. The function

$$\frac{F(x)}{f(x)}$$

is in this case only apparently fractional; in reality it is equal to the integral function Q.

The formula

$$\frac{x^m - 1}{x - 1} = x^{m-1} + x^{m-2} + x^{m-3} + \cdots + 1$$

gives a simple example of this.

For divisibility of functions the same laws hold as for divisibility of numbers, in particular the following:

1. If the function $F(x)$ is divisible by the function $f(x)$, and $f(x)$ is divisible by a third function $\varphi(x)$, then $F(x)$ is also divisible by $\varphi(x)$.

For if

$$F = Qf, \qquad f = q\varphi,$$

where Q and q are integral rational functions, then

$$F = Qq\varphi,$$

and since Qq is an integral rational function, F is divisible by φ.

2. If $F(x)$ is divisible by $f(x)$ and Q is any integral rational function, then $QF(x)$ is also divisible by $f(x)$.

3. If $F(x)$ and $f(x)$ are divisible by $\varphi(x)$, then $F(x) \pm f(x)$ is also divisible by $\varphi(x)$, or more generally:

4. If $F_1, F_2 \ldots$ are divisible by $f(x)$ and $Q_1, Q_2 \ldots$ are arbitrary integral rational functions, then

$$Q_1F_1 + Q_2F_2 + \cdots$$

is also divisible by $f(x)$.

The last theorem includes the two preceding ones and is proved simply as follows.

If $F_1, F_2 \dots$ are divisible by f, then the integral rational functions $\Phi_1, \Phi_2 \dots$ can be determined so that

$$F_1 = \Phi_1 f, \qquad F_2 = \Phi_2 f, \dots$$

and consequently

$$Q_1 F_1 + Q_2 F_2 + \cdots = (Q_1 \Phi_1 + Q_2 \Phi_2 + \cdots) f.$$

Since $Q_1 \Phi_1 + Q_2 \Phi_2 + \cdots$ is an integral rational function, the theorem is proved.

5. Every function is divisible by itself.

With respect to divisibility or indivisibility of functions nothing is changed if the functions are multiplied by arbitrary factors independent of x.

A non-zero quantity independent of x may be regarded as a function of degree zero. If we call such a quantity a constant, we may say:

6. Every function is divisible by every constant.

If a function is divisible by another, then the degree of the quotient is equal to the difference between the degree of the dividend and the degree of the divisor. The former therefore cannot be smaller than the latter. If the degrees are equal, the quotient is a constant, and hence:

7. If, of two integral rational functions of the same degree, one is divisible by the other, then they differ from one another only by a constant factor, and the second is also divisible by the first.

8. By § 4, the necessary and sufficient condition that the function $f(x)$ be divisible by the linear function $x - \alpha$ is that $f(\alpha) = 0$.

§ 6. Greatest common divisor.

It is a problem of fundamental importance to decide whether two integral rational functions have, besides the constants, any further common divisor. The solution is found by the algorithm of the greatest common divisor in exactly the same way as the corresponding question concerning divisibility of integers is answered. See the Introduction.

Let

$$f(x) = A, \qquad \varphi(x) = A' \tag{1}$$

be two given functions; the degree of $\varphi(x)$ is supposed to be lower, or at least not higher, than that of $f(x)$.

We divide A by A' and denote the remainder, whose degree is lower than the degree of A', by A'', thus:

$$A = Q' A' + A'';$$

then we divide A' by A'' and denote the remainder by A''', and continue in this way in forming the sequence of functions

$$A, A', A'', A''', \dots, \tag{2}$$

whose degrees $n, n', n'', n''' \dots$ always decrease and therefore, after a finite number of divisions, necessarily go down to zero. Let the last remainder of degree zero, and therefore a constant, be $A^{(\nu)}$. Then we have the chain of equations

$$\begin{aligned} A &= Q' A' + A'', \\ A' &= Q'' A'' + A''', \\ &\cdots \\ A^{(\nu-3)} &= Q^{(\nu-2)} A^{(\nu-2)} + A^{(\nu-1)}, \\ A^{(\nu-2)} &= Q^{(\nu-1)} A^{(\nu-1)} + A^{(\nu)}, \end{aligned} \tag{3}$$

where $A^{(\nu)}$ is a constant.

If now A, A' have any common divisor, then by the theorems of the preceding paragraph, as is seen from the equations (3), it is also a divisor of A'', A''', A'''' and so on down to $A^{(\nu)}$. If $A^{(\nu)}$ is a non-zero constant, then no divisor of $f(x)$ and $\varphi(x)$ depending on x can exist. Such functions are called *prime to one another*, or *relatively prime*. One also says, taking account only of divisors depending on x, that the functions have no common divisor.

The condition that the two functions have a common divisor is therefore

$$A^{(\nu)} = 0. \tag{4}$$

In this case $A^{(\nu-1)}$ is a divisor of $A^{(\nu-2)}$, as the last equation (3) shows, and by the preceding equation it is a divisor of $A^{(\nu-3)}$, and so on, hence also a divisor of A and A', that is, of f and φ. Conversely, every common divisor of A, A' is also a divisor of $A^{(\nu-1)}$; accordingly $A^{(\nu-1)}$ is called the *greatest common divisor* of f and φ. The algorithm (3) shows that $A^{(\nu)}$ or $A^{(\nu-1)}$ can be derived from the coefficients of f and φ by the rational operations of arithmetic, and in such a manner that division is always only by the coefficients of the highest powers of x in the functions $A, A', A'', \ldots$ which are not zero.

We shall apply these considerations to two examples.

First let

$$\begin{aligned} A &= a_0x^2 + a_1x + a_2, \\ B &= b_0x^2 + b_1x + b_2 \end{aligned} \tag{5}$$

be two functions of the second degree, so that a_0, b_0 are non-zero.

The first step is to form the equation

$$A = \frac{a_0}{b_0}B + C, \tag{6}$$

where

$$C = c_0x + c_1 \tag{7}$$

and

$$c_0 = \frac{a_1b_0 - b_1a_0}{b_0}, \qquad c_1 = \frac{a_2b_0 - b_2a_0}{b_0}. \tag{8}$$

If $c_0 = 0$, so that C is constant, then A, B have a common factor only if also $c_1 = 0$; and then A is divisible by B, that is, A and B differ only by a constant factor. If however c_0 is different from zero, we continue the calculation by putting

$$B = QC + D,$$

where, according to § 4, if there $\alpha = -c_1 : c_0$ is put,

$$D = \frac{b_0c_1^2 - b_1c_0c_1 + b_2c_0^2}{c_0^2}. \tag{9}$$

If D is non-zero, then A, B have no common divisor. If however $D = 0$, then C is the greatest common factor of A and B.

Substituting the values c_0, c_1 from (8) into (9), and suppressing the denominator $b_0c_0^2$, one obtains the condition for the existence of a common divisor C of A and B in the form

$$\begin{aligned} &a_0^2b_2^2 + a_2^2b_0^2 - 2a_0a_2b_0b_2 - a_1a_2b_0b_1 - a_0a_1b_1b_2 \\ &\qquad + a_0a_2b_1^2 + a_1^2b_0b_2 = 0 \end{aligned} \tag{10}$$

or

$$(a_0b_2 - b_0a_2)^2 + (a_0b_1 - a_1b_0)(a_2b_1 - a_1b_2) = 0. \tag{11}$$

This condition is also fulfilled when c_0 and c_1 are both zero, and is therefore the necessary and sufficient condition that A, B have a common divisor.

The left hand side of (10) or (11) is called the *resultant* of the functions A and B.

As a second example we take the one already chosen in § 3. Put

$$\begin{aligned} f(x) &= a_0x^3 + a_1x^2 + a_2x + a_3 = A, \\ f'(x) &= 3a_0x^2 + 2a_1x + a_2 = B, \end{aligned} \tag{12}$$

and suppose a_0 different from zero. Then by § 3 we have

$$A = QB + C, \tag{13}$$

$$C = c_0x + c_1, \tag{14}$$

where

$$c_0 = \frac{6a_0a_2 - 2a_1^2}{9a_0}, \qquad c_1 = \frac{9a_0a_3 - a_1a_2}{9a_0}. \tag{15}$$

If c_0 is zero, the algorithm is already closed. If c_1 is non-zero, then A and B have no common divisor; but if c_1 is also zero, then B itself is the greatest common divisor of A and B, that is, A is divisible by B. The conditions for this are therefore

$$3a_0a_2 - a_1^2 = 0, \qquad 9a_0a_3 - a_1a_2 = 0. \tag{16}$$

If c_0 is not zero, we go one step further and put

$$B = PC + D, \tag{17}$$

where D becomes constant and has the expression

$$D = \frac{a_2c_0^2 - 2a_1c_0c_1 + 3a_0c_1^2}{c_0^2}. \tag{18}$$

If this expression is non-zero, then A and B are prime to one another; if it is zero, then A and B have the greatest common divisor C. If for c_0, c_1 one substitutes the values (15), omits the denominator, still removes the non-vanishing factor $9a_0$, and changes the sign, this condition, after a simple calculation, assumes the form

$$a_1^2a_2^2 + 18a_0a_1a_2a_3 - 4a_0a_2^3 - 4a_1^3a_3 - 27a_0^2a_3^2 = 0. \tag{19}$$

It is, as one readily sees either by calculation or from (18), also fulfilled when the conditions (16) hold, and is therefore the necessary and sufficient condition that $f(x)$ and $f'(x)$ have a common divisor. The left hand side of (19), which is an integral rational and homogeneous function of the coefficients of $f(x)$, is called the *discriminant* of the function $f(x)$.

We derive one more theorem from the algorithm (3), which is often used.

From the first of these equations it follows that

$$A'' = A - Q'A', \tag{20}$$

and if this value of A'' is substituted in the following equation, one obtains

$$A''' = (1 + Q'Q'')A' - Q''A,$$

thus, denoting by p, p' integral rational functions,

$$A''' = pA + p'A'. \tag{21}$$

Substituting the expressions (20), (21) in the third equation (3), one obtains for A'''' again an expression of the form (21), and so one may continue, obtaining finally

$$A^{(\nu)} = PA + P'A', \tag{22}$$

where P, P' are integral rational functions whose coefficients are composed from the coefficients of A and A' by rational operations.

In the formula (22), $A^{(\nu)}$ is a constant. This theorem is especially important in the case where $A^{(\nu)}$ is non-zero, so that A, A' are relatively prime. If in this case we put

$$P = A^{(\nu)}F(x), \qquad P' = A^{(\nu)}\Phi(x),$$

then, after suppressing the factor $A^{(\nu)}$, the theorem can be expressed as follows:

I. If $f(x)$ and $\psi(x)$ are two integral functions without a common divisor, then two other integral functions $F(x)$ and $\Phi(x)$ can be determined which satisfy identically the equation

$$F(x)f(x) + \Phi(x)\psi(x) = 1. \tag{23}$$

The theorem can still be generalized. Multiplying (23) by an arbitrary integral function $\chi(x)$ gives

$$F(x)\chi(x)f(x) + \Phi(x)\chi(x)\psi(x) = \chi(x), \tag{24}$$

and by § 3 we can put

$$\Phi(x)\chi(x) = Q(x)f(x) + \varphi(x) \tag{25}$$

in such a way that $Q(x), \varphi(x)$ are integral rational functions of x and the degree of $\varphi(x)$ is lower than that of $f(x)$. Substituting this in (24), and putting in place of $F(x)\chi(x) + Q(x)\psi(x)$ again $F(x)$, we obtain

$$F(x)f(x) + \varphi(x)\psi(x) = \chi(x).$$

We can therefore generalize the preceding theorem as follows:

II. If $f(x), \psi(x), \chi(x)$ are given integral rational functions, and $f(x)$ and $\psi(x)$ are without a common divisor, then the integral rational functions $F(x), \varphi(x)$ can be so determined - with $\varphi(x)$ of lower degree than $f(x)$ - that the equation

$$F(x)f(x) + \varphi(x)\psi(x) = \chi(x) \tag{26}$$

is identically satisfied.

§ 7. Products of linear factors.

According to § 1, by multiplication of integral functions we again obtain such functions of higher degree, and the degree of the product is determined by the sum of the degrees of the individual factors.

If therefore we multiply n linear factors by one another, there arises a function of the nth degree, whose manner of formation we must examine somewhat more closely.

We shall take the linear factors in the simple form $x - \alpha_1$, $x - \alpha_2$, …, $x - \alpha_n$, and put, since the coefficient of the highest power is, by § 1, equal to 1,

$$\begin{aligned} f(x) &= (x - \alpha_1)(x - \alpha_2)\cdots(x - \alpha_n) \\ &= x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_n. \end{aligned} \tag{1}$$

A slight consideration shows the following law of formation of the coefficients $a_1, a_2 \ldots a_n$:

$-a_1$ is equal to the sum of the α's; a_2 is equal to the sum of the products of every two of the α's; $-a_3$ is the sum of the products of every three of the α's; in general $(-1)^\nu a_\nu$ is the sum of the products of every ν of the quantities α, or, in formulas,

$$\begin{aligned} -a_1 &= \sum \alpha_1, \\ +a_2 &= \sum \alpha_1\alpha_2, \\ &\cdots \\ (-1)^\nu a_\nu &= \sum \alpha_1\alpha_2\cdots\alpha_\nu, \\ &\cdots \\ (-1)^n a_n &= \alpha_1\alpha_2\cdots\alpha_n. \end{aligned} \tag{2}$$

But to see still more clearly the correctness of this law of formation, one uses complete induction.

One first verifies the correctness in the first cases by actually carrying out the multiplication:

$$(x-\alpha_1)(x-\alpha_2) = x^2 - (\alpha_1+\alpha_2)x + \alpha_1\alpha_2,$$

$$\begin{aligned} (x-\alpha_1)(x-\alpha_2)(x-\alpha_3) = x^3 &- (\alpha_1+\alpha_2+\alpha_3)x^2 \\ &+ (\alpha_1\alpha_2+\alpha_1\alpha_3+\alpha_2\alpha_3)x - \alpha_1\alpha_2\alpha_3. \end{aligned}$$

If we assume the law of formation proved for $n-1$ factors, and put

$$(x-\alpha_1)(x-\alpha_2)\cdots(x-\alpha_{n-1}) = x^{n-1} + a_1'x^{n-2} + \cdots + a_{n-1}',$$

then multiplication by $x-\alpha_n$ gives

$$\begin{aligned} a_1 &= a_1' - \alpha_n, \\ a_2 &= a_2' - \alpha_n a_1', \\ a_3 &= a_3' - \alpha_n a_2', \\ &\cdots \\ a_\nu &= a_\nu' - \alpha_n a_{\nu-1}', \\ &\cdots \\ a_n &= -\alpha_n a_{n-1}', \end{aligned} \tag{3}$$

and from this the correctness of the formulas (2) is immediately evident.

It is important to determine the number of terms which occur in each of the sums (2). We denote the number of terms occurring in the sum $(-1)^\nu a_\nu$ by $B_\nu^{(n)}$; it is the number of combinations without repetition of n elements to the νth class, that is, of every ν distinct ones among the n elements. To determine it, suppose first that the $B_{\nu-1}^{(n)}$ combinations to the $(\nu-1)$st class have been formed. From each of these combinations, by adding one at a time of the missing $n-\nu+1$ elements, one can derive $n-\nu+1$ combinations to the νth class. In this way however every combination is formed ν times, namely by addition of each of its elements; hence one obtains the relation

$$B_\nu^{(n)} = B_{\nu-1}^{(n)}\frac{n-\nu+1}{\nu}, \tag{4}$$

whereas $B_1^{(n)}$ plainly has the value n. Therefore, if one multiplies the equations following from (4),

$$\begin{aligned} B_1^{(n)} &= n, \\ B_2^{(n)} &= B_1^{(n)}\frac{n-1}{2}, \\ &\cdots \\ B_\nu^{(n)} &= B_{\nu-1}^{(n)}\frac{n-\nu+1}{\nu}, \end{aligned} \tag{5}$$

it follows that

$$B_\nu^{(n)} = \frac{n(n-1)(n-2)\cdots(n-\nu+1)}{1\cdot 2\cdot 3\cdots \nu}. \tag{6}$$

In what follows we shall often, when m is an arbitrary positive integer, use the symbol

$$\Pi(m) = 1\cdot 2\cdot 3\cdots m, \qquad \Pi(0) = 1, \tag{7}$$

so that

$$\Pi(m) = m\Pi(m-1). \tag{8}$$

With the help of this symbol the expression for $B_\nu^{(n)}$ can be displayed more conveniently as

$$B_\nu^{(n)} = B_{n-\nu}^{(n)} = \frac{\Pi(n)}{\Pi(\nu)\Pi(n-\nu)}, \tag{9}$$

which, if $B_0^{(n)} = 1$ is put, is also valid for $\nu = 0$ and $\nu = n$, and makes evident the invariance of $B_\nu^{(n)}$ under the interchange of ν with $n-\nu$.

The derivation of the formula (4), which we have just given according to the rules of the theory of combinations, is indeed perfectly correct and clear, but for its exact foundation it requires some reflection which cannot easily be compressed into a few words. We shall therefore subsequently prove its correctness once more by the method of complete induction.

The formulas (3), namely, give the following recurrence formulas:

$$\begin{aligned} B_1^{(n)} &= B_1^{(n-1)} + 1, \\ B_2^{(n)} &= B_2^{(n-1)} + B_1^{(n-1)}, \\ &\cdots \\ B_\nu^{(n)} &= B_\nu^{(n-1)} + B_{\nu-1}^{(n-1)}, \\ &\cdots \\ B_n^{(n)} &= B_{n-1}^{(n-1)}. \end{aligned} \tag{10}$$

Now the formula (6) can be confirmed very easily by counting for the first values of n. If one assumes it correct for $n-1$, then (10) gives

$$B_\nu^{(n)} = \frac{\Pi(n-1)}{\Pi(\nu)\Pi(n-\nu-1)} + \frac{\Pi(n-1)}{\Pi(\nu-1)\Pi(n-\nu)},$$

whence, by (8),

$$B_\nu^{(n)} = \frac{\Pi(n)}{\Pi(\nu)\Pi(n-\nu)},$$

and by this the general validity of formula (9) is proved.

From the meaning of $B_\nu^{(n)}$ it follows, and it is also proved from the formulas (10) by complete induction, that the $B_\nu^{(n)}$ are positive integers.

§ 8. The binomial theorem.

If in formula (1) of the preceding paragraph we set the previously arbitrary quantities $\alpha_1, \alpha_2 \dots \alpha_n$ equal to one another, we obtain the binomial theorem, which consists in nothing other than arranging the nth power of a binomial $x+y$ according to powers of x. Namely, if we put

$$\alpha_1 = \alpha_2 = \cdots = \alpha_n = -y,$$

then formula (2) gives

$$a_\nu = (-1)^\nu \sum \alpha_1\alpha_2\cdots\alpha_\nu = y^\nu B_\nu^{(n)},$$

where, according to the definition of the preceding paragraph, $B_\nu^{(n)}$ means the number of terms of the sum, which all become equal to one another and equal to $(-1)^\nu y^\nu$.

Accordingly one obtains

$$(x+y)^n = x^n + B_1^{(n)} x^{n-1} y + B_2^{(n)} x^{n-2} y^2 + \cdots + B_n^{(n)} y^n, \tag{1}$$

or, written out,

$$\begin{aligned}(x+y)^n &= x^n + n x^{n-1} y + \frac{n(n-1)}{1\cdot 2} x^{n-2} y^2 + \cdots + y^n \\ &= \Pi(n) \sum \frac{x^{n-\nu} y^\nu}{\Pi(n-\nu)\Pi(\nu)} = \Pi(n) \sum \frac{x^\alpha y^\beta}{\Pi(\alpha)\Pi(\beta)},\end{aligned}$$

the last sum being extended over all non-negative integral values α, β which satisfy the condition $\alpha + \beta = n$.

Accordingly the coefficients $B_\nu^{(n)}$ are called the binomial coefficients.

For reference we put here a small table of the first values of the binomial coefficients:

$n=1$	1	1						
$n=2$	1	2	1					
$n=3$	1	3	3	1				
$n=4$	1	4	6	4	1			
$n=5$	1	5	10	10	5	1		
$n=6$	1	6	15	20	15	6	1	
$n=7$	1	7	21	35	35	21	7	1

Among the various properties of the binomial coefficients we shall derive two, of which we shall afterwards make an interesting application.

From (1), replacing x, y by $1, x$ and taking $n = 0, 1, 2, 3 \ldots$, the formulas

$$\begin{aligned} 1 &= B_0^{(0)}, \\ 1 + x &= B_0^{(1)} + B_1^{(1)} x, \\ (1+x)^2 &= B_0^{(2)} + B_1^{(2)} x + B_2^{(2)} x^2, \\ &\cdots \\ (1+x)^n &= B_0^{(n)} + B_1^{(n)} x + B_2^{(n)} x^2 + \cdots + B_n^{(n)} x^n \end{aligned} \tag{2}$$

are obtained.

We now use the well-known summation formula for the geometric series:

$$1 + (1+x) + (1+x)^2 + \cdots + (1+x)^n = \frac{(1+x)^{n+1} - 1}{x},$$

where, if $(1+x)^{n+1}$ is again expanded by the binomial formula, replacing n by $n+1$ in the last formula (2) and setting $B_0^{(n+1)} = 1$,

$$\frac{(1+x)^{n+1} - 1}{x} = B_1^{(n+1)} + B_2^{(n+1)} x + \cdots + B_{n+1}^{(n+1)} x^n. \tag{3}$$

Comparing this with the sum of the right hand sides of (2), and, according to the rules of § 1, setting equal the coefficients of corresponding powers of x, one obtains

$$\begin{aligned} B_0^{(0)} + B_0^{(1)} + B_0^{(2)} + \cdots + B_0^{(n)} &= B_1^{(n+1)}, \\ B_1^{(1)} + B_1^{(2)} + \cdots + B_1^{(n)} &= B_2^{(n+1)}, \\ &\cdots \\ B_n^{(n)} &= B_{n+1}^{(n+1)}, \end{aligned} \tag{4}$$

or in general

$$B_\nu^{(\nu)} + B_\nu^{(\nu+1)} + \cdots + B_\nu^{(n)} = B_{\nu+1}^{(n+1)}. \tag{5}$$

If however the equations (2) are multiplied in order by

$$B_0^{(n)}, \quad -B_1^{(n)}, \quad +B_2^{(n)}, \ldots, \quad \pm B_n^{(n)},$$

where the upper sign is to be taken for even n and the lower for odd n, the sum of the left hand sides gives

$$B_0^{(n)} - B_1^{(n)}(1+x) + B_2^{(n)}(1+x)^2 - \cdots \pm B_n^{(n)}(1+x)^n = [1-(1+x)]^n = (-x)^n,$$

and equating the coefficients of equal powers on the right and left hand sides yields the system of formulas

$$\begin{aligned} 0 &= B_0^{(0)}B_0^{(n)} - B_0^{(1)}B_1^{(n)} + B_0^{(2)}B_2^{(n)} - \cdots \pm B_0^{(n)}B_n^{(n)}, \\ 0 &= -B_1^{(1)}B_1^{(n)} + B_1^{(2)}B_2^{(n)} - \cdots \pm B_1^{(n)}B_n^{(n)}, \\ &\cdots \\ \pm 1 &= \pm B_n^{(n)}B_n^{(n)}, \end{aligned} \tag{6}$$

and this system of formulas can also be written in the reversed order, using the relation $B_\nu^{(n)} = B_{n-\nu}^{(n)}$, as

$$\begin{aligned} 1 &= B_0^{(n)}B_0^{(n)}, \\ 0 &= B_1^{(n)}B_0^{(n)} - B_0^{(n-1)}B_1^{(n)}, \\ 0 &= B_2^{(n)}B_0^{(n)} - B_1^{(n-1)}B_1^{(n)} + B_0^{(n-2)}B_2^{(n)}, \\ &\cdots \\ 0 &= B_n^{(n)}B_0^{(n)} - B_{n-1}^{(n-1)}B_1^{(n)} + B_{n-2}^{(n-2)}B_2^{(n)} - \cdots \pm B_0^{(0)}B_n^{(n)}, \end{aligned} \tag{7}$$

or, in a summarizing notation,

$$\sum_{\nu=0}^{\mu} (-1)^\nu B_{\mu-\nu}^{(n-\nu)} B_\nu^{(n)} = \begin{cases} 1, & \mu = 0, \\ 0, & \mu = 1, 2, \ldots, n. \end{cases} \tag{8}$$

Volume I. First Section. Rational Functions.

§ 9. Interpolation.

We shall make an application of the last obtained formulae to a problem from the theory of interpolation.

A whole rational function of degree n is to be determined which, for $n+1$ given values of the variable x, has prescribed values.

Thus the problem is the determination of the $n+1$ coefficients $a_0, a_1, \dots, a_n$ in

$$f(x) = a_0 x^n + a_1 x^{n-1} + \cdots + a_{n-1} x + a_n \tag{1}$$

from the $n+1$ equations

$$f(\alpha_0) = A_0, \quad f(\alpha_1) = A_1, \quad \dots, \quad f(\alpha_n) = A_n, \tag{2}$$

when $\alpha_0, \alpha_1, \dots, \alpha_n$ are the given values of x, and $A_0, A_1, \dots, A_n$ are the corresponding values of $f(x)$.

Equations (2) form a system of equations of the first degree for the unknowns $a_0, a_1, \dots, a_n$, which in general can be solved by determinants, as we shall see in the next section. We shall later return to this problem, but treat it now under the special supposition that the given values of x are the first $n+1$ whole numbers $0, 1, 2, \dots, n$.

The binomial coefficients $B_\nu^{(x)}$, as they were defined by § 7, (6), keep their good sense even when x is not a whole number, as was previously presupposed, but an arbitrary variable quantity. Then

$$B_\nu^{(x)} = \frac{x(x-1)\cdots(x-\nu+1)}{1 \cdot 2 \cdot 3 \cdots \nu} \tag{3}$$

is a whole rational function of degree ν in x. [8] The coefficient of x^ν is different from zero, and therefore x^ν can also be expressed by $B_\nu^{(x)}$ and by lower powers of x. Thus every whole function of degree n in x can also be linearly expressed by $B_0^{(x)}, B_1^{(x)}, \dots, B_n^{(x)}$ in the form

$$f(x) = M_0 B_0^{(x)} + M_1 B_1^{(x)} + \cdots + M_n B_n^{(x)}, \tag{4}$$

where $M_0, M_1, \dots, M_n$ are constants; and the function $f(x)$ is determined when these constants are determined.

Now, by our supposition, $f(0), f(1), f(2), \dots, f(n)$ are given. Since, by (3), $B_\nu^{(x)}$ always vanishes when x has one of the values $0, 1, 2, \dots, \nu-1$, the following linear equations for the unknowns M are obtained from (4):

$$\begin{aligned} f(0) &= M_0 B_0^{(0)}, \\ f(1) &= M_0 B_0^{(1)} + M_1 B_1^{(1)}, \\ f(2) &= M_0 B_0^{(2)} + M_1 B_1^{(2)} + M_2 B_2^{(2)}, \\ &\cdots \\ f(n) &= M_0 B_0^{(n)} + M_1 B_1^{(n)} + \cdots + M_n B_n^{(n)}. \end{aligned} \tag{5}$$

These equations are now to be solved with respect to $M_0, M_1, \dots, M_n$, which is done very easily with the help of the final equations (8) of the preceding paragraph. The first equation (5) gives directly

$$M_0 = f(0). \tag{6}$$

[8] The meaning of these generalized binomial coefficients for the expansion of powers of a binomial does not belong here, but to analysis.

If the first equation (5) is multiplied by $B_0^{(1)}$, the second by $-B_1^{(1)}$, and they are added, one obtains, by the formula system mentioned (applied to $n = 1$),

$$-M_1 = B_0^{(1)} f(0) - B_1^{(1)} f(1),$$

and so in general, if the first ν equations (5) are multiplied in order by

$$B_0^{(\nu)}, \quad -B_1^{(\nu)}, \quad +B_2^{(\nu)}, \quad \dots, \quad \pm B_\nu^{(\nu)}$$

and then added,

$$\pm M_\nu = B_0^{(\nu)} f(0) - B_1^{(\nu)} f(1) + B_2^{(\nu)} f(2) - \cdots \pm B_\nu^{(\nu)} f(\nu), \tag{7}$$

whereby, according to (4), the function $f(x)$ is determined and the problem is solved. It is clear that, so long as we make no special supposition about the values $f(0), f(1), \dots, f(n)$, every arbitrary whole rational function of x can be represented in this form.

§ 10. Solution of the interpolation problem by differences.

The definition of the whole rational function

$$B_\nu^{(x)} = \frac{x(x-1)\cdots(x-\nu+1)}{1 \cdot 2 \cdot 3 \cdots \nu} \tag{1}$$

gives the relation, already proved earlier for the special case of a positive integral x,

$$B_\nu^{(x+1)} = B_\nu^{(x)} + B_{\nu-1}^{(x)}, \qquad B_0^{(x+1)} = B_0^{(x)} = 1. \tag{2}$$

From this we obtain for the coefficients $M_0, M_1, \dots, M_n$ a mode of determination which is much more convenient for practical calculation than the application of the formulae of the preceding paragraph.

Namely, by the formulae (4) and (6) of § 9,

$$f(x) = f(0) + M_1 B_1^{(x)} + M_2 B_2^{(x)} + \cdots + M_n B_n^{(x)}, \tag{3}$$

and if in this expression we replace x by $x + 1$ and form the difference

$$\Delta_x = f(x+1) - f(x), \tag{4}$$

then, with regard to (2),

$$\Delta_x = M_1 + M_2 B_1^{(x)} + M_3 B_2^{(x)} + \cdots + M_n B_{n-1}^{(x)}, \tag{5}$$

from which follows (since (5) is an equation of the same kind as (3), except that $n - 1$ has taken the place of n)

$$M_1 = \Delta_0 = f(1) - f(0).$$

Put

$$\Delta'_x = \Delta_{x+1} - \Delta_x, \qquad \Delta''_x = \Delta'_{x+1} - \Delta'_x, \quad \dots, \tag{6}$$

then, accordingly,

$$M_0 = f(0), \quad M_1 = \Delta_0, \quad M_2 = \Delta'_0, \quad \dots, \quad M_n = \Delta_0^{(n-1)},$$

and formula (3) gives

$$f(x) = f(0) + \Delta_0 B_1^{(x)} + \Delta'_0 B_2^{(x)} + \Delta''_0 B_3^{(x)} + \cdots + \Delta_0^{(n-1)} B_n^{(x)}. \tag{7}$$

Likewise the functions $\Delta_x, \Delta'_x, \Delta''_x, \dots$ can be expressed:

$$\begin{aligned}\Delta_x &= \Delta_0 + \Delta'_0 B_1^{(x)} + \Delta''_0 B_2^{(x)} + \cdots + \Delta_0^{(n-1)} B_{n-1}^{(x)},\\ \Delta'_x &= \Delta'_0 + \Delta''_0 B_1^{(x)} + \cdots + \Delta_0^{(n-1)} B_{n-2}^{(x)},\end{aligned} \tag{8}$$

and so on.

The $\Delta_x, \Delta'_x, \Delta''_x, \dots, \Delta_x^{(n-1)}$ are whole rational functions of degrees $n-1, n-2, \dots, 0$; the last of them is therefore constant. In order to represent them all, one needs only the values

$$f(0), \quad \Delta_0, \quad \Delta'_0, \quad \Delta''_0, \quad \dots, \quad \Delta_0^{(n-1)},$$

which one computes most easily by arranging a table. For the case $n = 3$, for example, the table has the form

$$\begin{array}{llll} f(0) & & & \\ f(1) & \Delta_0 & & \\ f(2) & \Delta_1 & \Delta'_0 & \\ f(3) & \Delta_2 & \Delta'_1 & \Delta''_0 \end{array}$$

where the differences are to be formed diagonally.

§ 11. Arithmetic series of higher order.

The preceding considerations can also be stated in another form, which we shall need later. If

$$u_0, u_1, u_2, \dots \tag{1}$$

is a sequence of quantities, we call

$$\Delta_0 = u_1 - u_0, \Delta_1 = u_2 - u_1, \Delta_2 = u_3 - u_2, \dots \tag{2}$$

the sequence of its differences; and

$$\Delta'_0 = \Delta_1 - \Delta_0, \Delta'_1 = \Delta_2 - \Delta_1, \dots \tag{3}$$

the sequence of its second differences, and so on.

It is clear that the sequence (1) is completely determined when its first term and the sequence of its first differences are given; for

$$u_1 = u_0 + \Delta_0, \quad u_2 = u_0 + \Delta_0 + \Delta_1, \quad \dots,$$

$$u_m = u_0 + \Delta_0 + \Delta_1 + \cdots + \Delta_{m-1}.$$

Likewise the sequence (1) is completely determined when the first two terms and the sequence of its second differences are given, and so forth. The sequence (1) is called an arithmetic series of nth order when the sequence of its nth differences is constant, and hence the sequence of the $(n+1)$st differences consists entirely of zeros.

One obtains an arithmetic series of nth order by substituting the numbers $0, 1, 2, 3, \dots$ for x in a whole function $f(x)$ of degree n. For if one puts

$$\Delta_x = f(x+1) - f(x), \quad \Delta_x^{(n-1)} = \Delta_{x+1}^{(n-2)} - \Delta_x^{(n-2)},$$

then Δ_x is of degree $n-1$, Δ'_x of degree $n-2$, and so on with respect to x, and therefore $\Delta_x^{(n-1)}$ is constant.

If now

$$u_0, u_1, u_2, \dots$$

is an arithmetic series of nth order, then the whole series is completely determined when the n values $u_0, u_1, u_2, \ldots, u_{n-1}$ and, besides these, the constant nth difference are given. But this latter is determined if also the $(n+1)$st term u_n is given. We can therefore state the theorem:

An arithmetic series of nth order is completely determined when its first $n+1$ terms are given.

Since a function $f(x)$ of degree n is likewise completely determined by the arbitrarily given values

$$f(0), f(1), f(2), \ldots, f(n),$$

it follows that from the whole rational functions of degree n all arithmetic series of nth order are produced when in them the series of natural numbers is substituted for x.

The expression for the general term is then given by formula (7) of the preceding paragraph.

The sums of the first $m+1$ terms of an arithmetic series of nth order,

$$s_m = u_0 + u_1 + \cdots + u_m,$$

form an arithmetic series of $(n+1)$st order, since their first differences

$$s_{m+1} - s_m = u_{m+1}$$

form an arithmetic series of nth order.

Thus, with the help of formula (7) of § 10, the sum s_m can be determined in general when $s_0, s_1, \ldots, s_{n+1}$ are assumed known.

To find the generating function $F(x)$ of s_m, when $f(x)$ is the generating function of u_m, put

$$F(0) = f(0), \quad F(1) = f(0) + f(1), \quad F(2) = f(0) + f(1) + f(2), \ldots$$

and in formula (7), § 10,

$$F(x) = F(0) + D_0 B_1^{(x)} + D_0' B_2^{(x)} + \cdots$$

one has to put

$$\begin{aligned} D_0 &= F(1) - F(0) = f(1), \quad D_0' = f(2) - f(1) = \Delta_1, \\ D_1 &= F(2) - F(1) = f(2), \quad D_1' = f(3) - f(2) = \Delta_2, \\ D_2 &= F(3) - F(2) = f(3), \quad D_0'' = \Delta_2 - \Delta_1 = \Delta_1', \end{aligned}$$

and so on. Thus one obtains

$$F(x) = f(0) + f(1) B_1^{(x)} + \Delta_1 B_2^{(x)} + \Delta_1' B_3^{(x)} + \cdots. \tag{4}$$

For example, take $f(x) = x^2$; then $F(m)$ gives the sum of the first m square numbers. Here

$$\Delta_x = 2x + 1, \qquad \Delta_x' = 2,$$

and therefore

$$F(x) = x + 3\frac{x(x-1)}{1 \cdot 2} + 2\frac{x(x-1)(x-2)}{1 \cdot 2 \cdot 3} = \frac{x(x+1)(2x+1)}{6}.$$

For $f(x) = x^3$ the same calculation gives

$$F(x) = \left(\frac{x(x+1)}{2}\right)^2.$$

Thus the sum of the first m cubes is equal to the square of the mth triangular number.

§ 12. The polynomial theorem.

In § 8 the following form of the binomial theorem was derived:

$$(x+y)^n = \Pi(n) \sum^{\alpha,\beta} \frac{x^\alpha y^\beta}{\Pi(\alpha)\Pi(\beta)}, \tag{1}$$

where the sum extends over all combinations of two numbers α, β, neither of which is negative, and which satisfy the condition

$$\alpha + \beta = n. \tag{2}$$

This form permits, first by induction, a generalization to the nth power of a polynomial:

$$(x+y+z+\cdots)^n = \Pi(n) \sum^{\alpha,\beta,\gamma,\ldots} \frac{x^\alpha y^\beta z^\gamma \cdots}{\Pi(\alpha)\Pi(\beta)\Pi(\gamma)\cdots}, \tag{3}$$

with the determination that $\alpha, \beta, \gamma, \ldots$ run through all positive or vanishing integral values which satisfy

$$\alpha + \beta + \gamma + \cdots = n. \tag{4}$$

But to prove the correctness of this formula in general, we assume that it is proved when the polynomial contains one term fewer, as is indeed the case when the polynomial contains only two terms.

We then put

$$u = y + z + \cdots \tag{5}$$

and apply formula (1) to $(x+u)$, from which there follows

$$(x+y+z+\cdots)^n = \Pi(n) \sum^{\alpha,\nu} \frac{x^\alpha u^\nu}{\Pi(\alpha)\Pi(\nu)}, \tag{6}$$

with the restriction

$$\alpha + \nu = n. \tag{7}$$

By the assumption, however, it has already been proved that

$$u^\nu = \Pi(\nu) \sum^{\beta,\gamma,\ldots} \frac{y^\beta z^\gamma \cdots}{\Pi(\beta)\Pi(\gamma)\cdots}, \tag{8}$$

where

$$\beta + \gamma + \cdots = \nu. \tag{9}$$

When this is inserted in (6), formula (3) is obtained immediately, and (7) passes over into (4).

The coefficients

$$P^{(n)}_{\alpha,\beta,\gamma,\ldots} = \frac{\Pi(n)}{\Pi(\alpha)\Pi(\beta)\Pi(\gamma)\cdots}, \tag{10}$$

which, according to their meaning, are whole numbers, are called the polynomial coefficients.

For example, for the third power of the trinomial one obtains

$$\begin{aligned}(x+y+z)^3 = x^3 + y^3 + z^3 + 3x^2y + 3xy^2 + 3x^2z + 3xz^2 \\ + 3y^2z + 3yz^2 + 6xyz.\end{aligned} \tag{11}$$

Volume I. First Section. Rational Functions.

§ 13. Derived Functions.

Let

$$f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_n \tag{1}$$

be an entire rational function of order n.

If in it we replace x by a binomial $x + y$, then we can apply the binomial theorem to each separate term, and can arrange the result according to descending or according to ascending powers of x or of y. We shall carry out the arrangement according to ascending powers of y. The highest power of y which occurs is the nth, and the coefficient of the zeroth power of y is the function $f(x)$ itself, as one sees by putting $y = 0$. We therefore put, denoting the other coefficients by

$$f'(x), \qquad \frac{f''(x)}{\Pi(2)}, \qquad \frac{f'''(x)}{\Pi(3)}, \dots :$$

$$f(x+y) = f(x) + yf'(x) + \frac{y^2}{1\cdot 2}f''(x) + \cdots = \sum_{0,n}^{\nu} \frac{y^\nu}{\Pi(\nu)} f^{(\nu)}(x). \tag{2}$$

The functions $f'(x), f''(x), f'''(x), \dots$ are called the first, second, third, ... derivative or derived function of $f(x)$. They are entire functions of x, and $f^{(\nu)}(x)$ cannot exceed the degree $n - \nu$, since the sum of the exponents of x and y in no term exceeds the degree n.

The first derivative, which is thus the coefficient of the first power of y in the expansion of $f(x + y)$ according to ascending powers of y, is obtained by applying the binomial theorem to (1):

$$f'(x) = na_0x^{n-1} + (n-1)a_1x^{n-2} + (n-2)a_2x^{n-3} + \cdots . \tag{3}$$

The principal theorem on derived functions follows from (2) if we transform x into $x + z$ or y into $y + z$:

$$f(x+y+z) = \sum_{0,n}^{\nu} \frac{y^\nu}{\Pi(\nu)} f^{(\nu)}(x+z) = \sum_{0,n}^{\nu} \frac{(y+z)^\nu}{\Pi(\nu)} f^{(\nu)}(x). \tag{4}$$

If by $f^{(\nu,\mu)}(x)$ we denote the μth derivative of $f^{(\nu)}(x)$, then by (2)

$$f^{(\nu)}(x+z) = \sum_{0,n-\nu}^{\mu} \frac{z^\mu}{\Pi(\mu)} f^{(\nu,\mu)}(x), \tag{5}$$

and by the binomial theorem

$$\frac{(y+z)^\nu}{\Pi(\nu)} = \sum^{\beta,\gamma} \frac{y^\beta z^\gamma}{\Pi(\beta)\Pi(\gamma)}, \qquad \beta + \gamma = \nu. \tag{6}$$

Substituting this in (4), one obtains

$$\sum_{0,n}^{\nu} \sum_{0,n-\nu}^{\mu} \frac{y^\nu z^\mu}{\Pi(\nu)\Pi(\mu)} f^{(\nu,\mu)}(x) = \sum^{\beta,\gamma} \frac{y^\beta z^\gamma}{\Pi(\beta)\Pi(\gamma)} f^{(\beta+\gamma)}(x). \tag{7}$$

The last sum is to be extended over all non-negative numbers β, γ whose sum does not exceed the degree n of $f(x)$. But the same number-combinations are also run through by the exponents ν, μ on the left-hand side, and comparison of the coefficients of equal powers and products gives (by § 1)

$$f^{(\nu,\mu)}(x) = f^{(\nu+\mu)}(x), \tag{8}$$

and therefore the theorem:

The μth derivative of the νth derivative is the $(\nu+\mu)$th derivative of the original function.

Thus all the higher derivatives are obtained by forming from each preceding one, according to the rule (3), the first derivative:

$$\begin{aligned} f(x) &= a_0x^n + a_1x^{n-1} + a_2x^{n-2} + a_3x^{n-3} + \cdots , \\ f'(x) &= na_0x^{n-1} + (n-1)a_1x^{n-2} + (n-2)a_2x^{n-3} + \cdots , \\ f''(x) &= n(n-1)a_0x^{n-2} + (n-1)(n-2)a_1x^{n-3} + \cdots . \end{aligned} \tag{9}$$

These derivatives assume a somewhat simpler form if one uses another notation, which is often in use and for certain purposes almost indispensable, and which we shall discuss in connection with this.

Because of the indeterminacy of the coefficients $a_0, a_1, \ldots, a_n$, it is clearly no restriction if we write an entire rational function of degree n in the form

$$f(x) = a_0x^n + B_1^{(n)}a_1x^{n-1} + B_2^{(n)}a_2x^{n-2} + \cdots + a_n, \tag{10}$$

or explicitly

$$f(x) = a_0x^n + na_1x^{n-1} + \frac{n(n-1)}{1\cdot 2}a_2x^{n-2} + \cdots . \tag{11}$$

When a function $f(x)$ is represented in this way, we shall say that it is "written with the binomial coefficients."

A greater agreement is already shown by the formulae (9), which then read:

$$\begin{aligned} f(x) &= a_0x^n + na_1x^{n-1} + \frac{n(n-1)}{1\cdot 2}a_2x^{n-2} + \cdots , \\ \frac{1}{n}f'(x) &= a_0x^{n-1} + (n-1)a_1x^{n-2} + \frac{(n-1)(n-2)}{1\cdot 2}a_2x^{n-3} + \cdots , \\ \frac{1}{n(n-1)}f''(x) &= a_0x^{n-2} + (n-2)a_1x^{n-3} + \frac{(n-2)(n-3)}{1\cdot 2}a_2x^{n-4} + \cdots , \end{aligned} \tag{12}$$

wherein the right-hand sides all likewise appear written with the binomial coefficients. We shall later become more acquainted with the usefulness of this notation, but already here we must emphasize that the choice of the one or the other manner of representation is nevertheless not wholly indifferent, and that the first often deserves preference. Especially in those cases in which the coefficients are numbers and the number-theoretic behavior of these coefficients is at issue, one must not overlook that through the binomial coefficients a numerical element foreign to the matter is introduced. That Gauss in the theory of quadratic forms (in the Disq. ar.) uses the notation with binomial coefficients when he represents the quadratic forms by $ax^2 + 2bxy + cy^2$, and that this notation has gained general entry, has led in number theory to an unnecessary and very regrettable complication.

§ 14. Derivative of a Product.

The derived functions which we have considered here are no others than the differential quotients known from differential calculus; however, here, where only entire rational functions are involved, we have obtained the concept without using infinitesimal calculus, from the coefficients of development of the powers of y in the function $f(x+y)$. If we denote the νth derivative of $f(x)$ by $D_\nu f$, then by (2), § 13:

$$f(x+y) = f(x) + yD_1f + \frac{y^2}{1\cdot 2}D_2f + \frac{y^3}{1\cdot 2\cdot 3}D_3f + \cdots , \tag{1}$$

and from this follow at once the two fundamental rules expressed in the formulae

$$D_\nu(Cf) = CD_\nu f; \tag{2}$$

$$D_\nu(f+\varphi) = D_\nu f + D_\nu \varphi, \tag{3}$$

where C is a constant, and φ a second entire rational function of x.

A generalization of formula (2) gives the representation of the derivatives of the product $f\varphi$. Namely, if after (1) one writes, for abbreviation,

$$\begin{aligned} f(x+y) &= u_0 + yu_1 + y^2u_2 + \cdots + y^n u_n, \\ \varphi(x+y) &= v_0 + yv_1 + y^2v_2 + \cdots + y^m v_m, \end{aligned} \tag{4}$$

so that

$$u_\nu = \frac{D_\nu f}{\Pi(\nu)}, \qquad v_\nu = \frac{D_\nu \varphi}{\Pi(\nu)}, \tag{5}$$

then carrying out the multiplication of the two formulae (4), and seeking in the product the coefficient of y^ν, gives

$$\frac{D_\nu(f\varphi)}{\Pi(\nu)} = u_\nu v_0 + u_{\nu-1}v_1 + u_{\nu-2}v_2 + \cdots + u_0 v_\nu, \tag{6}$$

or

$$D_\nu(f\varphi) = \varphi D_\nu f + B_1^{(\nu)} D_{\nu-1} f D_1 \varphi + B_2^{(\nu)} D_{\nu-2} f D_2 \varphi + \cdots, \tag{7}$$

where $B_1^{(\nu)}, B_2^{(\nu)}, \ldots$ are the binomial coefficients. In a similar way one can, by using the polynomial coefficients, form the derivatives of a product of more than two factors.

We shall form the first derivative, which we now denote by D instead of by D_1, for a product of n factors. For two factors we obtain from (7)

$$D(f\varphi) = \varphi Df + fD\varphi = \varphi f'(x) + f\varphi'(x),$$

and generally, if we denote the factors by $u_1, u_2, \ldots, u_n$ and their derivatives by $u_1', u_2', \ldots, u_n'$,

$$D(u_1u_2\cdots u_n) = u_1'u_2\cdots u_n + u_1u_2'\cdots u_n + \cdots + u_1u_2\cdots u_n', \tag{8}$$

or more briefly,

$$\frac{D(u_1u_2\cdots u_n)}{u_1u_2\cdots u_n} = \sum \frac{Du_\nu}{u_\nu}.$$

Thus if we consider a product of linear factors

$$f(x) = (x-\alpha_1)(x-\alpha_2)\cdots(x-\alpha_n), \tag{9}$$

we obtain, since the first derivatives of $x-\alpha_1$, $x-\alpha_2, \ldots, x-\alpha_n$ are all equal to 1,

$$\begin{aligned} f'(x) = {} & (x-\alpha_2)(x-\alpha_3)\cdots(x-\alpha_n) \\ & + (x-\alpha_1)(x-\alpha_3)\cdots(x-\alpha_n) + \cdots \\ & + (x-\alpha_1)(x-\alpha_2)\cdots(x-\alpha_{n-1}), \end{aligned} \tag{10}$$

for which one can also write

$$f'(x) = \frac{f(x)}{x-\alpha_1} + \frac{f(x)}{x-\alpha_2} + \cdots + \frac{f(x)}{x-\alpha_n}. \tag{11}$$

A very important result follows from this if x is put equal to one of the values $\alpha_1, \alpha_2, \ldots, \alpha_n$, namely

$$\begin{aligned} f'(\alpha_1) &= (\alpha_1-\alpha_2)(\alpha_1-\alpha_3)\cdots(\alpha_1-\alpha_n), \\ f'(\alpha_2) &= (\alpha_2-\alpha_1)(\alpha_2-\alpha_3)\cdots(\alpha_2-\alpha_n), \\ &\cdots \\ f'(\alpha_n) &= (\alpha_n-\alpha_1)(\alpha_n-\alpha_2)\cdots(\alpha_n-\alpha_{n-1}). \end{aligned} \tag{12}$$

§ 15. Entire Functions of Several Variables: Forms.

Up to now we have chiefly considered entire rational functions of one variable; but we cannot always restrict ourselves to this, and indeed we have already used functions of several variables above. By an entire rational function of degree n in several variables $F(x, y, z, \ldots)$ we understand a sum of terms

$$\sum A_{\alpha,\beta,\gamma,\ldots} x^\alpha y^\beta z^\gamma \cdots ,$$

where $\alpha, \beta, \gamma, \ldots$ are non-negative integral exponents whose sum $\alpha + \beta + \gamma + \cdots$ does not exceed the value n, and in at least one term actually attains it. Thus the degree is determined by the greatest value assumed by the sum $\alpha + \beta + \gamma + \cdots$.

If the sum of the exponents $\alpha + \beta + \gamma + \cdots$ has the same value in all terms, then the function is called homogeneous.

A fundamental property of homogeneous functions of degree n is this: if all variables are multiplied by the same factor, the result is the same as if the function were multiplied by the nth power; in signs, if t denotes an arbitrary variable,

$$F(tx, ty, tz, \ldots) = t^n F(x, y, z, \ldots); \tag{1}$$

for if in the product $x^\alpha y^\beta z^\gamma \cdots$ the variables are replaced by $tx, ty, tz, \ldots$, it receives the factor

$$t^{\alpha+\beta+\gamma+\cdots}.$$

If now $\alpha + \beta + \gamma + \cdots$ has in all terms the same value n, the factor t^n can be taken out before the sum. But if the sum $\alpha + \beta + \gamma + \cdots$ has different values in the separate terms, no such common factor can be taken out, at least not without leaving different powers of t in the separate terms.

By increasing the number of variables one can transform every non-homogeneous function into a homogeneous one of the same degree. If $m - 1$ is the number of variables in a non-homogeneous function of degree n, then we put

$$x = \frac{x_1}{x_m}, \qquad y = \frac{x_2}{x_m}, \qquad z = \frac{x_3}{x_m}, \ldots,$$

and obtain in

$$x_m^n F\left(\frac{x_1}{x_m}, \frac{x_2}{x_m}, \frac{x_3}{x_m}, \ldots\right)$$

an entire homogeneous function of degree n in the variables $x_1, x_2, \ldots, x_m$, which we denote by

$$\Phi(x_1, x_2, \ldots, x_m).$$

It is sometimes advisable to write homogeneous functions of several variables with the polynomial coefficients. We therefore put

$$\Phi(x_1, x_2, \ldots, x_m) = \sum \frac{\Pi(n)}{\Pi(\alpha_1)\Pi(\alpha_2)\cdots\Pi(\alpha_m)} A_{\alpha_1,\alpha_2,\ldots,\alpha_m} x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_m^{\alpha_m}, \tag{2}$$

where the sum is extended over all non-negative numbers satisfying the condition

$$\alpha_1 + \alpha_2 + \cdots + \alpha_m = n. \tag{3}$$

This notation, without the restriction (3), is also applicable to non-homogeneous functions.

But the homogeneous function can also be represented as

$$\Phi(x_1, x_2, \ldots, x_m) = \sum A_{\nu_1,\nu_2,\ldots,\nu_n} x_{\nu_1} x_{\nu_2} \cdots x_{\nu_n}, \tag{4}$$

where each of the indices $\nu_1, \nu_2, \ldots, \nu_n$, independently of the others, runs through the value-row $1, 2, \ldots, m$. Thus the sum (4) consists of m^n terms, but not all of them are different from one another. The product $x_{\nu_1}x_{\nu_2}\cdots x_{\nu_n}$ remains unchanged if the indices $\nu_1, \nu_2, \ldots, \nu_n$ are permuted among themselves in any way. The number of permutations of n elements is $\Pi(n)$. If among these elements $\alpha_1, \alpha_2, \ldots$ respectively are equal to one another, then the number of permutations is reduced to

$$\frac{\Pi(n)}{\Pi(\alpha_1)\Pi(\alpha_2)\cdots},$$

whence it follows that in (4) any product $x_1^{\alpha_1}x_2^{\alpha_2}\cdots$ occurs exactly

$$\frac{\Pi(n)}{\Pi(\alpha_1)\Pi(\alpha_2)\cdots}$$

times. If one further fixes that $A_{\nu_1,\nu_2,\ldots,\nu_n}$ shall not change when the indices are permuted in any way, then the notations (2) and (4) prove to be identical if, by collecting equal factors,

$$x_{\nu_1}x_{\nu_2}\cdots x_{\nu_n} = x_1^{\alpha_1}x_2^{\alpha_2}\cdots x_m^{\alpha_m}$$

and

$$A_{\nu_1,\nu_2,\ldots,\nu_n} = A_{\alpha_1,\alpha_2,\ldots,\alpha_m}$$

are put.

If the number of terms that occur in the function Φ [according to (2)] is denoted by (m, n), then, by first counting the terms which have the factor x_1 and then the remaining ones, which form a homogeneous function of order n of the remaining $m-1$ variables, one finds the recursion formula

$$(m, n) = (m, n-1) + (m-1, n), \tag{5}$$

with whose aid one proves by complete induction the expression

$$(m, n) = \frac{m(m+1)\cdots(m+n-1)}{1\cdot 2\cdots n} = \frac{\Pi(m+n-1)}{\Pi(n)\Pi(m-1)} \tag{6}$$

to be correct.

Entire homogeneous functions are also called forms. According to the number of variables one distinguishes unary forms (simple powers), binary, ternary, quaternary forms. It is the binary forms that especially interest us here; their theory is essentially identical with the theory of entire rational functions of one variable. One passes from the binary forms back to these functions by regarding one of the homogeneous variables as constant, for example by giving it the value 1.

§ 16. The Derivatives of Functions of Several Variables.

In § 13 we defined the derived functions of an entire rational function of one variable. The concept is immediately transferable to functions of several variables, by forming the derivatives with respect to each variable for itself, as though it were the only one. Thus, if one writes for instance as in § 15

$$F(x, y, z, \ldots) = \sum A_{\alpha,\beta,\gamma,\ldots}x^\alpha y^\beta z^\gamma \cdots, \tag{1}$$

one obtains the first derivative with respect to x:

$$F'(x) = \sum \alpha A_{\alpha,\beta,\gamma,\ldots}x^{\alpha-1}y^\beta z^\gamma \cdots, \tag{2}$$

or with respect to y:

$$F'(y) = \sum \beta A_{\alpha,\beta,\gamma,\ldots}x^\alpha y^{\beta-1}z^\gamma \cdots \tag{3}$$

and so on. From these functions one can again form, according to the same rules, the derivatives with respect to the different variables, and thus obtains the higher derivatives.

In order to display the results more clearly, let $\Phi(x_1, x_2, \ldots, x_m)$ be an entire rational function of order n in the m variables $x_1, x_2, \ldots, x_m$. We replace these variables by binomials

$$x_1 + \xi_1, \quad x_2 + \xi_2, \ldots, \quad x_m + \xi_m$$

and expand, in each term of the function

$$\Phi(x_1 + \xi_1, x_2 + \xi_2, \ldots, x_m + \xi_m) = \Phi(x + \xi),$$

by carrying out the multiplication

$$(x_1 + \xi_1)^{\mu_1}(x_2 + \xi_2)^{\mu_2} \cdots (x_m + \xi_m)^{\mu_m}$$

according to powers of $\xi_1, \xi_2, \ldots, \xi_m$. If equal powers and products of the variables ξ are collected each into one term, then for $\Phi(x + \xi)$, in the notation (2) of § 15, one obtains a representation which in differential calculus is called Taylor's expansion:

$$\Phi(x + \xi) = \sum \frac{\xi_1^{\alpha_1} \xi_2^{\alpha_2} \cdots \xi_m^{\alpha_m}}{\Pi(\alpha_1)\Pi(\alpha_2) \cdots \Pi(\alpha_m)} D_{\alpha_1, \alpha_2, \ldots, \alpha_m} \Phi. \tag{4}$$

The coefficients, which we denote by

$$D_{\alpha_1, \alpha_2, \ldots, \alpha_m} \Phi,$$

are functions of the variables x and are called, when

$$\alpha_1 + \alpha_2 + \cdots + \alpha_m = \nu,$$

the derivatives of order ν of the function Φ.

They are also written, according to the notation usual in differential calculus, as

$$D_{\alpha_1, \alpha_2, \ldots, \alpha_m} \Phi = \frac{\partial^\nu \Phi}{\partial x_1^{\alpha_1} \partial x_2^{\alpha_2} \cdots \partial x_m^{\alpha_m}}. \tag{5}$$

The law of formation of the derivatives can be summarized in the following propositions, where for brevity we omit the indices at the sign D.

I. If C is a constant, then

$$D(C\Phi) = CD\Phi.$$

II. If Φ and Ψ are any two functions, then

$$D(\Phi + \Psi) = D\Phi + D\Psi.$$

Both follow immediately from (4).

Thus we can easily form the derived functions in general if we know them for the special case in which Φ is a product of powers, that is, if we know

$$D_{\alpha_1, \alpha_2, \ldots, \alpha_m}(x_1^{\mu_1} x_2^{\mu_2} \cdots x_m^{\mu_m}),$$

where the μ are arbitrary non-negative exponents.

But

$$(x_1 + \xi_1)^{\mu_1} = \sum^{\alpha_1} \frac{\Pi(\mu_1)}{\Pi(\alpha_1)\Pi(\mu_1 - \alpha_1)} \xi_1^{\alpha_1} x_1^{\mu_1 - \alpha_1},$$

and therefore

$$\begin{aligned} &(x_1 + \xi_1)^{\mu_1} \cdots (x_m + \xi_m)^{\mu_m} \\ &= \sum^{\alpha_1 \ldots \alpha_m} \frac{\Pi(\mu_1) \cdots \Pi(\mu_m) x_1^{\mu_1 - \alpha_1} \cdots x_m^{\mu_m - \alpha_m}}{\Pi(\mu_1 - \alpha_1) \cdots \Pi(\mu_m - \alpha_m)\Pi(\alpha_1) \cdots \Pi(\alpha_m)} \xi_1^{\alpha_1} \cdots \xi_m^{\alpha_m}. \end{aligned} \tag{6}$$

Comparison with (4) and (6) therefore gives

$$D_{\alpha_1,\ldots,\alpha_m}(x_1^{\mu_1}\cdots x_m^{\mu_m}) = \frac{\Pi(\mu_1)\cdots\Pi(\mu_m)}{\Pi(\mu_1-\alpha_1)\cdots\Pi(\mu_m-\alpha_m)} x_1^{\mu_1-\alpha_1}\cdots x_m^{\mu_m-\alpha_m}, \tag{7}$$

as long as $\alpha_1 \leq \mu_1, \ldots, \alpha_m \leq \mu_m$. On the other hand,

$$D_{\alpha_1,\ldots,\alpha_m}(x_1^{\mu_1}\cdots x_m^{\mu_m}) = 0 \tag{8}$$

as soon as one of the indices α is greater than the corresponding exponent μ.

Now let $\beta_1, \beta_2, \ldots, \beta_m$ denote a second system of indices, and form from the function (7) the derivative $D_{\beta_1,\beta_2,\ldots,\beta_m}$. By a repeated application of the same formulae (7) and (8) one obtains

$$D_{\beta_1,\ldots,\beta_m} D_{\alpha_1,\ldots,\alpha_m}(x_1^{\mu_1}\cdots x_m^{\mu_m}) = D_{\beta_1+\alpha_1,\ldots,\beta_m+\alpha_m}(x_1^{\mu_1}\cdots x_m^{\mu_m}), \tag{9}$$

and from this, by II, follows the general validity of the theorem

$$\text{III.} \qquad D_{\beta_1,\ldots,\beta_m} D_{\alpha_1,\ldots,\alpha_m}\Phi = D_{\beta_1+\alpha_1,\ldots,\beta_m+\alpha_m}\Phi,$$

which is a generalization of the theorem (8), § 13; hence the higher derivatives can be formed by continued differentiation of the lower ones.

For the first derivatives of a function Φ,

$$D_{1,0,\ldots,0}\Phi, \quad D_{0,1,\ldots,0}\Phi, \ldots, D_{0,0,\ldots,1}\Phi,$$

we also use the shorter signs

$$\Phi'(x_1), \quad \Phi'(x_2), \ldots, \Phi'(x_m),$$

and likewise for the second derivatives

$$D_{2,0,\ldots,0}, \quad D_{1,1,\ldots,0}, \ldots$$

the signs

$$\Phi''(x_1, x_1), \quad \Phi''(x_1, x_2) = \Phi''(x_2, x_1), \ldots.$$

This notation too can be generalized and would lead to an expression corresponding to formula (4), § 15.

Because of their frequent use, we mention the formulae for the quadratic forms especially. Let

$$\Phi(x) = \sum a_{i,k} x_i x_k, \tag{10}$$

where i, k independently run through the row of numbers $1, 2, \ldots, m$ and $a_{i,k} = a_{k,i}$. Then

$$\begin{aligned} \frac{1}{2}\Phi'(x_1) &= a_{1,1}x_1 + a_{1,2}x_2 + \cdots + a_{1,m}x_m, \\ \frac{1}{2}\Phi'(x_2) &= a_{2,1}x_1 + a_{2,2}x_2 + \cdots + a_{2,m}x_m, \\ &\cdots \\ \frac{1}{2}\Phi'(x_m) &= a_{m,1}x_1 + a_{m,2}x_2 + \cdots + a_{m,m}x_m, \end{aligned} \tag{11}$$

and we further put

$$\Phi(x, \xi) = \Phi(\xi, x) = \sum \xi_i \Phi'(x_i) = \sum x_i \Phi'(\xi_i). \tag{12}$$

Then

$$\Phi(x + \xi) = \Phi(x) + \Phi(x, \xi) + \Phi(\xi). \tag{13}$$

The function $\Phi(x, \xi)$ is called the polar of Φ. It is linear and homogeneous both with respect to the x and with respect to the ξ.

It can be expressed by

$$\Phi(x, \xi) = 2\sum a_{ik}\xi_i x_k \tag{14}$$

and satisfies the condition

$$\Phi(x, x) = 2\Phi(x). \tag{15}$$

§ 17. Euler's Theorem on Homogeneous Functions.

From the preceding developments one can easily derive a fundamental theorem on homogeneous functions, discovered by Euler and named after him.

We obtain it most simply from formula (4) of the preceding paragraph if, denoting by t an arbitrary variable, we put

$$\xi_1 = tx_1, \quad \xi_2 = tx_2, \ldots, \quad \xi_m = tx_m \tag{1}$$

and then apply the fundamental formula § 15 (1) for homogeneous functions. We thus obtain first:

$$(1+t)^n \Phi(x) = \sum \frac{(tx_1)^{\alpha_1}(tx_2)^{\alpha_2}\cdots(tx_m)^{\alpha_m}}{\Pi(\alpha_1)\Pi(\alpha_2)\cdots\Pi(\alpha_m)} D_{\alpha_1,\alpha_2,\ldots,\alpha_m}\Phi. \tag{2}$$

If one applies the binomial theorem to the left-hand side of (2) and then equates on both sides the coefficients of equal powers of t, one obtains for every $\nu = 1, 2, \ldots, n$:

$$\frac{\Pi(n)}{\Pi(n-\nu)}\Phi(x_1, x_2, \ldots, x_m) = \sum^{\alpha} \frac{\Pi(\nu)}{\Pi(\alpha_1)\Pi(\alpha_2)\cdots\Pi(\alpha_m)} x_1^{\alpha_1} x_2^{\alpha_2}\cdots x_m^{\alpha_m} D_{\alpha_1,\alpha_2,\ldots,\alpha_m}\Phi, \tag{3}$$

where the sum is extended over all systems of values of the α satisfying the condition

$$\alpha_1 + \alpha_2 + \cdots + \alpha_m = \nu. \tag{4}$$

In this form the theorem to be proved is contained in its full generality. For the special case $\nu = 1$ we obtain the formula

$$n\Phi(x_1, x_2, \ldots, x_m) = x_1\Phi'(x_1) + x_2\Phi'(x_2) + \cdots + x_m\Phi'(x_m), \tag{5}$$

of which formula (15) of the preceding paragraph is a special case, and for $\nu = 2$:

$$n(n-1)\Phi(x_1, x_2, \ldots, x_m) = \sum_{i,k}^{m} x_i x_k \Phi''(x_i, x_k), \tag{6}$$

where the sum is to be extended from $i = 1$ to $i = m$ and from $k = 1$ to $k = m$, so that every term with unequal i, k occurs twice in the sum.

If, when the α are subject to the condition (4), we put

$$\Phi_\nu(\xi, x) = \sum \frac{\xi_1^{\alpha_1}\xi_2^{\alpha_2}\cdots\xi_m^{\alpha_m}}{\Pi(\alpha_1)\Pi(\alpha_2)\cdots\Pi(\alpha_m)} D_{\alpha_1,\alpha_2,\ldots,\alpha_m}\Phi, \tag{7}$$

then by (4) of § 16,

$$\Phi(x+\xi) = \Phi(x) + \Phi_1(x,\xi) + \Phi_2(x,\xi) + \cdots + \Phi_n(x,\xi), \tag{8}$$

and since the left-hand side remains unchanged when x and ξ are interchanged, the relation follows:

$$\Phi_{n-\nu}(x,\xi) = \Phi_\nu(\xi, x), \tag{9}$$

and therefore in particular

$$\Phi_n(x,\xi) = \Phi(\xi). \tag{10}$$

The function $\Phi_\nu(x,\xi)$, regarded as a function of x, is called the νth polar of the function Φ for the value-system ξ.

We shall still specialize formula (3) to the case of a binary form ($m = 2$). We denote the variables by x, y and put, for abbreviation,

$$\Phi(x,y) = u, \qquad D_{h,\nu-h}\Phi = u_h.$$

From (4) we obtain

$$\frac{\Pi(n)}{\Pi(n-\nu)} u = \sum_{0,\nu}^{h} \frac{\Pi(\nu)}{\Pi(h)\Pi(\nu-h)} u_h x^h y^{\nu-h}, \tag{9}$$

where ν can assume any value not greater than n.

Second Section.

Determinants.

§ 18. Permutations of n elements.

We consider a system of n distinct elements of any kind, for example the n digits

$$1, 2, 3, \ldots, n,$$

whose complex, in this definite arrangement, we shall denote by $\mathfrak{A}$. The elements of $\mathfrak{A}$ can be arranged in different ways, for example

$$2, 1, 3, \ldots, n.$$

The passage from one arrangement to another is called a permutation.

If we denote by $\Pi(n)$ the number of the different arrangements, which can depend only on the number n of the elements, then first

$$\Pi(1) = 1, \qquad \Pi(2) = 2.$$

To determine the number in general, we think of an n-th element as added to $n-1$ elements. In every arrangement of the $n-1$ elements the n-th element can now be put in n different places, namely before the first, between the first and the second, between the second and the third, and so on, finally after the $(n-1)$-st; and all arrangements thus obtained are distinct. Hence

$$\Pi(n) = n\Pi(n-1), \tag{1}$$

from which, by complete induction,

$$\Pi(n) = 1 \cdot 2 \cdot 3 \cdots n \tag{2}$$

is obtained, so that here the sign $\Pi(n)$ has the same meaning as in the first section (§ 7).

Any arrangement of the system $\mathfrak{A}$ will be denoted by $\mathfrak{A}'$, or more fully, if $\alpha_1, \alpha_2, \ldots, \alpha_n$ are the digits $1, 2, \ldots, n$ in some order, by

$$\mathfrak{A}' = \alpha_1, \alpha_2, \ldots, \alpha_n. \tag{3}$$

One can derive any arrangement from any other in very different ways, that is, one can carry out a permutation in different ways. Among the several possibilities, those carried out by so-called transpositions, that is, by successive interchange of only two elements, are of special interest for us. By several transpositions performed successively one can derive every arrangement, for example $\mathfrak{A}'$, from $\mathfrak{A}$. For this purpose one may proceed, for instance, by first interchanging in $\mathfrak{A}$ the element 1 with that which stands in the first place in $\mathfrak{A}'$, that is, with α_1 (unless $\alpha_1 = 1$); then, if $\alpha_2 \neq 2$, interchanging 2 with α_2, and so on.

Thus, for example, to pass from $(1, 2, 3, 4)$ to $(4, 3, 2, 1)$, one forms the arrangements

$$(1, 2, 3, 4), \qquad (4, 2, 3, 1), \qquad (4, 3, 2, 1).$$

If we briefly denote a transposition by the two interchanged digits, and therefore the interchange of 1 with 2 by $(1, 2)$, then here we have performed successively the transpositions

$$(1, 4), \quad (2, 3).$$

It is to be observed that the passage from one arrangement to a definite other one can be reached in infinitely many different ways by successive transpositions. Thus the arrangement $(1, 2, 3, 4)$ also passes into $(4, 3, 2, 1)$ by the transpositions

$$(1, 2), \quad (1, 3), \quad (2, 4), \quad (1, 2).$$

§ 19. Permutations of the first and second kind.

The $\Pi(n)$ arrangements of n elements can be separated into two kinds according to the following point of view.

From the n elements of our system one can form

$$\frac{n(n-1)}{2}$$

and no more pairs. We now assign to the n elements $1, 2, \ldots, n$, in a definite way, n real numerical values $a_1, a_2, \ldots, a_n$, and from these numerical values form the $\frac{n(n-1)}{2}$ differences

$$a_1 - a_2, \quad a_1 - a_3, \ldots,$$

where the lower index is to belong to the minuend. The product of differences

$$P = (a_1 - a_2)(a_1 - a_3) \cdots (a_1 - a_n)(a_2 - a_3) \cdots (a_2 - a_n) \cdots (a_{n-1} - a_n) \tag{1}$$

will, when $a_1, a_2, \ldots, a_n$ are mutually distinct numerical values, have a value different from zero, for example a positive one if $a_1 > a_2 > a_3 > \cdots > a_n$ was assumed.

If now we interchange the indices $1, 2, 3, \ldots, n$ in some way, and so pass, say, from $\mathfrak{A}$ to $\mathfrak{A}'$, then P passes into

$$\begin{aligned} P' = (a_{\alpha_1} - a_{\alpha_2})(a_{\alpha_1} - a_{\alpha_3}) \cdots (a_{\alpha_1} - a_{\alpha_n}) \\ \cdot (a_{\alpha_2} - a_{\alpha_3}) \cdots (a_{\alpha_2} - a_{\alpha_n}) \cdots (a_{\alpha_{n-1}} - a_{\alpha_n}), \end{aligned} \tag{2}$$

and, apart from the sign, this product consists of the same factors as P. Thus P' is either equal to P or opposite to P.

I. We count the arrangement $\mathfrak{A}'$, and therefore also the permutation which transforms $\mathfrak{A}$ into $\mathfrak{A}'$, as of the first or of the second kind according as P' is equal to P or opposite to P; hence $\mathfrak{A}$ itself belongs to the first kind.

II. By a simple transposition (h, k), in which h, k denote any two of the digits $1, 2, \ldots, n$, both P and P' change their sign.

For the factors which contain neither h nor k are not affected by this transposition; furthermore, in P and P' there is the factor $\pm(a_h - a_k)$, and the pairs of factors

$$\pm(a_h - a_\nu)(a_k - a_\nu),$$

where ν runs through the series $1, 2, \ldots, n$ with the exception of h, k. The first factor changes its sign, while the pair of factors remains unchanged under the transposition (h, k). Hence follows:

III. The permutations of the first kind are composed of an even number of transpositions, and those of the second kind of an odd number.

The so-called identical permutation, which leaves $\mathfrak{A}$ unchanged, is therefore also to be counted as of the first kind.

From this one obtains further the conclusion: in however many different ways one may derive $\mathfrak{A}'$ from $\mathfrak{A}$ by transpositions, the number of these transpositions is in all these ways consistently even or odd, according as $\mathfrak{A}'$ belongs to the first or to the second kind.

If in all arrangements

$$\mathfrak{A}, \mathfrak{A}', \mathfrak{A}'', \ldots$$

of the n elements we perform one transposition, say $(1, 2)$, then each of these arrangements passes into a definite other one, say $\mathfrak{A}$ into $\mathfrak{B}$, $\mathfrak{A}'$ into $\mathfrak{B}'$, $\mathfrak{A}''$ into $\mathfrak{B}''$, and so on; and if the same transposition is repeated once more, then $\mathfrak{B}$ again passes into $\mathfrak{A}$, $\mathfrak{B}'$ again into $\mathfrak{A}'$, and so on. Hence the arrangements $\mathfrak{B}, \mathfrak{B}', \mathfrak{B}'', \ldots$ are all distinct from one another, and consequently in their totality agree with the totality of the $\mathfrak{A}$'s. Since, as we have seen above, all difference-products

$$P, P', P'', \ldots$$

which are formed from P with the different arrangements $\mathfrak{A}, \mathfrak{A}', \mathfrak{A}'', \ldots$ change their sign under a transposition, it follows that to each $\mathfrak{A}$ of the first kind there corresponds a $\mathfrak{B}$ of the second kind, and to each $\mathfrak{A}$ of the second kind a $\mathfrak{B}$ of the first kind.

IV. Accordingly the number of arrangements of the first kind is just as large as the number of arrangements of the second kind, namely

$$\frac{1}{2}\Pi(n).$$

For $n = 3$ we have the following six arrangements, of which the first horizontal row forms the first kind:

$$\begin{array}{lll} (1,2,3), & (2,3,1), & (3,1,2), \\ (3,2,1), & (2,1,3), & (1,3,2). \end{array}$$

These theorems have here been obtained from the consideration of the product P, hence from a numerical quantity. How one can arrive at the same results without using such a function will be seen in the fourteenth section.

§ 20. Determinants.

We now consider a system of n^2 arbitrary quantities, with which the rational operations of arithmetic can be performed. For a simple notation of these quantities we choose one letter with a double index $a_i^{(k)}$, where both i and k are to run through the series of digits $1, 2, 3, \ldots, n$. For better survey we arrange these quantities in a square, so that all a's with the same upper index stand in a horizontal row, and all a's with the same lower index in a vertical row, and denote this square by Δ:

$$\Delta = \begin{vmatrix} a_1^{(1)} & a_2^{(1)} & a_3^{(1)} & \cdots & a_n^{(1)} \\ a_1^{(2)} & a_2^{(2)} & a_3^{(2)} & \cdots & a_n^{(2)} \\ a_1^{(3)} & a_2^{(3)} & a_3^{(3)} & \cdots & a_n^{(3)} \\ \vdots & \vdots & \vdots & & \vdots \\ a_1^{(n)} & a_2^{(n)} & a_3^{(n)} & \cdots & a_n^{(n)} \end{vmatrix}. \tag{1}$$

For brevity one calls the horizontal rows rows, and the vertical rows columns.

But by the square enclosed between vertical lines we shall understand not only the complex of the quantities a, but a definite arithmetical combination of these quantities, which can be calculated as soon as the a's are given numerically, and which we shall now describe.

Form the product of the terms standing in the diagonal running from upper left to lower right:

$$M = a_1^{(1)} a_2^{(2)} a_3^{(3)} \cdots a_n^{(n)}. \tag{2}$$

From this derive $\Pi(n)$ products

$$M, M', M'', \ldots$$

by permuting the lower indices, and give to each product so obtained the positive or negative sign according as the applied permutation belongs to the first or to the second kind; thus, in the notation of the preceding paragraph,

$$M' = \pm a_{\alpha_1}^{(1)} a_{\alpha_2}^{(2)} \cdots a_{\alpha_n}^{(n)}. \tag{3}$$

The sum of these products is to be Δ. The quantity Δ is called the determinant of the n^2 elements $a_i^{(k)}$, more precisely, when the distinction is necessary, an n-rowed determinant (also a determinant of the n-th degree or n-th order). The term M of this sum, that is, the product of all elements standing in the diagonal of the square, is called the principal term.

For example, if $n = 2$, we obtain

$$\Delta = \begin{vmatrix} a_1^{(1)} & a_2^{(1)} \\ a_1^{(2)} & a_2^{(2)} \end{vmatrix} = a_1^{(1)} a_2^{(2)} - a_2^{(1)} a_1^{(2)}, \tag{4}$$

and for $n = 3$:

$$\begin{aligned} \Delta = a_1^{(1)} a_2^{(2)} a_3^{(3)} + a_2^{(1)} a_3^{(2)} a_1^{(3)} + a_3^{(1)} a_1^{(2)} a_2^{(3)} \\ - a_1^{(1)} a_3^{(2)} a_2^{(3)} - a_2^{(1)} a_1^{(2)} a_3^{(3)} - a_3^{(1)} a_2^{(2)} a_1^{(3)}. \end{aligned} \tag{5}$$

In another notation this is

$$\begin{vmatrix} a & b & c \\ a' & b' & c' \\ a'' & b'' & c'' \end{vmatrix} = ab'c'' + bc'a'' + ca'b'' - ac'b'' - ba'c'' - cb'a''. \tag{6}$$

§ 21. Principal theorems on determinants.

From the concept of the determinant the first theorems useful for application are easily obtained.

If in the product

$$M' = \pm a_{\alpha_1}^{(1)} a_{\alpha_2}^{(2)} \cdots a_{\alpha_n}^{(n)} \tag{1}$$

we rearrange the factors, its value is not changed. We can therefore also order the factors in such a way that the lower indices occur in their natural order $1, 2, \ldots, n$. The upper indices will then occur in a certain permuted order; thus M' will receive the form

$$\pm a_1^{(\beta_1)} a_2^{(\beta_2)} \cdots a_n^{(\beta_n)}, \tag{2}$$

where

$$(\beta_1, \beta_2, \ldots, \beta_n) = \mathfrak{B}$$

is, just as $(\alpha_1, \ldots, \alpha_n)$, an arrangement of the digits $1, 2, \ldots, n$. One obtains the arrangement $\mathfrak{B}$ by undoing, in the factors of M', the transpositions which led to $\mathfrak{A}$, beginning with the last, in order to recover the original arrangement in the row of the lower indices. The resulting order of the upper indices is then the arrangement $\mathfrak{B}$. It follows that $\mathfrak{B}$ belongs to the first or the second kind according as $\mathfrak{A}$ belongs to the first or the second kind, since both arise through the same number of transpositions. The totality of the $\mathfrak{B}$'s represents, just like the totality of the $\mathfrak{A}$'s, all permutations of the n elements, since two different $\mathfrak{A}$'s can never lead to the same $\mathfrak{B}$. Thus is proved:

I. The determinant Δ can also be formed by permuting the upper indices in the principal term $a_1^{(1)} a_2^{(2)} \cdots a_n^{(n)}$ in all possible ways, assigning to every product so formed the positive or negative sign according as the applied permutation belongs to the first or second kind, and then taking the sum of all these products.

In the representation (1) of § 20 the rows are characterized by the upper indices, the columns by the lower indices. Hence this theorem may also be expressed as follows:

II. A determinant is not changed when the rows are made columns and the columns are made rows.

If, in all arrangements $\mathfrak{A}, \mathfrak{A}', \mathfrak{A}'', \ldots$ of the n elements, we interchange any two elements with one another, the totality of these arrangements remains unchanged, but every arrangement of the first kind passes into an arrangement of the second kind, and conversely. If, therefore, in the terms $M, M', M'', \ldots$ of which Δ is composed, any two lower indices are interchanged, every term with positive sign passes into another which in Δ bore the negative sign, and conversely; therefore Δ changes its sign. With the help of II follows the theorem:

III. If in Δ two lower or two upper indices are interchanged with one another, the determinant changes only its sign.

Expressed somewhat differently: if two rows or two columns are interchanged with one another, the determinant changes only its sign; and from this more generally:

IV. If in a determinant the rows or the columns are permuted, the absolute value is not changed, and the sign is not changed or passes into the opposite sign according as the applied permutation belongs to the first or second kind.

From III one obtains the following fundamental theorem:

V. If in two rows or in two columns the terms standing in the same place are equal to one another (more briefly, if two rows are equal to one another), then the determinant has the value zero.

For the interchange of the two rows changes the sign by III, but on the other hand, since both rows are identical, cannot change anything; thus for Δ only the value zero remains.

One only expresses theorem V differently when one says:

VI. One obtains a vanishing determinant if the elements of one row are replaced by the corresponding elements of another row, or, shortly, if one lower or upper index is replaced by another.

§ 22. Subdeterminants.

In every term of the expanded determinant

$$\sum \pm a_{\alpha_1}^{(1)} a_{\alpha_2}^{(2)} \cdots a_{\alpha_n}^{(n)},$$

whose value we now wish to denote by A, every one of the numbers $1, 2, \ldots, n$ occurs once and only once as a lower index. Thus a certain complex of terms will contain the factor $a_1^{(1)}$, another complex the factor $a_1^{(2)}$, and so on, finally a complex the factor $a_1^{(n)}$; every term of the determinant occurs in one and only one of these complexes.

If, therefore, we denote the first of these complexes by $a_1^{(1)} A_1^{(1)}$, the second by $a_1^{(2)} A_1^{(2)}$, and the last by $a_1^{(n)} A_1^{(n)}$, then the determinant can be represented in the following way:

$$A = a_1^{(1)} A_1^{(1)} + a_1^{(2)} A_1^{(2)} + \cdots + a_1^{(n)} A_1^{(n)}. \tag{1}$$

Instead of the lower index 1, we could equally well single out any other index r, and hence put

$$A = a_r^{(1)} A_r^{(1)} + a_r^{(2)} A_r^{(2)} + \cdots + a_r^{(n)} A_r^{(n)}. \tag{2}$$

Here the product $a_r^{(\nu)} A_r^{(\nu)}$ denotes the complex of all terms of the determinant which contain the factor $a_r^{(\nu)}$.

Since the same rules hold for upper as for lower indices, the determinant can also be written in the following way:

$$A = a_1^{(\mu)} A_1^{(\mu)} + a_2^{(\mu)} A_2^{(\mu)} + \cdots + a_n^{(\mu)} A_n^{(\mu)}, \tag{3}$$

where μ, too, may denote any one of the indices $1, 2, \ldots, n$.

The quantities $A_r^{(\nu)}$ thereby completely defined are called the subdeterminants of the determinant A. To become precisely acquainted with their mode of formation, let us first consider the complex $a_1^{(1)} A_1^{(1)}$. It is obtained by leaving the lower index 1 unchanged in the product

$$a_1^{(1)} a_2^{(2)} \cdots a_n^{(n)}$$

and permuting only the remaining indices $2, 3, \ldots, n$ in all ways, and forming the sum of the terms so obtained with regard to the rule of signs. Hence $A_1^{(1)}$ is the $(n-1)$-rowed determinant

$$A_1^{(1)} = \begin{vmatrix} a_2^{(2)} & a_3^{(2)} & \cdots & a_n^{(2)} \\ a_2^{(3)} & a_3^{(3)} & \cdots & a_n^{(3)} \\ \vdots & \vdots & & \vdots \\ a_2^{(n)} & a_3^{(n)} & \cdots & a_n^{(n)} \end{vmatrix}, \tag{4}$$

or the determinant obtained from A by deleting, in the square representing A, the first row and the first column.

From this one easily obtains the meaning of $A_\mu^{(\nu)}$. By making $\nu-1$ interchanges of rows one can bring the ν-th row to the first, and by adding $\mu-1$ interchanges of columns one can bring the μ-th column to the first; apart from this the rows remain unchanged in their order. The determinant itself has taken the factor $(-1)^{\mu+\nu}$, and has remained unchanged in absolute value. In the row order thus changed the element $a_\mu^{(\nu)}$ has taken the place of the element $a_1^{(1)}$, and from this one infers the following rule of formation:

One obtains the subdeterminant $A_\mu^{(\nu)}$ by deleting from the square representing the determinant the two rows which cross in $a_\mu^{(\nu)}$, and adjoining the factor $(-1)^{\mu+\nu}$.

For example, for the determinant of three rows, one obtains the representation

$$\begin{vmatrix} a & b & c \\ a' & b' & c' \\ a'' & b'' & c'' \end{vmatrix} = a(b'c''-c'b'')+b(c'a''-a'c'')+c(a'b''-b'a''). \tag{5}$$

Since the lower index ν does not occur at all in $A_\mu^{(\nu)}$, $A_\mu^{(\nu)}$ is not changed if the lower index ν is replaced by another. But then, by § 21, VI, the determinant vanishes. We therefore obtain from (2) the important relation, in which μ,ν can be any two distinct digits $1,2,\dots,n$:

$$0 = a_1^{(\mu)}A_1^{(\nu)} + a_2^{(\mu)}A_2^{(\nu)} + \cdots + a_n^{(\mu)}A_n^{(\nu)}, \tag{6}$$

and likewise from (3)

$$0 = a_\mu^{(1)}A_\nu^{(1)} + a_\mu^{(2)}A_\nu^{(2)} + \cdots + a_\mu^{(n)}A_\nu^{(n)}. \tag{7}$$

For instance from (5), if a,b,c are replaced by a',b',c', there follows

$$a'(b'c''-c'b'')+b'(c'a''-a'c'')+c'(a'b''-b'a'')=0, \tag{8}$$

a formula of whose correctness one is convinced by the simplest calculation.

If we multiply the relation (6) by an arbitrary factor λ and add it to (2), we obtain the formula

$$\begin{aligned} A = (a_r^{(1)}+\lambda a_s^{(1)})A_r^{(1)} + (a_r^{(2)}+\lambda a_s^{(2)})A_r^{(2)} \\ + \cdots + (a_r^{(n)}+\lambda a_s^{(n)})A_r^{(n)}, \end{aligned} \tag{9}$$

which expresses for us the following theorem:

VII. The determinant does not change its value if to the elements of one row one adds the corresponding elements of another row multiplied by an arbitrary common factor.

The same theorem holds also for the columns. It is often used with advantage for the simplification and numerical calculation of determinants. We add the following theorems, which can be read off at once from the representations (2), (3).

VIII. If all elements of a row or of a column have a common factor, then this factor can be omitted from the row or column and put as a factor before the determinant.

For by (2) it is

$$pa_r^{(1)}A_r^{(1)} + pa_r^{(2)}A_r^{(2)} + \cdots + pa_r^{(n)}A_r^{(n)} = pA.$$

IX. If in a row or in a column all elements but one vanish, then the determinant reduces to the product of this one element and the corresponding subdeterminant.

For if $a_r^{(1)}, a_r^{(2)}, \dots, a_r^{(n)}$, with the exception of $a_r^{(\nu)}$, vanish, then by (2)

$$A = a_r^{(\nu)}A_r^{(\nu)}.$$

The value of A is then wholly independent of the $a_1^{(\nu)}, a_2^{(\nu)}, \dots, a_n^{(\nu)}$ with the exception of $a_r^{(\nu)}$.

To make an application of these theorems, we determine the value of the determinant

$$\Delta = \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix},$$

where a, b, c are arbitrary quantities. If we multiply the second column by a and subtract it from the third, then multiply the first by a and subtract it from the second, then by VII

$$\Delta = \begin{vmatrix} 1 & 0 & 0 \\ 1 & b-a & b(b-a) \\ 1 & c-a & c(c-a) \end{vmatrix},$$

and by IX and VIII

$$\Delta = (b-a)(c-a) \begin{vmatrix} 1 & b \\ 1 & c \end{vmatrix} = (b-a)(c-a)(c-b). \tag{10}$$

In the same way one can also treat the n-rowed determinant

$$\begin{vmatrix} 1 & a_1 & a_1^2 & \cdots & a_1^{n-1} \\ 1 & a_2 & a_2^2 & \cdots & a_2^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & a_n & a_n^2 & \cdots & a_n^{n-1} \end{vmatrix}$$

and finds its value equal to

$$(a_2 - a_1)(a_3 - a_1) \cdots (a_n - a_1)(a_3 - a_2) \cdots (a_n - a_2) \cdots (a_n - a_{n-1}). \tag{11}$$

If one orders the columns in the reverse sequence, then, according as n is even or odd,

$$\frac{n}{2}(n-1) \quad \text{or} \quad \frac{n-1}{2}n$$

interchanges are necessary; in either case this amounts to distinguishing the determinant so ordered from Δ by the factor

$$(-1)^{n(n-1)/2}.$$

It comes to the same thing if, in the $\frac{n(n-1)}{2}$ factors of the product (11), the opposite sign is given. Thus, along with (11), the equality also holds:

$$\begin{vmatrix} a_1^{n-1} & a_1^{n-2} & \cdots & a_1 & 1 \\ a_2^{n-1} & a_2^{n-2} & \cdots & a_2 & 1 \\ \vdots & \vdots & & \vdots & \vdots \\ a_n^{n-1} & a_n^{n-2} & \cdots & a_n & 1 \end{vmatrix} = \prod_{1 \le i < k \le n} (a_i - a_k). \tag{12}$$

We still mention here a notation for the subdeterminants which is borrowed from differential calculus and is often used with advantage, especially when derivatives are to be formed. If the quantities $a_\mu^{(\nu)}$ are regarded as independent variables, then the determinant A is only of the first degree with respect to each of them. The derivative or differential quotient taken with respect to $a_\mu^{(\nu)}$ is therefore equal to the coefficient of $a_\mu^{(\nu)}$, hence

$$\frac{\partial A}{\partial a_\mu^{(\nu)}} = A_\mu^{(\nu)}.$$

If therefore, for example, the $a_\mu^{(\nu)}$ are functions of a variable t, then A is also a function of t, and by the first rules of differential calculus one obtains the derivative of A with respect to t in the form

$$A'(t) = \sum A_\mu^{(\nu)} \frac{\partial a_\mu^{(\nu)}}{\partial t},$$

where $\partial a_\mu^{(\nu)}/\partial t$ denotes the derivative of $a_\mu^{(\nu)}$ with respect to t.

§ 23. The subdeterminants in the wider sense

We can now generalize the considerations of the preceding paragraph in the following way. Just as there we started from the problem of seeking out all terms in the developed determinant A which contain the factor $a_1^{(1)}$, so now we seek all terms which contain the factor

$$a_1^{(1)} a_2^{(2)} \cdots a_\nu^{(\nu)},$$

where ν may be any number not exceeding n. These terms are obtained from the principal term

$$a_1^{(1)} a_2^{(2)} \cdots a_\nu^{(\nu)} a_{\nu+1}^{(\nu+1)} \cdots a_n^{(n)}$$

if, in permuting the lower indices, one leaves $1, 2, \ldots, \nu$ unchanged and permutes only $\nu+1, \ldots, n$ in all possible ways, with observance of the rule of signs. Thus the aggregate of the required terms is

$$a_1^{(1)} a_2^{(2)} \cdots a_\nu^{(\nu)} \begin{vmatrix} a_{\nu+1}^{(\nu+1)} & \cdots & a_n^{(\nu+1)} \\ \vdots & & \vdots \\ a_{\nu+1}^{(n)} & \cdots & a_n^{(n)} \end{vmatrix}. \tag{1}$$

I. The determinant of $n-\nu$ rows which occurs here as a factor, and which we denote by

$$A_{1,2,\ldots,\nu}^{1,2,\ldots,\nu},$$

arises from A by suppressing the first ν rows and columns.

We shall generalize this result as follows. Choose any ν elements

$$a_{\alpha_1}^{(\beta_1)}, a_{\alpha_2}^{(\beta_2)}, \ldots, a_{\alpha_\nu}^{(\beta_\nu)} \tag{3}$$

so that no two of them occur in the same row or in the same column, that is, so that neither a lower nor an upper index is repeated. We denote by

$$A_{\alpha_1,\alpha_2,\ldots,\alpha_\nu}^{\beta_1,\beta_2,\ldots,\beta_\nu} \tag{2}$$

the aggregate of those terms of the determinant which contain the product of these elements as a factor.

By interchanging rows and columns, whereby at most the sign of the determinant is changed, one can always bring the elements (3) into the places of

$$a_1^{(1)}, a_2^{(2)}, \ldots, a_\nu^{(\nu)}.$$

Then rule I applies, and one obtains:

II. Apart from sign, $A_{\alpha_1,\ldots,\alpha_\nu}^{\beta_1,\ldots,\beta_\nu}$ is obtained as an $(n-\nu)$-rowed determinant by deleting from A all rows and all columns which meet in one of the elements (3), and by leaving the remaining rows and columns in their order.

For the determination of the sign one uses the following rule. Arrange the lower and upper indices $1, 2, \ldots, n$ in the orders

$$\alpha_1, \alpha_2, \ldots, \alpha_\nu, \alpha_{\nu+1}, \ldots, \alpha_n, \tag{4}$$

$$\beta_1, \beta_2, \ldots, \beta_\nu, \beta_{\nu+1}, \ldots, \beta_n, \tag{5}$$

where the missing $\alpha_{\nu+1}, \ldots, \alpha_n$, and likewise $\beta_{\nu+1}, \ldots, \beta_n$, are taken in increasing order.

III. The determinant of $n-\nu$ rows described in II receives the positive or negative sign according as the two arrangements (4), (5) of the digits $1, 2, \ldots, n$ both belong to the same kind or to different kinds.

Indeed, the determinant changes its sign through each interchange of two lower or two upper indices. In order to reduce the general case (2) to the special case (1), one has to make just so many transpositions of upper and lower indices that the permutations (4) and (5) both pass into the original arrangement $1, 2, 3, \ldots, n$; the same number of changes of sign has then taken place.

The quantities so defined are called the ν-th subdeterminants, or subdeterminants of the ν-th order. They are represented by determinants of $n - \nu$ rows. From III follows the theorem:

IV. The subdeterminant

$$A^{\beta_1,\ldots,\beta_\nu}_{\alpha_1,\ldots,\alpha_\nu}$$

changes only its sign if two of its lower or two of its upper indices are interchanged; more generally, it remains unaltered in absolute value if the order of the indices is replaced by any other order, and changes sign or not according as this permutation is of the second or of the first kind.

If $\bar{\alpha}_1, \bar{\alpha}_2, \ldots, \bar{\alpha}_\nu$ denotes any ordering of $\alpha_1, \alpha_2, \ldots, \alpha_\nu$, then the determinant A also contains the complex of terms

$$\pm a^{(\beta_1)}_{\bar{\alpha}_1} a^{(\beta_2)}_{\bar{\alpha}_2} \cdots a^{(\beta_\nu)}_{\bar{\alpha}_\nu} A^{\beta_1,\ldots,\beta_\nu}_{\bar{\alpha}_1,\ldots,\bar{\alpha}_\nu}.$$

Collecting all these terms gives the complex

$$A^{\beta_1,\ldots,\beta_\nu}_{\alpha_1,\ldots,\alpha_\nu} \sum \pm a^{(\beta_1)}_{\alpha_1} a^{(\beta_2)}_{\alpha_2} \cdots a^{(\beta_\nu)}_{\alpha_\nu}. \tag{6}$$

The ν-rowed determinant which occurs in (6) is called the subdeterminant complementary to $A^{\beta_1,\ldots,\beta_\nu}_{\alpha_1,\ldots,\alpha_\nu}$, and is denoted by

$$B^{\beta_1,\ldots,\beta_\nu}_{\alpha_1,\ldots,\alpha_\nu}.$$

It contains exactly the rows and columns missing in $A^{\beta_1,\ldots,\beta_\nu}_{\alpha_1,\ldots,\alpha_\nu}$, and, apart from sign, agrees with a subdeterminant of order $n - \nu$. Thus the complex of terms (6) is denoted by

$$A^{\beta_1,\ldots,\beta_\nu}_{\alpha_1,\ldots,\alpha_\nu} B^{\beta_1,\ldots,\beta_\nu}_{\alpha_1,\ldots,\alpha_\nu}. \tag{7}$$

If now, for $\alpha_1, \ldots, \alpha_\nu$, one chooses each combination of ν of the digits $1, 2, \ldots, n$, while $\beta_1, \ldots, \beta_\nu$ is held fixed, one obtains the same number of complexes of the form (7), and every term of the determinant A occurs in one and only one of them.

V. Consequently, by summing all expressions (7), one obtains the determinant A:

$$A = \sum A^{\beta_1,\ldots,\beta_\nu}_{\alpha_1,\ldots,\alpha_\nu} B^{\beta_1,\ldots,\beta_\nu}_{\alpha_1,\ldots,\alpha_\nu}. \tag{8}$$

Of course one may also keep the combination of the α's fixed and sum with respect to the β's. This is Laplace's theorem.

Another representation of the determinant A by the first and second subdeterminants is obtained as follows. Choose in A any two rows which intersect in an element, say $a^{(\mu)}_\nu$. In every term of A there occurs one element with lower index ν and one with upper index μ. Hence one may write

$$A = a^{(\mu)}_\nu A^{(\mu)}_\nu + \sum a^{(i)}_\nu a^{(\mu)}_k A^{\mu i}_{\nu k}, \tag{9}$$

or, according to IV,

$$A = a^{(\mu)}_\nu A^{(\mu)}_\nu - \sum a^{(i)}_\nu a^{(\mu)}_k A^{\mu i}_{\nu k}. \tag{10}$$

Here i denotes every index different from μ, and k every index different from ν. The quantity $A^{\mu i}_{\nu k}$ is at the same time the first subdeterminant, corresponding to the element $a^{(i)}_k$, of the $(n-1)$-rowed determinant $A^{(\mu)}_\nu$.

For the bordered determinant

$$U = \begin{vmatrix} a^{(1)}_1 & a^{(1)}_2 & \cdots & a^{(1)}_n & u_1 \\ a^{(2)}_1 & a^{(2)}_2 & \cdots & a^{(2)}_n & u_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a^{(n)}_1 & a^{(n)}_2 & \cdots & a^{(n)}_n & u_n \\ v_1 & v_2 & \cdots & v_n & q \end{vmatrix} \tag{11}$$

one obtains, by developing with respect to the elements of the last row and column,

$$U = qA - \sum u_i v_k A_k^{(i)}. \tag{12}$$

In the higher subdeterminants it is sometimes useful to denote them by differential quotients; for example

$$A_{\nu k}^{\mu i} = \frac{\partial^2 A}{\partial a_\nu^{(\mu)}\, \partial a_k^{(i)}}. \tag{13}$$

§ 24. Linear homogeneous equations

The principal application of determinants, from which the whole theory took its origin, is the solution of linear equations. We treat the problem at once in its most general form, since the special form gives scarcely any simplification and is easily derived afterwards from the general result.

Consider a system of m equations of the first degree in which n unknowns $x_1, x_2, \ldots, x_n$ occur homogeneously:

$$\begin{aligned} a_1^{(1)} x_1 + a_2^{(1)} x_2 + \cdots + a_n^{(1)} x_n &= 0, \\ a_1^{(2)} x_1 + a_2^{(2)} x_2 + \cdots + a_n^{(2)} x_n &= 0, \\ &\vdots \\ a_1^{(m)} x_1 + a_2^{(m)} x_2 + \cdots + a_n^{(m)} x_n &= 0. \end{aligned} \tag{1}$$

The coefficients $a_i^{(k)}$ are regarded as given quantities. For the present we make no assumption about the numbers m, n, but ask generally for all systems of values of $x_1, x_2, \ldots, x_n$ satisfying (1).

One solution is immediate, whatever the coefficients may be:

$$x_1 = 0,\ x_2 = 0, \ldots, x_n = 0. \tag{2}$$

There is also the other extreme case in which all coefficients vanish; then the equations are satisfied for arbitrary values of the unknowns.

We write the system of coefficients as the rectangular scheme

$$\begin{matrix} a_1^{(1)} & a_2^{(1)} & \cdots & a_n^{(1)} \\ a_1^{(2)} & a_2^{(2)} & \cdots & a_n^{(2)} \\ \vdots & \vdots & & \vdots \\ a_1^{(m)} & a_2^{(m)} & \cdots & a_n^{(m)}. \end{matrix} \tag{3}$$

Such a scheme, which by itself has no numerical meaning, is called a matrix insofar as it is regarded as the source of a greater number of determinants. The determinants arising from the matrix are obtained by deleting arbitrary rows and columns, in such a number that the elements left form a square, and then regarding this square as a determinant.

Assume that among the ν-rowed determinants of the matrix at least one is different from zero, while all $(\nu+1)$-rowed and therefore all higher determinants, if such exist, vanish. Excluding the uninteresting case in which all coefficients vanish, such a number ν always exists. Without loss of generality, we may suppose that the non-vanishing determinant is

$$A = \begin{vmatrix} a_1^{(1)} & a_2^{(1)} & \cdots & a_\nu^{(1)} \\ a_1^{(2)} & a_2^{(2)} & \cdots & a_\nu^{(2)} \\ \vdots & \vdots & & \vdots \\ a_1^{(\nu)} & a_2^{(\nu)} & \cdots & a_\nu^{(\nu)} \end{vmatrix}. \tag{4}$$

The subdeterminants of A are denoted, as before, by $A_i^{(k)}$, where i, k range from 1 to ν.

I. If first $\nu = n$, which presupposes that m is not smaller than n, then the equations (1) have no other solution than (2).

For take the first n equations. If these are multiplied successively by $A_\mu^{(1)}, A_\mu^{(2)}, \dots, A_\mu^{(n)}$, where μ is any of the indices $1, 2, \dots, n$, and then added, the relations of the preceding paragraph give

$$Ax_\mu = 0,$$

and since, by hypothesis, $A \neq 0$, it follows that $x_\mu = 0$.

II. In particular, if a system of n linear homogeneous equations with n unknowns has a determinant different from zero, all unknowns have the value zero. Equivalently: if such a system has a solution in which not all unknowns vanish, then the determinant of the system vanishes.

We now suppose that $\nu < n$. Since $m \geq \nu$, choose the first ν equations and write them in the form

$$\begin{aligned} a_1^{(1)} x_1 + \cdots + a_\nu^{(1)} x_\nu &= -a_{\nu+1}^{(1)} x_{\nu+1} - \cdots - a_n^{(1)} x_n, \\ &\vdots \\ a_1^{(\nu)} x_1 + \cdots + a_\nu^{(\nu)} x_\nu &= -a_{\nu+1}^{(\nu)} x_{\nu+1} - \cdots - a_n^{(\nu)} x_n. \end{aligned} \tag{6}$$

If these equations are multiplied by the corresponding subdeterminants $A_\mu^{(1)}, \dots, A_\mu^{(\nu)}$, and added, one obtains

$$Ax_\mu = -\sum_{h=\nu+1}^{n} C_{\mu h} x_h, \tag{7}$$

where, for abbreviation,

$$C_{\mu h} = \sum_{i=1}^{\nu} a_h^{(i)} A_\mu^{(i)}, \qquad h = \nu+1, \dots, n. \tag{8}$$

By the theorem on subdeterminants, $C_{\mu h}$ is the determinant obtained from (4) by replacing the μ-th column by $a_h^{(1)}, a_h^{(2)}, \dots, a_h^{(\nu)}$. Thus (7), since $A \neq 0$, expresses $x_1, \dots, x_\nu$ linearly in terms of $x_{\nu+1}, \dots, x_n$ and the known coefficients.

III. The expressions (7) satisfy all equations (1), whatever values are assigned to $x_{\nu+1}, \dots, x_n$; hence $n - \nu$ of the unknowns remain arbitrary and the other ν are dependent on them according to (7).

Indeed, writing (7) in the form

$$Ax_\mu = -\sum_{h=\nu+1}^{n} \sum_{i=1}^{\nu} x_h a_h^{(i)} A_\mu^{(i)}, \tag{9}$$

and multiplying by $a_\mu^{(k)}$ and summing over $\mu = 1, \dots, \nu$, then adding the remaining terms, gives

$$A \sum_{\mu=1}^{n} a_\mu^{(k)} x_\mu = \sum_{h=\nu+1}^{n} x_h \left(A a_h^{(k)} - \sum_{i=1}^{\nu} \sum_{\mu=1}^{\nu} a_h^{(i)} a_\mu^{(k)} A_\mu^{(i)} \right). \tag{11}$$

The factor of x_h on the right is, by (12) of the preceding paragraph, a $(\nu+1)$-rowed determinant of the matrix (3), and therefore vanishes by hypothesis; if $k \leq \nu$ it vanishes because two rows coincide. Thus

$$\sum_{\mu=1}^{n} a_\mu^{(k)} x_\mu = 0, \qquad k = 1, 2, \dots, m, \tag{13}$$

as was required.

A useful special case is $m = n - 1$, $\nu = n - 1$. Then only one of the unknowns remains arbitrary, and the ratios of the unknowns are fully determined. If the $(n-1)$-rowed determinants of the matrix

$$\begin{array}{cccc} a_1^{(1)} & a_2^{(1)} & \cdots & a_n^{(1)} \\ a_1^{(2)} & a_2^{(2)} & \cdots & a_n^{(2)} \\ \vdots & \vdots & & \vdots \\ a_1^{(n-1)} & a_2^{(n-1)} & \cdots & a_n^{(n-1)} \end{array} \tag{14}$$

taken with alternating sign are denoted by $A_1, A_2, \ldots, A_n$, and if at least one of them is different from zero, the solution of

$$\begin{gathered} a_1^{(1)}x_1 + a_2^{(1)}x_2 + \cdots + a_n^{(1)}x_n = 0, \\ \vdots \\ a_1^{(n-1)}x_1 + a_2^{(n-1)}x_2 + \cdots + a_n^{(n-1)}x_n = 0 \end{gathered} \tag{15}$$

is given by the ratios

$$x_1 : x_2 : \cdots : x_n = A_1 : A_2 : \cdots : A_n. \tag{16}$$

For $n = 3$, the system

$$ax + by + cz = 0, \qquad a'x + b'y + c'z = 0 \tag{17}$$

has the solution in ratios

$$x : y : z = bc' - cb' : ca' - ac' : ab' - ba'. \tag{18}$$

§ 25. Elimination from linear equations

It sometimes happens that, for a given system of linear equations, the question is not so much the actual determination of the unknowns as the decision whether a solution is possible; that is, the setting up of the condition equations which must hold between the coefficients if solutions, or solutions of a prescribed kind, are to exist. The setting up of these condition equations is called elimination. The solution of this problem has already been contained implicitly in what precedes; nevertheless we return explicitly to a few questions belonging here.

Let again a system of m linear homogeneous equations with n unknowns be given, and ask when the system has a solution in which not all unknowns vanish. This is always the case when $n > m$. If, however, $n \le m$, the necessary and sufficient condition for such a solution is that all n-rowed determinants of the matrix vanish. For if one of them does not vanish, the unknowns must all be zero by §24, II; but if they all vanish, one can find a number $\nu < n$ such that all $(\nu+1)$-rowed determinants vanish while at least one ν-rowed determinant does not vanish, and then §24, III gives a solution of the required kind.

From the matrix one can form, when $n \le m$,

$$\frac{m(m-1)\cdots(m-n+1)}{1 \cdot 2 \cdots n}$$

n-rowed determinants, and this would be the number of conditions. If $n = m$, this number is 1. In general, however, although all these conditions must be fulfilled, their number is greater than necessary, because some are necessary consequences of the others.

To obtain a system of necessary, sufficient, and mutually independent conditions, make the question more precise: what are the conditions that from a system of m linear homogeneous equations with n unknowns, ν of the unknowns can be completely determined by the remaining $n - \nu$, which remain arbitrary?

Assume that $x_{\nu+1}, x_{\nu+2}, \ldots, x_n$ are to remain arbitrary and that $x_1, x_2, \ldots, x_\nu$ are to be determined by them, and assume

$$A = \begin{vmatrix} a_1^{(1)} & a_2^{(1)} & \cdots & a_\nu^{(1)} \\ \vdots & \vdots & & \vdots \\ a_1^{(\nu)} & a_2^{(\nu)} & \cdots & a_\nu^{(\nu)} \end{vmatrix} \ne 0. \tag{1}$$

The unknowns $x_1, \ldots, x_\nu$ are then calculated by §24, (9), and one forms the sums §24, (11), whose vanishing expresses that the given system is actually satisfied. The conditions are therefore the vanishing of the determinants

$$\begin{vmatrix} a_1^{(1)} & \cdots & a_\nu^{(1)} & a_h^{(1)} \\ \vdots & & \vdots & \vdots \\ a_1^{(\nu)} & \cdots & a_\nu^{(\nu)} & a_h^{(\nu)} \\ a_1^{(k)} & \cdots & a_\nu^{(k)} & a_h^{(k)} \end{vmatrix} = 0, \tag{2}$$

or, written otherwise,

$$A a_h^{(k)} - \sum_{i=1}^{\nu} \sum_{\mu=1}^{\nu} a_h^{(i)} a_\mu^{(k)} A_\mu^{(i)} = 0. \tag{3}$$

For $h = \nu+1, \ldots, n$ and $k = \nu+1, \ldots, m$ these give

$$(n-\nu)(m-\nu) \tag{5}$$

conditions. These conditions are mutually independent. For the left sides of (3) can, by suitable assumptions about the coefficients a, be made to take arbitrary values for each index combination, as is seen by setting all $a_h^{(i)}$, $a_\mu^{(k)}$, except the one in question, equal to zero.

§ 26. Non-homogeneous linear equations

We have simplified the problem of solving linear equations, and at the same time led it back to more general laws, by supposing the equations to be homogeneous in the unknowns. In applications, however, the unknowns often do not occur homogeneously. Although in principle the one case is not essentially different from the other, we consider the non-homogeneous case separately.

We can derive it from the homogeneous case by adding the requirement that one of the unknowns shall have the value 1. If this unknown belongs among those which the problem leaves arbitrary, there is no difficulty; if it belongs among those determined by the rest, further conditions have to be fulfilled. We treat the question independently, with a slightly changed notation, and restrict ourselves to the most important case in which the number of unknowns equals the number of equations.

Let the following system be solved for $x_1, x_2, \ldots, x_n$:

$$\begin{aligned} a_1^{(1)} x_1 + a_2^{(1)} x_2 + \cdots + a_n^{(1)} x_n &= y_1, \\ a_1^{(2)} x_1 + a_2^{(2)} x_2 + \cdots + a_n^{(2)} x_n &= y_2, \\ &\vdots \\ a_1^{(n)} x_1 + a_2^{(n)} x_2 + \cdots + a_n^{(n)} x_n &= y_n. \end{aligned} \tag{1}$$

The coefficients and the independent terms are regarded as given. Let

$$A = \sum \pm a_1^{(1)} a_2^{(2)} \cdots a_n^{(n)} \tag{2}$$

be the determinant of the system, and let $A_\mu^{(k)}$ denote its first subdeterminants. Multiplying the equations successively by $A_\mu^{(1)}, A_\mu^{(2)}, \ldots, A_\mu^{(n)}$ and adding, one obtains

$$A x_\mu = \sum_{k=1}^{n} y_k A_\mu^{(k)}. \tag{4}$$

If $A \neq 0$, the values of the unknowns are uniquely determined and satisfy (1). If we put $A_\mu^{(k)} = A_{k\mu}$, the solution can be written in the form

$$\begin{aligned} Ax_1 &= A_{11}y_1 + A_{12}y_2 + \cdots + A_{1n}y_n, \\ Ax_2 &= A_{21}y_1 + A_{22}y_2 + \cdots + A_{2n}y_n, \\ &\vdots \\ Ax_n &= A_{n1}y_1 + A_{n2}y_2 + \cdots + A_{nn}y_n, \end{aligned} \tag{5}$$

which displays the analogy with the original system.

If $A = 0$, the equations (4) give no values for the x; they only give conditions which the y's must necessarily satisfy if the equations are to be soluble at all. Suppose that, besides A, all subdeterminants up to a certain order vanish, and denote by

$$B = \begin{vmatrix} a_1^{(1)} & a_2^{(1)} & \cdots & a_\nu^{(1)} \\ \vdots & \vdots & & \vdots \\ a_1^{(\nu)} & a_2^{(\nu)} & \cdots & a_\nu^{(\nu)} \end{vmatrix} \tag{6}$$

a non-vanishing ν-rowed subdeterminant. If all $(\nu + 1)$-rowed subdeterminants of A vanish, multiplication of the first ν equations by the subdeterminants $B_\mu^{(1)}, \dots, B_\mu^{(\nu)}$ gives

$$Bx_\mu = \sum_{i=1}^{\nu} B_\mu^{(i)} y_i - \sum_{h=\nu+1}^{n} x_h \sum_{i=1}^{\nu} B_\mu^{(i)} a_h^{(i)}. \tag{7}$$

Thus ν of the unknowns are determined by the remaining ones. Substitution into (1) shows that the equations are satisfied if and only if, for $\lambda = \nu + 1, \dots, n$,

$$By_\lambda = \sum_{i=1}^{\nu} \sum_{\mu=1}^{\nu} B_\mu^{(i)} a_\mu^{(\lambda)} y_i. \tag{10}$$

These are $n - \nu$ independent conditions for the y's. If one of them is not fulfilled, the given system of equations has no solution.

If in (1) the x's are regarded not as unknowns but as variables, the y's also become variable quantities. In this interpretation the system (1) is called a linear substitution, since it mediates the passage from one system of variables to another. Only when the determinant A, now called the determinant of the substitution, is different from zero can the y's be considered independent variables. In that case (5) gives the representation of the variables x by the variables y, or the substitution inverse to (1).

§ 27. Multiplication of determinants.

The theorem which we still have to prove teaches how one can represent the product of two determinants with the same number of rows by a single determinant with the same number of rows. One is led to it most simply by considering the solution of two systems of linear equations.

Let the coefficients $a_i^{(k)}, b_h^{(k)}$, when i, h run through the sequence of numbers $1, 2, \dots, n$, be arbitrary variable quantities; likewise let the quantities x_i, y_i, z_i be bound only by the relations

$$\sum_i a_i^{(k)} x_i = y_k, \tag{1}$$

$$\sum_k b_h^{(k)} y_k = z_h. \tag{2}$$

If the problem is now posed of determining the variables x by the variables z, this problem can be solved in two ways. By § 26, (4) one may solve equations (1) with respect to x, equations (2) with respect to y, and insert the latter expressions into the former. If A, B denote the determinants of the two systems of equations, that is,

$$A = \begin{vmatrix} a_1^{(1)} & a_2^{(1)} & \cdots & a_n^{(1)} \\ a_1^{(2)} & a_2^{(2)} & \cdots & a_n^{(2)} \\ \cdots & \cdots & & \cdots \\ a_1^{(n)} & a_2^{(n)} & \cdots & a_n^{(n)} \end{vmatrix}, \qquad B = \begin{vmatrix} b_1^{(1)} & b_2^{(1)} & \cdots & b_n^{(1)} \\ b_1^{(2)} & b_2^{(2)} & \cdots & b_n^{(2)} \\ \cdots & \cdots & & \cdots \\ b_1^{(n)} & b_2^{(n)} & \cdots & b_n^{(n)} \end{vmatrix},$$

and if $A_i^{(k)}, B_i^{(k)}$ denote the first subdeterminants, one obtains

$$Ax_k = \sum_i A_k^{(i)} y_i, \tag{3}$$

$$By_i = \sum_h B_i^{(h)} z_h, \tag{4}$$

and, by substitution of (4) in (3),

$$ABx_k = \sum_h z_h \sum_i A_k^{(i)} B_i^{(h)}. \tag{5}$$

But one may also proceed by inserting from (1) the expressions for y into (2), whereby, if

$$c_i^{(h)} = \sum_s a_i^{(s)} b_h^{(s)} \tag{6}$$

is put, one obtains

$$\sum_i c_i^{(h)} x_i = z_h. \tag{7}$$

Now put

$$C = \begin{vmatrix} c_1^{(1)} & c_2^{(1)} & \cdots & c_n^{(1)} \\ c_1^{(2)} & c_2^{(2)} & \cdots & c_n^{(2)} \\ \cdots & \cdots & & \cdots \\ c_1^{(n)} & c_2^{(n)} & \cdots & c_n^{(n)} \end{vmatrix}. \tag{8}$$

If $C_i^{(k)}$ denotes the subdeterminants of C, the solution of (7) gives

$$Cx_k = \sum_h z_h C_k^{(h)}, \tag{9}$$

and comparison of (5) with (9) gives

$$AB = C, \tag{10}$$

and for the subdeterminants

$$C_k^{(h)} = \sum_i A_k^{(i)} B_i^{(h)}. \tag{11}$$

In this conclusion, however, there is still a gap, caused by the doubt whether (5) and (9) might not differ by a factor common to both sides. To remove this doubt we shall prove equation (10) directly; the correctness of (11) then follows.

We shall form the principal term of the determinant C from (6). Let $s_1, s_2, \ldots, s_n$ be mutually independent summation letters which run from 1 to n. Then

$$\begin{aligned} c_1^{(1)} c_2^{(2)} \cdots c_n^{(n)} &= \sum_{s_1} a_1^{(s_1)} b_1^{(s_1)} \sum_{s_2} a_2^{(s_2)} b_2^{(s_2)} \cdots \sum_{s_n} a_n^{(s_n)} b_n^{(s_n)} \\ &= \sum_{s_1, s_2, \ldots, s_n} a_1^{(s_1)} a_2^{(s_2)} \cdots a_n^{(s_n)} b_1^{(s_1)} b_2^{(s_2)} \cdots b_n^{(s_n)}. \end{aligned} \tag{12}$$

The permutation of the lower indices of the c corresponds to the permutation of the lower indices of the a, and if, with regard to the rule of signs, one therefore forms the determinant C, one obtains

$$\sum \pm c_1^{(1)} c_2^{(2)} \cdots c_n^{(n)} = \sum_{s_1, s_2, \ldots, s_n} b_1^{(s_1)} b_2^{(s_2)} \cdots b_n^{(s_n)} \sum \pm a_1^{(s_1)} a_2^{(s_2)} \cdots a_n^{(s_n)}. \tag{13}$$

Now

$$\sum \pm a_1^{(s_1)} a_2^{(s_2)} \cdots a_n^{(s_n)}$$

is, by § 21, VI, always zero if the same digit occurs twice among $s_1, s_2, \ldots, s_n$; hence only those terms in (13) keep a non-zero value for which $s_1, s_2, \ldots, s_n$ is an arrangement of the indices $1, 2, \ldots, n$. This value is $+A$ or $-A$ according as this arrangement belongs to the first or the second kind (§ 21, IV).

Therefore the right-hand side of (13) becomes

$$A \sum_{s_1, s_2, \ldots, s_n} \pm b_1^{(s_1)} b_2^{(s_2)} \cdots b_n^{(s_n)},$$

where the sum is extended over all permutations $s_1, s_2, \ldots, s_n$. But this sum is precisely the determinant B, and hence the formula

$$C = AB$$

is proved.

The law of formation of the elements $c_i^{(h)}$ contained in formula (6) can be expressed in words as follows. To obtain the elements of the determinant C which is the product of the two determinants A, B, one multiplies the elements of each column of A by the corresponding elements of a column of B and adds the products.

By the theorems on determinants the form of C can be varied in many ways. We record the following. If two columns in A or in B are interchanged, the elements $c_i^{(h)}$ do not change, but only interchange among themselves. If, however, two rows in A or in B are interchanged, then the $c_i^{(h)}$ change because the factors in the individual products of the sum are grouped differently; but if corresponding interchanges are made simultaneously in the rows of A and of B, the $c_i^{(h)}$ remain unchanged, since only the individual terms of the sum are thereby interchanged.

By turning the rows into columns in A or in B, or in both at once, one obtains three further different ways of forming the product of two determinants in determinant form. This can also be expressed as follows. To form the product of two determinants, one may multiply the elements of the separate rows or columns of one factor by the corresponding elements of the rows or columns of the other factor, add the products, and regard these sums of products as the elements of a new determinant.

The multiplication rule can be applied to a product of two determinants with different numbers of elements by transforming the determinant with the smaller number of rows, by the theorem § 22, IX, into another with more rows.

§ 28. Determinants of subdeterminants.

We immediately make an application here of the multiplication law of determinants. Let, as before,

$$A = \begin{vmatrix} a_1^{(1)} & a_2^{(1)} & \cdots & a_n^{(1)} \\ a_1^{(2)} & a_2^{(2)} & \cdots & a_n^{(2)} \\ \cdots & \cdots & & \cdots \\ a_1^{(n)} & a_2^{(n)} & \cdots & a_n^{(n)} \end{vmatrix}, \tag{1}$$

and let

$$\begin{matrix} A_1^{(1)} & A_2^{(1)} & \cdots & A_n^{(1)} \\ A_1^{(2)} & A_2^{(2)} & \cdots & A_n^{(2)} \\ \cdots & \cdots & & \cdots \\ A_1^{(n)} & A_2^{(n)} & \cdots & A_n^{(n)} \end{matrix} \tag{2}$$

be the system of subdeterminants. If we form from (2) the determinant which we shall denote by Δ, we may apply the multiplication rule to the product $A\Delta$. By § 22, (3) and (7), this gives

$$A\Delta = \begin{vmatrix} A & 0 & \cdots & 0 \\ 0 & A & \cdots & 0 \\ \cdots & \cdots & & \cdots \\ 0 & 0 & \cdots & A \end{vmatrix} = A^n,$$

and, by division by A,

$$\Delta = A^{n-1}. \tag{3}$$

Thus Δ is the $(n-1)$st power of A. In this derivation it has indeed first been assumed that A is different from zero. But since (3) is an identity with respect to the elements $a_i^{(h)}$, that is, since it also holds when these quantities are independent variables, it follows that the formula (3) still holds in the exceptional case; that is, if A vanishes, then Δ also vanishes.

This result is a special case of a more general theorem by which any determinant of the matrix (2) may be formed. Consider the ν-rowed subdeterminant

$$\Delta_\nu = \begin{vmatrix} A_1^{(1)} & A_2^{(1)} & \cdots & A_\nu^{(1)} \\ A_1^{(2)} & A_2^{(2)} & \cdots & A_\nu^{(2)} \\ \cdots & \cdots & & \cdots \\ A_1^{(\nu)} & A_2^{(\nu)} & \cdots & A_\nu^{(\nu)} \end{vmatrix}, \tag{4}$$

from which all other ν-rowed subdeterminants can be derived by permutation of the upper and lower indices. One may apply the multiplication rule by transforming Δ_ν, according to the concluding remark of the last paragraph, into an n-rowed determinant.

Put

$$\Delta_\nu = \begin{vmatrix} A_1^{(1)} & \cdots & A_\nu^{(1)} & A_{\nu+1}^{(1)} & \cdots & A_n^{(1)} \\ \cdots & & \cdots & \cdots & & \cdots \\ A_1^{(\nu)} & \cdots & A_\nu^{(\nu)} & A_{\nu+1}^{(\nu)} & \cdots & A_n^{(\nu)} \\ 0 & \cdots & 0 & 1 & \cdots & 0 \\ \cdots & & \cdots & \cdots & & \cdots \\ 0 & \cdots & 0 & 0 & \cdots & 1 \end{vmatrix}, \tag{5}$$

where $n-\nu$ rows and columns have been adjoined, the former containing only zeros except in the diagonal terms. If one now forms the product $A\Delta_\nu$, it follows that

$$A\Delta_\nu = A^\nu \begin{vmatrix} a_{\nu+1}^{(\nu+1)} & \cdots & a_n^{(\nu+1)} \\ \cdots & & \cdots \\ a_{\nu+1}^{(n)} & \cdots & a_n^{(n)} \end{vmatrix}, \tag{6}$$

and this is by Theorem IX, § 22. Dividing here by A and applying the notation of § 23, one obtains

$$\Delta_\nu = A^{\nu-1} A_{1,2,\ldots,\nu}^{1,2,\ldots,\nu}. \tag{7}$$

For $\nu = 2$ this gives the special result

$$A_1^{(1)} A_2^{(2)} - A_1^{(2)} A_2^{(1)} = A A_{1,2}^{1,2}. \tag{8}$$

We shall use formula (8) often later. In the notation by differential quotients it can be represented as

$$A\frac{\partial^2 A}{\partial a_1^{(1)}\partial a_2^{(2)}} = \frac{\partial A}{\partial a_1^{(1)}}\frac{\partial A}{\partial a_2^{(2)}} - \frac{\partial A}{\partial a_1^{(2)}}\frac{\partial A}{\partial a_2^{(1)}}. \tag{9}$$

It is especially important in the case where A is a symmetric determinant, so that $a_i^{(k)} = a_k^{(i)}$; then also

$$\frac{\partial A}{\partial a_1^{(2)}} = \frac{\partial A}{\partial a_2^{(1)}},$$

where in the differentiation no account is taken of the dependence $a_i^{(k)} = a_k^{(i)}$. Then formula (9) becomes

$$A\frac{\partial^2 A}{\partial a_1^{(1)}\partial a_2^{(2)}} = \frac{\partial A}{\partial a_1^{(1)}}\frac{\partial A}{\partial a_2^{(2)}} - \left(\frac{\partial A}{\partial a_1^{(2)}}\right)^2. \tag{10}$$

§ 29. Interpolation.

In § 9, under a special assumption, we already solved the problem of determining an integral rational function which assumes prescribed values for a sufficient number of prescribed values of the argument. We shall now solve this problem generally by applying determinants. First, however, we shall put forward the following observations.

In § 22, (12), the value of the determinant

$$\begin{vmatrix} \alpha_1^{n-1} & \alpha_1^{n-2} & \cdots & \alpha_1 & 1 \\ \alpha_2^{n-1} & \alpha_2^{n-2} & \cdots & \alpha_2 & 1 \\ \cdots & \cdots & & \cdots & \cdots \\ \alpha_n^{n-1} & \alpha_n^{n-2} & \cdots & \alpha_n & 1 \end{vmatrix}$$

was found to be equal to the product of differences

$$(\alpha_1-\alpha_2)(\alpha_1-\alpha_3)\cdots(\alpha_1-\alpha_n)(\alpha_2-\alpha_3)\cdots(\alpha_2-\alpha_n)\cdots(\alpha_{n-1}-\alpha_n). \tag{1}$$

For brevity we now denote this product of differences by

$$[\alpha_1, \alpha_2, \alpha_3, \ldots, \alpha_n], \tag{2}$$

so that the quantity $[\alpha_1, \alpha_2, \alpha_3, \ldots, \alpha_n]$ changes its sign when two of the elements $\alpha_1, \alpha_2, \ldots, \alpha_n$ are interchanged.

We further define an integral rational function $f(x)$ of degree n in the variable x by the equation

$$f(x) = (x-\alpha_1)(x-\alpha_2)\cdots(x-\alpha_n), \tag{3}$$

so that, by § 13,

$$\begin{aligned} f'(\alpha_1) &= (\alpha_1-\alpha_2)(\alpha_1-\alpha_3)\cdots(\alpha_1-\alpha_n), \\ f'(\alpha_2) &= (\alpha_2-\alpha_1)(\alpha_2-\alpha_3)\cdots(\alpha_2-\alpha_n), \\ &\vdots \\ f'(\alpha_n) &= (\alpha_n-\alpha_1)(\alpha_n-\alpha_2)\cdots(\alpha_n-\alpha_{n-1}). \end{aligned} \tag{4}$$

In the product of all these expressions each factor of the product (1) occurs twice with opposite sign; since the number of factors of (1) is $\frac{n(n-1)}{2}$, we obtain the equation

$$f'(\alpha_1)f'(\alpha_2)\cdots f'(\alpha_n) = (-1)^{\frac{n(n-1)}{2}}[\alpha_1, \alpha_2, \ldots, \alpha_n]^2. \tag{5}$$

But with the aid of the relations (4) we may also represent our product (1) as follows:

$$\begin{aligned}[\alpha_1, \alpha_2, \ldots, \alpha_n] &= f'(\alpha_1)[\alpha_2, \alpha_3, \ldots, \alpha_n] \\ &= -f'(\alpha_2)[\alpha_1, \alpha_3, \ldots, \alpha_n] \\ &= \cdots = (-1)^{n-1} f'(\alpha_n)[\alpha_1, \alpha_2, \ldots, \alpha_{n-1}],\end{aligned} \tag{6}$$

where on the right-hand side each bracketed expression contains one element fewer than the bracketed expression on the left.

Now pose the problem of determining an integral rational function $\varphi(x)$ of degree $(n-1)$ which, for the argument values $x = \alpha_1, \alpha_2, \ldots, \alpha_n$, assumes in order the prescribed values $\varphi(\alpha_1), \varphi(\alpha_2), \ldots, \varphi(\alpha_n)$.

If one puts

$$\varphi(x) = a_1 x^{n-1} + a_2 x^{n-2} + \cdots + a_{n-1} x + a_n, \tag{7}$$

then the n unknown coefficients $a_1, a_2, \ldots, a_n$ are to be determined from the linear equations

$$\begin{aligned}\varphi(\alpha_1) &= a_1 \alpha_1^{n-1} + a_2 \alpha_1^{n-2} + \cdots + a_{n-1}\alpha_1 + a_n, \\ \varphi(\alpha_2) &= a_1 \alpha_2^{n-1} + a_2 \alpha_2^{n-2} + \cdots + a_{n-1}\alpha_2 + a_n, \\ &\vdots \\ \varphi(\alpha_n) &= a_1 \alpha_n^{n-1} + a_2 \alpha_n^{n-2} + \cdots + a_{n-1}\alpha_n + a_n.\end{aligned} \tag{8}$$

The determinant of this system is precisely the quantity (2), and the problem is therefore always soluble by § 26 when this quantity is different from zero, that is, when no two of the n quantities $\alpha_1, \alpha_2, \ldots, \alpha_n$ are equal.

Instead of solving the equations (8) by § 26 and substituting the expressions so found into (7), it is preferable, by § 24, II, to eliminate the homogeneous quantities $1, a_1, a_2, \ldots, a_n$ from the $n+1$ equations (7) and (8). This gives the determinant equation

$$\begin{vmatrix} \varphi(x) & x^{n-1} & x^{n-2} & \cdots & x & 1 \\ \varphi(\alpha_1) & \alpha_1^{n-1} & \alpha_1^{n-2} & \cdots & \alpha_1 & 1 \\ \cdots & \cdots & \cdots & & \cdots & \cdots \\ \varphi(\alpha_n) & \alpha_n^{n-1} & \alpha_n^{n-2} & \cdots & \alpha_n & 1 \end{vmatrix} = 0. \tag{9}$$

We now expand this determinant according to the elements of the first column. For the subdeterminants which occur there we can use the notation (2), and we obtain

$$\begin{aligned}\varphi(x)[\alpha_1, \alpha_2, \ldots, \alpha_n] - \varphi(\alpha_1)[x, \alpha_2, \alpha_3, \ldots, \alpha_n] + \varphi(\alpha_2)[x, \alpha_1, \alpha_3, \ldots, \alpha_n] \\ + \cdots + (-1)^n \varphi(\alpha_n)[x, \alpha_1, \alpha_2, \ldots, \alpha_{n-1}] = 0.\end{aligned} \tag{10}$$

Now by (1)

$$[x, \alpha_2, \alpha_3, \ldots, \alpha_n] = (x-\alpha_2)(x-\alpha_3)\cdots(x-\alpha_n)[\alpha_2, \alpha_3, \ldots, \alpha_n].$$

Dividing (10) by $[\alpha_1, \alpha_2, \ldots, \alpha_n]$ and taking account of the several expressions (6), one finally obtains

$$\varphi(x) = f(x)\left\{\frac{\varphi(\alpha_1)}{(x-\alpha_1)f'(\alpha_1)} + \frac{\varphi(\alpha_2)}{(x-\alpha_2)f'(\alpha_2)} + \cdots + \frac{\varphi(\alpha_n)}{(x-\alpha_n)f'(\alpha_n)}\right\}. \tag{11}$$

This completely determines the function $\varphi(x)$. This expression for $\varphi(x)$ is called the interpolation formula of Lagrange. That it satisfies the stipulated requirements can be verified afterwards very easily by putting $x = \alpha_1, \alpha_2, \ldots, \alpha_n$. The denominators $x - \alpha_1, \ldots, x - \alpha_n$ are present in it only apparently.

Third Section. The roots of algebraic equations.

§ 30. Concept of roots. Multiple roots.

After, in the first two sections, the algebraic magnitudes had been considered more from the formal side, where the matter was identical transformations of letter-expressions in which the letters throughout could be conceived as symbols for variable magnitudes, the numerical magnitudes now come more into the foreground.

Here, in accordance with what was fixed in the Introduction, by numbers we understand real or imaginary magnitudes of the form $a + bi$, and for illustration we represent the real magnitudes by points of a straight line and the imaginary magnitudes by points of a plane. By the absolute value of an imaginary magnitude $a + bi$ we understand $\sqrt{a^2 + b^2}$ and denote it, following Weierstrass, by

$$|a + bi|.$$

Let now

$$f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_n \tag{1}$$

be an integral function of x, in which the coefficients are arbitrary real or imaginary numbers, and the first, a_0, is assumed different from zero. If α is a number which, when put for x, makes the function $f(x)$ equal to zero, that is, which satisfies the condition $f(\alpha) = 0$, then α is called a root of the equation

$$f(x) = 0.$$

We also say briefly that α is a root of $f(x)$. By § 4, $f(x)$ may then be divided by $x - \alpha$ without remainder, so that one may put

$$f(x) = (x - \alpha)f_1(x), \tag{2}$$

where $f_1(x)$ is only of the $(n-1)$st degree and has the same first coefficient a_0 as $f(x)$, thus

$$f_1(x) = a_0x^{n-1} + \cdots .$$

The coefficients of $f_1(x)$ are given in § 4, (6).

Every root of $f_1(x)$ is therefore at the same time a root of $f(x)$; and conversely every root of $f(x)$ is either $= \alpha$ or is a root of $f_1(x)$. If one root α of $f(x)$ is known, then the task of finding the remaining ones is reduced to the solution of an equation of the $(n-1)$st degree.

The aim of the considerations of this section consists in proving that every function $f(x)$ of the nth degree has at least one root. This theorem is called the fundamental theorem of algebra. First we draw from (2) the important consequence:

I. An equation of the nth degree cannot have more than n roots.

For if $f(x)$ had more than n roots, then $f_1(x)$, which is only of the $(n-1)$st degree, would have more than $n - 1$ roots. But an equation of the first degree has not more than one root, whence the correctness of our theorem follows by complete induction.

Sometimes one also gives it the expression:

II. If a function $f(x)$ of the nth degree has more than n roots, then all its coefficients must be zero.

If β is a root of $f_1(x)$, then one may likewise put

$$f_1(x) = (x - \beta)f_2(x),$$

where $f_2(x) = a_0x^{n-2} + \cdots$ is only of the $(n-2)$nd degree. Thus, if one assumes that each of the functions obtained by this division, $f_1(x), f_2(x), \ldots$, has at least one root, one finally obtains

$$f(x) = a_0(x - \alpha)(x - \beta)\cdots(x - \nu), \tag{3}$$

and we can therefore also express the theorem which we earlier described as the aim of our considerations as follows. It is to be proved:

A function of the nth degree can be decomposed into n linear factors.

The magnitudes $\alpha, \beta, \ldots, \nu$ are then all roots of $f(x)$, and their number is therefore n. It is, however, not excluded that the same number occurs more than once among $\alpha, \beta, \ldots, \nu$. Then $f(x)$ would have fewer than n roots, while the theorem would still remain that $f(x)$ is decomposable into n linear factors. To establish agreement, it has been agreed that, if $x - \alpha$ goes into $f(x)$ several times, α is to be counted several times among the roots, and hence that one speaks of simple, double, triple, etc. roots.

By the concept of the derived functions [§ 13, (2)], if α is an arbitrary magnitude,

$$f(x) = f(\alpha) + (x-\alpha)f'(\alpha) + \frac{(x-\alpha)^2}{1\cdot 2}f''(\alpha) + \frac{(x-\alpha)^3}{1\cdot 2\cdot 3}f'''(\alpha) + \cdots . \tag{4}$$

If it is now again assumed that $f(\alpha)$ vanishes, then there results

$$f_1(x) = f'(\alpha) + \frac{x-\alpha}{1\cdot 2}f''(\alpha) + \frac{(x-\alpha)^2}{1\cdot 2\cdot 3}f'''(\alpha) + \cdots ,$$

and hence

$$f_1(\alpha) = f'(\alpha).$$

Thus α is a double root of $f(x)$ when $f'(\alpha)$ vanishes simultaneously with $f(\alpha)$. Formula (4) also shows that only under this assumption is $f(x)$ divisible by $(x-\alpha)^2$. This conclusion may be carried further and leads to the theorem:

III. The necessary and sufficient condition that α be an m-fold root of $f(x)$ is that $f(\alpha), f'(\alpha), f''(\alpha), \ldots, f^{(m-1)}(\alpha)$ vanish, while $f^{(m)}(\alpha)$ does not vanish.

§ 31. Continuity of integral functions.

If, as before,

$$f(x) = a_0x^n + a_1x^{n-1} + \cdots + a_n$$

is an integral function of x, we first prove the following theorem:

IV. $f(x)$ becomes infinitely large at the same time as x.

This means that, if C is an arbitrarily given positive number, the positive number R can be so chosen that

$$|f(x)| > C$$

as soon as the absolute value of x becomes greater than R; or, expressed geometrically: one can describe in the x-plane a circle around the origin such that outside this circle the absolute value of $f(x)$ no longer falls below C, however large C is assumed. The theorem is evident for the case of a simple power x^n; for if r is the absolute value of x, then r^n is the absolute value of x^n, and r^n, as soon as r has become greater than 1, is greater than r and hence grows with r into the infinite.

To prove the theorem in general, let us denote the absolute values of $x, a_0, a_1, a_2, \ldots, a_n$ by $r, c_0, c_1, c_2, \ldots, c_n$. Then

$$|f(x)| = r^n\left|a_0 + \frac{a_1}{x} + \frac{a_2}{x^2} + \cdots + \frac{a_n}{x^n}\right|, \tag{2}$$

and, since by the theorems proved in the Introduction the absolute value of a sum cannot be smaller than the difference and cannot be greater than the sum of the absolute values of the summands,

$$\left|a_0 + \frac{a_1}{x} + \frac{a_2}{x^2} + \cdots + \frac{a_n}{x^n}\right| \geq c_0 - \left|\frac{a_1}{x} + \frac{a_2}{x^2} + \cdots + \frac{a_n}{x^n}\right| \geq c_0 - \frac{c_1}{r} - \frac{c_2}{r^2} - \cdots - \frac{c_n}{r^n}.$$

Since $c_0, c_1, \ldots, c_n$ are fixed constants, one can choose R so large that, as soon as $r > R$,

$$\frac{c_1}{r} + \frac{c_2}{r^2} + \cdots + \frac{c_n}{r^n}$$

becomes arbitrarily small, for example smaller than $\frac{1}{2}c_0$. Then

$$\left|a_0 + \frac{a_1}{x} + \frac{a_2}{x^2} + \cdots + \frac{a_n}{x^n}\right| > \tfrac{1}{2}c_0,$$

and hence by (2)

$$|f(x)| > \tfrac{1}{2}c_0 R^n,$$

so that, if one assumes

$$R^n > \frac{2C}{c_0}, \tag{3}$$

one has

$$|f(x)| > C.$$

To this is attached the theorem:

V. $f(x)$ is a continuous function of x.

By this the following is meant. If the absolute value of x is smaller than an arbitrary finite magnitude R, then, however small the positive magnitude ω is assumed, one can find a positive magnitude ε such that

$$|f(x+h) - f(x)| < \omega \tag{4}$$

as long as the absolute value ρ of h remains smaller than ε, and indeed in such a way that ε depends only on the choice of ω, not on x.

By § 13, (2),

$$f(x+h) - f(x) = hf'(x) + \frac{h^2}{1 \cdot 2}f''(x) + \cdots + h^n a_0. \tag{5}$$

If now we assume the absolute value of h smaller than 1, then the absolute value of the magnitude in parentheses

$$f'(x) + \frac{h}{1 \cdot 2}f''(x) + \cdots + h^{n-1}a_0 \tag{6}$$

is smaller than a certain non-zero number K which one obtains if in (6) one replaces h by 1, the coefficients $a_0, a_1, \ldots, a_{n-1}$ by their absolute values and x by R, because thereby every individual term of the sum (6) is replaced by its absolute value or by a larger number. If one chooses ε so small that

$$\varepsilon K < \omega,$$

then by (5) the requirement (4) is fulfilled as long as $\rho < \varepsilon$.

We can give the theorem of the continuity of the function $f(x)$ a somewhat different expression, which is important for what follows. By the theorem that the absolute value of a sum of two terms lies, in magnitude, between the sum and the difference of the absolute values of the terms (cf. Introduction, p. 19), we obtain from (5), if we put

$$\varphi(h) = hf'(x) + \frac{h^2}{1 \cdot 2}f''(x) + \cdots + h^n a_0,$$

$$|f(x)| - |\varphi(h)| < |f(x+h)| < |f(x)| + |\varphi(h)|,$$

or

$$-|\varphi(h)| < |f(x+h)| - |f(x)| < |\varphi(h)|.$$

Now, if $\rho < \varepsilon$, then $|\varphi(h)| < \omega$, and we can therefore also say:

VI. The absolute value of the difference

$$|f(x+h)| - |f(x)|$$

is smaller than an arbitrarily given magnitude ω as soon as the absolute value of h is smaller than a sufficiently small ε.

From this we can still conclude, if we put $x = y + iz$ and assume h either real or purely imaginary, that $|f(y+zi)|$ is, for fixed z, a continuous function of y, and for fixed y, a continuous function of z.

If the coefficients $a_0, a_1, \ldots, a_n$ and the variable x are real, then x may be chosen, in absolute value, so large that the sign of $f(x)$ agrees with the sign of the first term $a_0 x^n$; thus, for positive a_0 and even n, it becomes positive, while for odd n it becomes negative for negative x and positive for positive x.

For in

$$f(x) = x^n \left(a_0 + \frac{a_1}{x} + \frac{a_2}{x^2} + \cdots + \frac{a_n}{x^n} \right)$$

one can choose x so large in absolute value that the absolute value of

$$\frac{a_1}{x} + \frac{a_2}{x^2} + \cdots + \frac{a_n}{x^n}$$

falls below that of a_0, and therefore the expression in parentheses

$$a_0 + \frac{a_1}{x} + \frac{a_2}{x^2} + \cdots + \frac{a_n}{x^n}$$

has the sign of a_0, which it then retains for every absolutely larger x.

§ 32. Change of sign of $f(x)$. Roots of equations of odd degree and of pure equations.

On the basis of the theorems of the preceding paragraph we can prove the following important theorem.

VIII. If the coefficients of $f(x)$ are real and there exist two real values a, b of x such that $f(a)$ and $f(b)$ have opposite signs, then between a and b there is at least one root ξ of the equation $f(x) = 0$.

We shall assume that $a < b$ and that $f(a)$ is negative, $f(b)$ positive. We divide the numbers between a and b into two parts $\mathfrak{A}$ and $\mathfrak{B}$ as follows: a number α belongs to $\mathfrak{A}$ if, between α and a, including the endpoints, there is no x for which $f(x)$ becomes positive; and a number β belongs to $\mathfrak{B}$ if between β and a there is at least one value of x for which $f(x)$ becomes positive. A third possibility is plainly not possible, and every number between a and b is therefore placed in $\mathfrak{A}$ or in $\mathfrak{B}$. We also count the numbers smaller than a as belonging to $\mathfrak{A}$, and those greater than b as belonging to $\mathfrak{B}$.

If α belongs to $\mathfrak{A}$, then every x smaller than α also belongs to $\mathfrak{A}$; and if β belongs to $\mathfrak{B}$, every larger x belongs to $\mathfrak{B}$. Thus every β is greater than every α, and the two numerical domains $\mathfrak{A}, \mathfrak{B}$ form a cut (Introduction, p. 5), generated by a number ξ, such that every number α smaller than ξ belongs to $\mathfrak{A}$, and every number β greater than ξ belongs to $\mathfrak{B}$.

If now, for some number x_0, the value of $f(x_0)$ is negative or positive, then by V of the preceding paragraph a positive magnitude ε can be so determined that for every x between $x_0 - \varepsilon$ and $x_0 + \varepsilon$ the values of $f(x)$ are always negative or always positive. It follows from this that $f(\xi)$ can be neither negative nor positive and must therefore be zero. For if $f(\xi)$ were negative, then $f(x)$ would also be negative between ξ and $\xi + \varepsilon$; these values would therefore belong to $\mathfrak{A}$, although they are greater than ξ. If $f(\xi)$ were positive, then $f(x)$ would be positive as long as x lay between $\xi - \varepsilon$

and ξ; these values, although smaller than ξ, would therefore belong to $\mathfrak{B}$. Both alternatives are impossible by the definition of ξ. The same conclusion may be drawn in quite the same way when $f(a)$ is positive and $f(b)$ negative, and so our theorem is proved.

From this we draw two important consequences. If $f(\xi)$ vanishes, then an interval

$$\xi - \varepsilon < x < \xi + \varepsilon$$

can be so determined that $f(x)$ does not vanish in this interval except for $x = \xi$. Then four cases are possible with respect to the signs of $f(x)$ in the two intervals $\xi - \varepsilon < x < \xi$ and $\xi < x < \xi + \varepsilon$, which are put together in the following scheme:

	$\xi - \varepsilon < x < \xi$	$\xi < x < \xi + \varepsilon$
1.	$+$	$-$
2.	$-$	$+$
3.	$+$	$+$
4.	$-$	$-$

If $f^{(\nu)}(x)$ is the first among the derivatives of $f(x)$ which is different from zero for $x = \xi$, then the formula [§ 13, (2)]

$$f(\xi + h) = \frac{h^\nu}{\Pi(\nu)} f^{(\nu)}(\xi) + \cdots$$

shows that these four cases are distinguished by the following signs:

$$\begin{array}{lll} 1. & \nu \text{ odd}, & f^{(\nu)}(\xi) < 0, \\ 2. & \nu \text{ odd}, & f^{(\nu)}(\xi) > 0, \\ 3. & \nu \text{ even}, & f^{(\nu)}(\xi) > 0, \\ 4. & \nu \text{ even}, & f^{(\nu)}(\xi) < 0. \end{array}$$

We say that in the first case $f(x)$ passes decreasing through zero, and in the second increasing through zero. In case 3, $f(\xi) = 0$ is a minimum, in case 4 a maximum.

In particular, if $f'(\xi)$ is different from zero, so that ξ is a simple root, then the positive or negative sign of $f'(\xi)$ is the characteristic mark for the increase or decrease of the function $f(x)$ as x passes through ξ.

With these aids we still prove two further important theorems.

IX. The equation

$$x^n - a = 0$$

has, when a is a positive number, one and only one positive root.

For if we put

$$f(x) = x^n - a, \tag{1}$$

then by § 31, VII, $f(x)$ is positive for sufficiently large positive x, and negative for $x = 0$; hence there is a positive value of x, denoted by $\sqrt[n]{a}$, for which $f(x)$ vanishes. That there can be only one such value follows from the fact that x^n, as long as x remains positive, increases continually with x.

If n is odd, then also for negative a there exists one and only one negative root of (1), which is proved in the same way.

X. An equation of odd degree with real coefficients always has at least one real root.

For by theorem VII, $f(x)$, if the degree is odd, can become both positive and negative; hence, by VIII, $f(x) = 0$ has a root.

Finally we prove:

XI. A pure equation always has a root.

By a pure equation we understand an equation of the form

$$x^n - a = 0, \tag{2}$$

where a is an arbitrary complex magnitude. That this equation always has a root for real positive a has already been proved above; and for $a = 0$ it is plainly satisfied by $x = 0$.

The general proof can be divided into two parts; it is sufficient to assume first $n = 2$, and then n odd. For if the magnitude $\sqrt{a}$ and $\sqrt[n]{a}$, or $\sqrt[n]{\alpha}$ for every α, exists, then also the existence of $\sqrt[n]{a}$ for every arbitrary n is secured. It would even suffice to prove the existence of $\sqrt[n]{a}$ for the case in which n is a prime number. But from this no particular advantage is at first to be drawn.

We denote the complex magnitude a by $b + ci$ and put, since x will in general also be complex,

$$x = y + iz.$$

Then we first have to show that

$$(y + iz)^2 = b + ic, \tag{3}$$

whatever real values b, c may have, can be satisfied by real y, z. Equation (3), however, is fulfilled if

$$y^2 - z^2 = b, \qquad 2yz = c, \tag{4}$$

and hence

$$(y^2 + z^2)^2 = b^2 + c^2,$$

therefore

$$y^2 + z^2 = \sqrt{b^2 + c^2}. \tag{5}$$

By IX there always exists a positive square root $\sqrt{b^2 + c^2}$. From (4) and (5) it now follows that

$$y^2 = \frac{b + \sqrt{b^2 + c^2}}{2}, \qquad z^2 = \frac{-b + \sqrt{b^2 + c^2}}{2},$$

and the two magnitudes $\pm b + \sqrt{b^2 + c^2}$ are plainly positive, since $b^2 + c^2 > b^2$, hence also $\sqrt{b^2 + c^2} > \sqrt{b^2}$. Thus we obtain

$$y = \pm\sqrt{\frac{b + \sqrt{b^2 + c^2}}{2}}, \qquad z = \pm\sqrt{\frac{-b + \sqrt{b^2 + c^2}}{2}},$$

but one sign is determined by the other through the condition

$$2yz = c;$$

one of the two signs remains arbitrary, so that we obtain not merely one, but two opposite solutions of (3).

Further let n be odd; we take the equation to be solved in the form

$$(y + iz)^n = b + ci. \tag{6}$$

If $c = 0$, so that a is real, then we can put $z = 0$ and by IX obtain a real value of x. But if we assume c different from zero, then from (6) (Introduction, p. 19) there follows

$$(y - iz)^n = b - ci. \tag{7}$$

Multiplying the two equations (6), (7) by one another gives

$$(y^2 + z^2)^n = b^2 + c^2,$$

and hence, by IX,

$$y^2 + z^2 = \sqrt[n]{b^2 + c^2}, \tag{8}$$

which determines the absolute value of x. Furthermore, from (6) and (7) one obtains

$$\varphi(y,z) = (y-iz)^n(b+ic) - (y+iz)^n(b-ci) = 0. \tag{9}$$

Now $\varphi(y,z)$ is an integral homogeneous function of the nth degree in the two variables y,z, and indeed with real coefficients, since replacing i by $-i$ changes nothing in $\varphi(y,z)$. If it is ordered by descending powers of y, the first term is $2ciy^n$, hence, by assumption, different from zero. Putting

$$y = \lambda z,$$

we obtain from (9)

$$\varphi(\lambda, 1) = 0, \tag{10}$$

therefore an equation of the nth degree for λ, which by X certainly has a root. Once λ is determined, it follows from (8) that

$$y = \lambda\sqrt{\frac{\sqrt[n]{b^2+c^2}}{1+\lambda^2}}, \qquad z = \sqrt{\frac{\sqrt[n]{b^2+c^2}}{1+\lambda^2}}. \tag{11}$$

Through this the equations (8), (9) are in fact satisfied; from (9), however, follows

$$\frac{(y+iz)^n}{b+ic} = \frac{(y-iz)^n}{b-ci},$$

and from (8) that the common value of these two fractions is $+1$. Therefore we have

$$(y+iz)^n = \pm(b+ci), \qquad (y-iz)^n = \pm(b-ci),$$

and thus the sign of the square root in (11) can be so determined that equations (6) and (7) are fulfilled.

§ 33. Solution of pure equations by trigonometric functions.

The solvability of a pure equation can be shown far more completely and simply if one presupposes the trigonometric functions and their simplest properties as known. These functions are, it is true, foreign to algebra proper, and for that reason it is more satisfactory to answer the questions of principle without their aid, as has been done in the preceding and will also be done later. For application and for a more convenient visualization, however, we shall not dispense with this aid.

If, when b,c are real numbers, we put

$$b = r\cos\varphi, \qquad c = r\sin\varphi, \tag{1}$$

then, if r is assumed positive, the angle φ is thereby determined up to a multiple of 2π. It will be completely determined if we fix an interval of length 2π in which it is to lie, for example

$$-\pi < \varphi \leq \pi.$$

In geometrical interpretation r,φ are the polar coordinates of the point whose rectangular coordinates are b,c. The absolute value of the complex magnitude $a = b+ci$ is r, and we shall call φ the phase of this complex magnitude.

The product of two complex magnitudes

$$a = r(\cos\varphi + i\sin\varphi), \qquad a' = r'(\cos\varphi' + i\sin\varphi')$$

is obtained in the form

$$aa' = rr'[\cos(\varphi+\varphi') + i\sin(\varphi+\varphi')], \tag{3}$$

and from this, if n is an arbitrary positive whole number, or even a negative one, one obtains the theorem named after Moivre,

$$(\cos\varphi + i\sin\varphi)^n = \cos n\varphi + i\sin n\varphi, \tag{4}$$

which has led to regarding the magnitude $\cos\varphi + i\sin\varphi$ as an exponential function with imaginary exponent, and hence to setting

$$\cos\varphi + i\sin\varphi = e^{i\varphi}.$$

Thereby formulas (3) and (4) take the form

$$e^{i\varphi}e^{i\varphi'} = e^{i(\varphi+\varphi')}, \qquad (e^{i\varphi})^n = e^{in\varphi}. \tag{5}$$

If accordingly we put

$$a = r(\cos\varphi + i\sin\varphi) \tag{6}$$

and by $\sqrt[n]{r}$ understand the positive value existing and completely determined by § 32, IX, whose nth power is r, then

$$x = \sqrt[n]{r}\left(\cos\frac{\varphi}{n} + i\sin\frac{\varphi}{n}\right) \tag{7}$$

is a magnitude whose nth power is a, hence a root of the equation

$$x^n = a, \tag{8}$$

where n is understood to be an arbitrary positive whole number. But every one of the values

$$x_k = \sqrt[n]{r}\left(\cos\frac{\varphi + 2k\pi}{n} + i\sin\frac{\varphi + 2k\pi}{n}\right) \tag{9}$$

satisfies the same requirement, when k is an arbitrary integer. In (9), however, not more than n different values are contained, namely those obtained by putting

$$k = 0, 1, 2, \ldots, n-1; \tag{10}$$

for if k is increased by a multiple of n, the value of (9) is not changed, while the n values (10) give only different values of x, because they have different phases.

Equation (8) consequently has not merely one, but n different roots. The geometrical images of the values x_k all lie on a circle with radius $\sqrt[n]{r}$, and are separated from one another by the angle $2\pi/n$. The first of these points is obtained by dividing the given angle φ into n equal parts. The radius $\sqrt[n]{r}$ is smaller or larger than r according as r is smaller or larger than 1. The adjacent figure shows these relations for $n = 3$ under the assumption $r > 1$.

One can obtain the different values x_k by multiplying the first of them,

$$x_0 = \sqrt[n]{r}\left(\cos\frac{\varphi}{n} + i\sin\frac{\varphi}{n}\right),$$

by the different values of

$$\xi_k = \cos\frac{2k\pi}{n} + i\sin\frac{2k\pi}{n}. \tag{11}$$

The magnitudes ξ_k,

$$\xi_k^n = 1, \tag{12}$$

are roots of the equation $x^n = 1$ and are called the nth roots of unity. There are n, and only n, different values of them; for odd n only one is real, namely for $k = 0$, and for even n two are real, namely for $k = 0$ and $k = n/2$. The geometrical images of the nth roots of unity are the vertices of a regular n-gon inscribed in the circle of radius 1. Their algebraic determination is the subject of the theory of cyclotomy. If we denote by ε the value of ξ_k for $k = 1$, thus put

$$\varepsilon = \cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n},$$

then, by Moivre's theorem,

$$\xi_k = \varepsilon^k,$$

so that all nth roots of unity are represented as powers of one among them.

§ 34. Removal of the second term of an equation.

The certainty of the existence of the roots of pure equations puts us in a position to prove the existence of roots for equations of the second, third, and fourth degrees, or, as one also says, for quadratic, cubic, and biquadratic equations, by reducing the determination of their roots to the solution of pure equations. We shall return to this question later from various more general points of view, but here we shall briefly discuss the older methods of solution, which, even if they give little insight into the general connection of this question, are nevertheless very simple in application. They arouse the appearance as if one were dealing with a method capable of generalization to higher equations, which is not the case. First the following general remark.

Let us, for the sake of simplicity, take in the function

$$f(x) = x^n + a_1 x^{n-1} + a_2 x^{n-2} + \cdots + a_n \tag{1}$$

the coefficient of x^n equal to 1, which in general is obtained by division by the coefficient of x^n. Then by the simple substitution

$$x = y - \frac{a_1}{n} \tag{2}$$

we can transform $f(x)$ into another function $\varphi(y)$ of the nth degree in which the second term y^{n-1} does not occur; for, by the binomial theorem,

$$x^n = y^n - a_1 y^{n-1} + \frac{n-1}{2n} a_1^2 y^{n-2} + \cdots ,$$

$$x^{n-1} = y^{n-1} - \frac{n-1}{n} a_1 y^{n-2} + \cdots , \qquad x^{n-2} = y^{n-2} - \cdots ,$$

and consequently

$$f(x) = y^n + \left(a_2 - \frac{n-1}{2n} a_1^2\right) y^{n-2} + \cdots .$$

If we actually carry out the transformation for the cases $n = 2, 3, 4$, we obtain

$$\begin{aligned} n &= 2, & \varphi(y) &= y^2 + a, \\ n &= 3, & \varphi(y) &= y^3 + ay + b, \\ n &= 4, & \varphi(y) &= y^4 + ay^2 + by + c, \end{aligned} \tag{3}$$

provided that we put

$$\begin{aligned} &\text{for } n = 2: & a &= -\frac{a_1^2}{4} + a_2, \\ &\text{for } n = 3: & a &= -\frac{a_1^2}{3} + a_2, \\ & & b &= \frac{2a_1^3}{27} - \frac{a_1 a_2}{3} + a_3, \\ &\text{for } n = 4: & a &= -\frac{3a_1^2}{8} + a_2, \\ & & b &= \frac{a_1^3}{8} - \frac{a_1 a_2}{2} + a_3, \\ & & c &= -\frac{3a_1^4}{256} + \frac{a_1^2 a_2}{16} - \frac{a_1 a_3}{4} + a_4. \end{aligned} \tag{4}$$

Thus the solution of the equation $f(x) = 0$ is reduced by (2) to the solution of $\varphi(y) = 0$. This first gives, for $n = 2$,

$$y = \pm\sqrt{-a},$$

and consequently

$$x = -\frac{a_1}{2} \pm \sqrt{\frac{a_1^2}{4} - a_2} = \frac{-a_1 \pm \sqrt{a_1^2 - 4a_2}}{2}.$$

We therefore have two roots

$$\alpha = \frac{-a_1 + \sqrt{a_1^2 - 4a_2}}{2}, \qquad \beta = \frac{-a_1 - \sqrt{a_1^2 - 4a_2}}{2}. \tag{5}$$

§ 35. Cubic equations. Cardano's formula.

We take the equation of the third degree in the reduced form

$$y^3 + ay + b = 0 \tag{1}$$

and introduce two new unknowns u, v by putting

$$y = u + v. \tag{2}$$

Substitution in (1) gives

$$u^3 + v^3 + (3uv + a)(u + v) + b = 0. \tag{3}$$

We determine one of the two quantities u, v by the other according to the equation

$$3uv = -a, \tag{4}$$

and obtain from (3)

$$u^3 + v^3 = -b. \tag{5}$$

But from (4) and (5), u^3 and v^3 can be determined by a square root. Namely one obtains

$$(u^3 - v^3)^2 = (u^3 + v^3)^2 - 4u^3v^3 = b^2 + \frac{4a^3}{27}. \tag{6}$$

Put, for abbreviation,

$$R = \frac{b^2}{4} + \frac{a^3}{27}; \tag{7}$$

then

$$u^3 - v^3 = 2\sqrt{R}.$$

We give $\sqrt{R}$ one of the two signs and, according to (5), obtain

$$\begin{aligned} u^3 &= -\frac{b}{2} + \sqrt{R}, \qquad & v^3 &= -\frac{b}{2} - \sqrt{R}, \\ u &= \sqrt[3]{-\frac{b}{2} + \sqrt{R}}, \qquad & v &= \sqrt[3]{-\frac{b}{2} - \sqrt{R}}, \end{aligned} \tag{8}$$

and therefore

$$y = \sqrt[3]{-\frac{b}{2} + \sqrt{R}} + \sqrt[3]{-\frac{b}{2} - \sqrt{R}}. \tag{9}$$

Multiplication of the two expressions (8) gives

$$u^3v^3 = -\frac{a^3}{27}$$

and shows that, if for u a determinate one among the three values of the cube root is taken,

$$v = -\frac{a}{3u} = -\frac{a}{3\sqrt[3]{-\frac{b}{2} + \sqrt{R}}} \tag{10}$$

is in any case one of the three values of $\sqrt[3]{-\frac{b}{2} - \sqrt{R}}$. But it follows from (4) that only when we take this value for v is equation (1) satisfied by $y = u + v$. Thus, corresponding to the three values of u, we obtain three roots of equation (1). From the existence of one root of the cubic equation, however, the existence of three already follows also from the fact that it has already been proved that the quadratic equation has two roots.

To derive the other two roots from the one root (9), put

$$\alpha = u + v$$

and divide $y^3 + ay + b$ by $y - \alpha$. The quotient whose roots are the two other roots β, γ of the cubic equation (1) is

$$y^2 + \alpha y + \alpha^2 + a = 0,$$

whence, by (4) of the preceding paragraph, one finds

$$y = \frac{-\alpha \pm \sqrt{-3\alpha^2 - 4a}}{2}.$$

If here one puts $\alpha = u + v$ and, by (4), $a = -3uv$, then it follows that

$$y = \frac{-(u+v) \pm \sqrt{-3(u+v)^2 + 12uv}}{2} = \frac{-(u+v) \pm \sqrt{-3}(u-v)}{2}.$$

We therefore put

$$\varepsilon = \frac{-1 + \sqrt{-3}}{2}, \tag{11}$$

from which

$$\varepsilon^3 = 1, \qquad \varepsilon^2 + \varepsilon + 1 = 0, \qquad \varepsilon^2 = \frac{-1 - \sqrt{-3}}{2} \tag{12}$$

follows. Thus one obtains the three roots of the cubic equation (1)

$$\begin{aligned} \alpha &= u + v, \\ \beta &= \varepsilon u + \varepsilon^2 v, \\ \gamma &= \varepsilon^2 u + \varepsilon v. \end{aligned} \tag{13}$$

Here ε is a third root of unity different from 1, expressed algebraically without the trigonometric functions. If one substitutes εu or $\varepsilon^2 u$ for u, and hence $\varepsilon^2 v$ or εv for v, the three quantities α, β, γ are interchanged with one another.

The expression (9) for the root of a cubic equation is called Cardano's formula.

§ 36. Cayley's expression for Cardano's formula.

Every cube root has, as we have seen, three different values, which one obtains from one of them by multiplication by $1, \varepsilon, \varepsilon^2$. Thus each of the two quantities u, v, as they are defined by (8), § 35, also has three values, and the sum $u + v$, if one takes account only of the conditions (8), has nine different values. Of these, however, only three give solutions of the cubic equation, and only by adjoining the relation (4), § 35,

$$3uv = -a \tag{1}$$

are the usable values singled out from among the nine. This is a defect of Cardano's formula, which accordingly by itself does not yet suffice to give the roots of the cubic equation unambiguously. Cayley remedied this defect by the following representation. Define two new quantities ξ, η by the equations

$$u = \xi^2 \eta, \qquad v = \xi \eta^2, \tag{2}$$

from which, by multiplication and use of (1), one obtains

$$\xi\eta = \sqrt[3]{-\frac{a}{3}}. \tag{3}$$

Consequently, if this value is inserted in (2) and the expressions § 35, (8) for u, v are substituted, then

$$\xi = \sqrt[3]{\frac{3b}{2a} + \sqrt{\frac{9b^2}{4a^2} + \frac{a}{3}}}, \qquad \eta = \sqrt[3]{\frac{3b}{2a} - \sqrt{\frac{9b^2}{4a^2} + \frac{a}{3}}}. \tag{4}$$

Thus, if one now puts the root y of the cubic equation in the form

$$y = \xi\eta(\xi + \eta), \tag{5}$$

one obtains an expression which, when ξ is replaced, independently, by $\xi, \varepsilon\xi, \varepsilon^2\xi$ and η by $\eta, \varepsilon\eta, \varepsilon^2\eta$, assumes not nine but only the three values

$$\begin{gathered}\xi^2\eta + \xi\eta^2, \\ \varepsilon\xi^2\eta + \varepsilon^2\xi\eta^2, \\ \varepsilon^2\xi^2\eta + \varepsilon\xi\eta^2.\end{gathered}$$

[9]

§ 37. The biquadratic equation.

A similar path to that used in solving the cubic equation by Cardano's formula may be taken for the solution of the equation of the fourth degree

$$y^4 + ay^2 + by + c = 0. \tag{1}$$

We put

$$2y = u + v + w$$

and obtain, if we write, for abbreviation,

$$s = u^2 + v^2 + w^2, \qquad t = v^2w^2 + w^2u^2 + u^2v^2, \tag{2}$$

then

$$\begin{gathered}4y^2 = s + 2(vw + wu + uv), \\ 16y^4 = s^2 + 4s(vw + wu + uv) + 4t + 8uvw(u + v + w).\end{gathered}$$

Substitution in (1) gives

$$s^2 + 4t + 4as + 16c + 8(uvw + b)(u + v + w) + 4(s + 2a)(vw + wu + uv) = 0.$$

This equation is satisfied by the assumptions

$$s + 2a = 0, \qquad uvw + b = 0, \qquad s^2 + 4t + 4as + 16c = 0.$$

With the help of the first of these three equations the third becomes

$$t = a^2 - 4c,$$

and, putting back the values (2) for s, t,

$$\begin{aligned} u^2 + v^2 + w^2 &= -2a, \\ v^2w^2 + w^2u^2 + u^2v^2 &= a^2 - 4c, \\ uvw &= -b. \end{aligned} \tag{3}$$

[9] Cayley, Phil. Mag. vol. XXI, 1861. Collected mathematical papers, vol. V, No. 310.

By § 7 these equations are satisfied if and only if u^2, v^2, w^2 are the roots of the cubic equation

$$z^3 + 2az^2 + (a^2 - 4c)z - b^2 = 0, \tag{4}$$

and if the signs of u, v, w are so chosen that the last of equations (3) is satisfied. This last condition still allows four different determinations of signs, so that the four roots of the biquadratic equation are obtained in the following way:

$$\begin{array}{rcl} 2\alpha & = & u + v + w, \\ 2\beta & = & u - v - w, \\ 2\gamma & = & -u + v - w, \\ 2\delta & = & -u - v + w. \end{array} \tag{5}$$

Thus the solution of the biquadratic equation has been reduced to that of the cubic equation (4).

This equation is called a cubic resolvent of the biquadratic equation.

§ 38. Proof of the fundamental theorem.

We now proceed to the proof of the fundamental theorem of algebra, namely that every equation $f(x) = 0$ has at least one root. We first put forward some general propositions.

1. If S denotes an arbitrary system of real numbers all greater than a certain positive or negative number C (that is, a system which contains no infinitely negative numbers), then there exists a lower bound for the numbers S.

By a lower bound g one is to understand a number which is not undercut by any number of the system S, but which is of such a nature that, if d is an arbitrarily given positive quantity, then between g and $g + d$ (including the endpoints) there always lies at least one number of the system S, however small d may be.

The proof follows directly from the possibility of a cut $(\mathfrak{A}, \mathfrak{B})$, constructed as follows: one throws a number α into $\mathfrak{A}$ if it is not undercut by any number of the system S, and a number β into $\mathfrak{B}$ if it is undercut by at least one number in S. The number g determined by this cut is not undercut by any number in S, for otherwise there would also be numbers smaller than g, and therefore belonging to $\mathfrak{A}$, which nevertheless are undercut by numbers in S; while, on the other hand, every number β lying even ever so little above g belongs to $\mathfrak{B}$ and is therefore undercut by a number in S. These are precisely the characteristic marks of the lower bound.

2. If S' is a part of S, then the numbers S' also have a lower bound g', and this is either equal to g or greater than g.

For it cannot be smaller than g, since otherwise numbers would occur in S', and therefore also in S, which lie below g.

Now let $f(x)$ be a real function of x which has only finite values in the interval

$$a \leq x \leq b \tag{1}$$

and which is also continuous in this interval. In agreement with § 31, V and VI, this means that

$$f(x \pm h) - f(x) \tag{2}$$

falls, in absolute value, below an arbitrarily prescribed number η throughout the interval whenever the positive h is smaller than a certain number ε. For $x = a$ we take in (2) only the upper sign, and for $x = b$ only the lower sign, so as not to pass outside the interval with $x \pm h$.

For the values of such a function in the interval there is, by 1., a lower bound g, and we now prove the proposition:

3. The function $f(x)$ assumes the value g for some value ξ of the interval, so that the lower bound becomes a minimum of the function $f(x)$ in the interval.

The proof is as follows. By 2., the lower bound of the function values in a part of the interval (1) is either equal to g or greater than g.

If now $f(a) = g$, what is to be proved is true. If, however, $f(a) > g$, then by the continuity of $f(x)$ one can give an interval $a \leq x \leq a + h$ in which $f(x) > g$ remains true, and hence the lower bound of $f(x)$ is greater than g.

We now construct in the interval (1) a cut $(\mathfrak{A}, \mathfrak{B})$ of the following kind: we count a value α of the interval (1) as belonging to $\mathfrak{A}$ if the lower bound of $f(x)$ in the interval $a \leq x \leq \alpha$ is greater than g, and a value β as belonging to $\mathfrak{B}$ if the lower bound in the interval $a \leq x \leq \beta$ is equal to g.

It is possible that $\mathfrak{B}$ consists only of the single value b; but then $f(b) = g$ must hold, and our assertion is fulfilled for $x = b$. For if $f(b) > g$, one could choose a quantity g' between $f(b)$ and g, and, by continuity, an interval $b - h \leq x \leq b$ such that in this interval all function values $f(x)$ are greater than g', so that their lower bound is equal to or greater than g', and hence certainly greater than g. But since $b - h$ belongs to $\mathfrak{A}$, the lower bound is also greater than g in the interval $a \leq x \leq b - h$, and consequently in the whole interval (1), contrary to the assumption.

If therefore $f(b) > g$, the cut $(\mathfrak{A}, \mathfrak{B})$ defines a number ξ in the interior of the interval (1), and we can now show that

$$f(\xi) = g \tag{3}$$

must hold.

If $f(\xi) > g$ and $f(\xi) > g' > g$, then, by the continuity of $f(x)$, we can determine an interval

$$\xi - h \leq x \leq \xi + h$$

in which all function values $f(x)$ are greater than g', and hence their lower bound is greater than g. But since $\xi - h$ belongs to $\mathfrak{A}$, the lower bound of $f(x)$ in the whole interval $a \leq x \leq \xi + h$ is also greater than g, while $\xi + h$ belongs to $\mathfrak{B}$, and this is the contradiction.

Let $f(x)$ now be an integral rational function with arbitrary complex or real coefficients, and let the variable x also be allowed to take complex values.

By § 31, IV, if C is an arbitrary positive value, the positive quantity R can be so chosen that

$$|f(x)| > C \tag{4}$$

whenever

$$|x| \geq R. \tag{5}$$

For visualization we represent the region (R) bounded by the inequality

$$|x| < R$$

for the variable $x = y + iz$ by a circular disc of radius R in the plane of the variable $x = y + iz$.

If we take C larger than any value of $|f(x)|$ in the interior of the region (R), for example larger than $|f(0)|$, then $|f(x)|$ will certainly become smaller in the interior of (R) than on its boundary. Since now in the whole interior of (R) the function $|f(x)|$, which for abbreviation we shall now denote by X, is not negative, there is a lower bound g for the values of X, and we have to prove the proposition:

4. There exists a value ξ of x in the interior of the region (R) such that

$$|f(\xi)| = g, \tag{6}$$

so that here also the lower bound is a minimum.

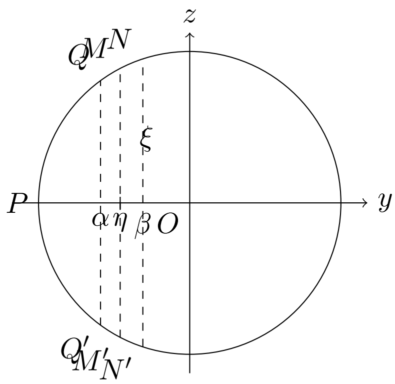
The lower bound of X in any part of the region is either equal to g or greater than g.

A domain of magnitudes determined by the inequalities

$$|x| \leq R, \qquad -R \leq y \leq \alpha \tag{7}$$

is represented in our Figure 2 by the segment $(PQ\alpha Q')$, which we shall call the segment (P, α).

Fig. 2.



We now first determine a value η of y by a cut $(\mathfrak{A}, \mathfrak{B})$ as follows: a number α between $-R$ and $+R$ is admitted into $\mathfrak{A}$ if the lower bound of X in the segment (P, α) is greater than g, and a value β is admitted into $\mathfrak{B}$ if the lower bound of X in the segment (P, β) is equal to g. This cut $(\mathfrak{A}, \mathfrak{B})$ defines a number η with the property that the lower bound of X in the region (P, γ) is greater than g if $\gamma < \eta$, and equal to g if $\gamma > \eta$.

Now, by § 31, VI,

$$|f(\eta + iz)|$$

is a continuous function of z in the interval

$$-\sqrt{R^2 - \eta^2} \leq z \leq \sqrt{R^2 - \eta^2}, \tag{8}$$

which is denoted in Figure 2 by M, M'.

Thus by 3. this function receives, for some value ζ of z in this interval, a minimum value γ, so that, if

$$\xi = \eta + i\zeta$$

is put,

$$|f(\xi)| = \gamma. \tag{9}$$

It is now further easy to see that $\gamma = g$ must be the case; for since g is the lower bound of all values of X within (R), γ cannot first of all be smaller than g.

But γ also cannot be greater than g; for all values which X assumes on the segment MM' are $\geq \gamma$. If, however, $\gamma > g$, then one can choose a quantity g' such that $\gamma > g' > g$, and, by the continuity of X proved in § 31, VI, numbers α, β can be so determined that

$$\alpha < \eta < \beta$$

and such that in the whole region $(P, \beta) - (P, \alpha) = (\alpha, \beta)$ ($QNN'Q'$ in Figure 2) X remains greater than g'. Consequently the lower bound of X in (α, β) is greater than g. Since now the lower bound of X in (P, α) is also greater than g, it is also greater than g in (P, β), which is composed of (P, α) and (α, β); but β belongs to $\mathfrak{B}$, and this would give a contradiction with the definition of $\mathfrak{B}$.

Thus there remains only the possibility that $\gamma = g$, and proposition 4 is thereby proved; namely, there is a value ξ in the interior of (R) for which

$$|f(\xi)| = g$$

holds, so that $|f(x)|$ attains a minimum value. We now prove:

5. If α is any value of x for which $f(\alpha)$ is different from zero, then h can be chosen so that

$$|f(\alpha + h)| < |f(\alpha)| \tag{10}$$

results.

It then follows that, if $f(\xi)$ were different from zero, g could not be the minimum of the function $|f(x)|$; but since this has been proved, one must have

$$f(\xi) = 0, \tag{11}$$

and ξ is a root of the equation $f(x) = 0$. The fundamental theorem will therefore be proved.

In the proof of 5. we make use of the theorem already proved (§ 32), that a pure equation always has a root.

Some of the derived functions $f'(\alpha), f''(\alpha), \dots$ may vanish; the last, $f^{(n)}(\alpha)$, which is equal to $\Pi(n)a_0$, is however different from zero. Thus suppose that $f'(\alpha), f''(\alpha), \dots, f^{(m-1)}(\alpha)$ vanish, but $f^{(m)}(\alpha)$ does not vanish. We then have, by § 13,

$$f(\alpha + h) = f(\alpha) + \frac{h^m}{\Pi(m)} f^{(m)}(\alpha) + \frac{h^{m+1}}{\Pi(m+1)} f^{(m+1)}(\alpha) + \cdots . \tag{12}$$

We choose, as is always possible by § 31, V, a positive number ε so that

$$\left| \frac{h}{m+1} \frac{f^{(m+1)}(\alpha)}{f^{(m)}(\alpha)} + \frac{h^2}{(m+1)(m+2)} \frac{f^{(m+2)}(\alpha)}{f^{(m)}(\alpha)} + \cdots \right| < 1 \tag{13}$$

whenever

$$|h| < \varepsilon, \tag{14}$$

and a positive proper fraction δ such that

$$\delta |f(\alpha)| < \left| \frac{\varepsilon^m f^{(m)}(\alpha)}{\Pi(m)} \right|, \tag{15}$$

which, if $f(\alpha)$ does not vanish, is likewise possible. We then determine h (on the basis of § 32) from the equation

$$\frac{h^m f^{(m)}(\alpha)}{\Pi(m)} = -\delta f(\alpha), \tag{16}$$

whence, by (15),

$$|h| < \varepsilon$$

follows. From (12), when the value from (16) is substituted for h^m, we now get

$$f(\alpha + h) = (1 - \delta) f(\alpha) - \delta f(\alpha) \left(\frac{h}{m+1} \frac{f^{(m+1)}(\alpha)}{f^{(m)}(\alpha)} + \frac{h^2}{(m+1)(m+2)} \frac{f^{(m+2)}(\alpha)}{f^{(m)}(\alpha)} + \cdots \right),$$

and hence

$$\begin{aligned} |f(\alpha + h)| &\le |f(\alpha)|(1 - \delta) \\ &\quad + \delta |f(\alpha)| \left| \frac{h}{m+1} \frac{f^{(m+1)}(\alpha)}{f^{(m)}(\alpha)} + \frac{h^2}{(m+1)(m+2)} \frac{f^{(m+2)}(\alpha)}{f^{(m)}(\alpha)} + \cdots \right|. \end{aligned} \tag{17}$$

By (13), therefore,

$$|f(\alpha + h)| < |f(\alpha)|. \tag{18}$$

Thus the proof of the fundamental theorem is completed.

§ 39. Algorithm for the calculation of the roots.

The proof given in the preceding paragraph for the existence of a root of an algebraic equation, although it leaves nothing to be desired in conclusiveness, nevertheless still has the defect that it does not make clear the steps by which a root can be calculated by a convergent computational procedure. We therefore add the following considerations, intended to remedy this defect, even if only theoretically. They also give a second proof of the fundamental theorem, due in its essentials to Lipschitz (*Lehrbuch der Analysis*, vol. I, § 61 ff.; I owe a simplification to a communication of Dedekind).

If the two integral rational functions $f(x)$ and $f'(x)$ have a common divisor, then this can be found and removed by rational operations according to § 6. We may therefore assume that $f(x)$ and $f'(x)$ have no common divisor, that is, that they do not vanish simultaneously for any value of x.

We now assume that the calculation of the roots has already succeeded for a function of degree $n-1$, and accordingly suppose $f'(x)$ resolved into linear factors. Thus, if

$$f(x) = x^n + a_1 x^{n-1} + a_2 x^{n-2} + \cdots, \tag{1}$$

then

$$f'(x) = nx^{n-1} + (n-1)a_1 x^{n-2} + \cdots = n(x-\beta_1)(x-\beta_2)\cdots(x-\beta_{n-1}), \tag{2}$$

where $\beta_1, \beta_2, \ldots, \beta_{n-1}$ are to be regarded as known numbers, some of which may also be identical. By our assumption, the absolute values

$$|f(\beta_1)|, |f(\beta_2)|, \ldots, |f(\beta_{n-1})|, \tag{3}$$

which we denote by

$$b_1, b_2, \ldots, b_{n-1}, \tag{4}$$

are all different from zero; let the notation be so chosen that b_1 is the smallest among them, or at least exceeds none of the others. By theorem 5 of § 38 a value α of x can then be determined so that the absolute value a of $f(\alpha)$ is smaller than b_1, hence

$$a < b_1, b_2, \ldots, b_{n-1}. \tag{5}$$

We now bound a domain G for the variable x in such a way that outside this domain the absolute value of $f(x)$ is always greater than a, so that G contains all points x for which $|f(x)| < a$ (but also other points). By § 31, IV this is possible in the following way. About the origin as centre draw a circle (R) of sufficiently large radius R containing the points $\alpha, \beta_1, \beta_2, \ldots, \beta_{n-1}$, and cut out of this circle the points $\beta_1, \beta_2, \ldots, \beta_{n-1}$, at which $|f(x)|$ is certainly greater than a, by circular envelopes of sufficiently small but non-zero radii $\rho_1, \rho_2, \ldots, \rho_{n-1}$, so that within all these circles $(\rho_1), (\rho_2), \ldots, (\rho_{n-1})$ the absolute value $|f(x)|$ remains greater than a (§ 31, V). None of these circles then encloses the point α, for at that point $|f(x)| = a$. The connected piece of surface bounded by the circles $(R), (\rho_1), \ldots, (\rho_{n-1})$ is the domain G.

We now determine a sequence of numbers

$$\alpha, \alpha', \alpha'', \alpha''', \ldots$$

in such a way that the absolute values of $f(\alpha), f(\alpha'), f(\alpha''), \ldots$, which we denote by

$$a, a', a'', a''', \ldots,$$

always decrease. The points so determined all remain in the interior of the domain G, since for them $|f(x)| < a$.

For this we use a procedure quite similar to that used for the proof of theorem 5 in § 38, but simplified by the fact that $f'(\alpha)$ is already different from zero. If δ denotes a positive proper fraction, still to be determined more precisely, we put in the expansion

$$f(\alpha + h) = f(\alpha) + hf'(\alpha) + \frac{h^2}{1 \cdot 2} f''(\alpha) + \cdots \tag{6}$$

$$h = -\delta \frac{f(\alpha)}{f'(\alpha)}, \tag{7}$$

and obtain

$$f(\alpha + h) = (1 - \delta) f(\alpha) + \frac{\delta^2 f(\alpha)^2 f''(\alpha)}{1 \cdot 2\, f'(\alpha)^2} - \frac{\delta^3 f(\alpha)^3 f'''(\alpha)}{1 \cdot 2 \cdot 3\, f'(\alpha)^3} + \cdots . \tag{8}$$

Now by (2)

$$f'(\alpha) = n(\alpha - \beta_1)(\alpha - \beta_2) \cdots (\alpha - \beta_{n-1});$$

therefore, if the absolute values of $\alpha - \beta_1, \alpha - \beta_2, \ldots, \alpha - \beta_{n-1}$ are denoted by $r_1, r_2, \ldots, r_{n-1}$,

$$|f'(\alpha)| = n r_1 r_2 \cdots r_{n-1}.$$

Since α lies outside the circle of radius ρ_i drawn about β_i, we have $r_1 > \rho_1, r_2 > \rho_2, \ldots$, and consequently

$$|f'(\alpha)| > n \rho_1 \rho_2 \cdots \rho_{n-1}.$$

Thus $|f'(\alpha)|$ is greater than a positive number k independent of the position of α in G. It follows that one can choose a sufficiently large positive number Q, likewise independent of the position of α and of the proper fraction δ, so that

$$\left| \frac{f(\alpha)^2 f''(\alpha)}{1 \cdot 2\, f'(\alpha)^2} - \delta \frac{f(\alpha)^3 f'''(\alpha)}{1 \cdot 2 \cdot 3\, f'(\alpha)^3} + \cdots \right| < Q. \tag{9}$$

One obtains Q, for example, by replacing in the denominators on the left of (9) $f'(\alpha)$ by k, replacing δ by 1, replacing all the other factors by their absolute values, replacing finally the absolute value of α by the greater value R, and then making the number so obtained arbitrarily larger. From (8) one obtains

$$|f(\alpha + h)| < (1 - \delta)|f(\alpha)| + \delta^2 Q. \tag{10}$$

Assume now Q so chosen that, for every α in G,

$$2Q > |f(\alpha)| = a,$$

which is plainly compatible with the above determination. We may then put

$$\delta = \frac{a}{2Q}, \qquad h = -\frac{a f(\alpha)}{2Q f'(\alpha)}, \tag{11}$$

and, denoting $\alpha + h$ by α', and denoting $|f(\alpha')|$ by a', obtain from (10)

$$a' < a\left(1 - \frac{a}{4Q}\right). \tag{12}$$

By the same procedure we derive from α', a' a second system α'', a'', then from α'', a'' a third system α''', a''', and so on; for the sequence of decreasing positive numbers $a, a', a'', \ldots$ we obtain the inequalities

$$a' < a\left(1 - \frac{a}{4Q}\right), \quad a'' < a'\left(1 - \frac{a'}{4Q}\right), \quad a''' < a''\left(1 - \frac{a''}{4Q}\right), \ldots . \tag{13}$$

We prove that this sequence approaches the limit zero. If this were not the case, a positive number ω could be given such that all $a^{(\nu)}$ remain above ω; the differences

$$\frac{a^{(\nu)}}{4Q}$$

would then all be greater than the positive proper fraction $\omega/(4Q)$, and from (13) it would follow that

$$a' < a\theta, \quad a'' < a'\theta, \quad a''' < a''\theta, \quad \ldots,$$

where $0 < \theta < 1$. By multiplication one gets

$$a^{(\nu)} < a\theta^{\nu}. \tag{14}$$

This is a contradiction, since θ^{ν} becomes infinitely small as ν increases without bound. The result so obtained is expressed by

$$\lim a^{(\nu)} = 0. \tag{15}$$

If the inequalities (13) are written in the form

$$a - a' > \frac{a^2}{4Q}, \qquad a' - a'' > \frac{a'^2}{4Q}, \qquad \ldots,$$

and added from the $(\mu+1)$-st to the $(\nu+1)$-st, one obtains

$$(a^{(\mu)})^2 + (a^{(\mu+1)})^2 + \cdots + (a^{(\nu)})^2 < 4Q\big(a^{(\mu)} - a^{(\nu+1)}\big),$$

and therefore, when μ and ν both tend to infinity,

$$\lim\big((a^{(\mu)})^2 + (a^{(\mu+1)})^2 + \cdots + (a^{(\nu)})^2\big) = 0. \tag{16}$$

For the sequence of numbers $\alpha, \alpha', \alpha'', \ldots$ the law of formation resulting from (11) is

$$\begin{aligned} \alpha' &= \alpha - \frac{af(\alpha)}{2Qf'(\alpha)}, \\ \alpha'' &= \alpha' - \frac{a'f(\alpha')}{2Qf'(\alpha')}, \\ &\vdots \\ \alpha^{(\nu)} &= \alpha^{(\nu-1)} - \frac{a^{(\nu-1)}f(\alpha^{(\nu-1)})}{2Qf'(\alpha^{(\nu-1)})}. \end{aligned} \tag{17}$$

If an arbitrary number of these equations are added, from the $(\mu+1)$-st to the $(\nu+1)$-st, then

$$\alpha^{(\mu)} - \alpha^{(\nu+1)} = \sum_{r=\mu}^{\nu} \frac{a^{(r)}f(\alpha^{(r)})}{2Qf'(\alpha^{(r)})}. \tag{18}$$

Since all the $|f'(\alpha^{(r)})|$ are at least equal to k, it follows for the absolute value of this difference that

$$|\alpha^{(\mu)} - \alpha^{(\nu+1)}| < \frac{(a^{(\mu)})^2 + (a^{(\mu+1)})^2 + \cdots + (a^{(\nu)})^2}{2Qk}, \tag{19}$$

and hence, by (16),

$$\lim |\alpha^{(\mu)} - \alpha^{(\nu+1)}| = 0. \tag{20}$$

This says (according to the concept of a sequence of numbers explained in the Introduction, p. 15) that the numbers $\alpha, \alpha', \alpha'', \ldots$ approach a definite limit, which we shall denote by ξ. From the continuity of the function $f(x)$ it then follows easily that $f(\xi) = 0$; for if $|f(\xi)| > 0$, then, since the $\alpha^{(\nu)}$ come arbitrarily close to ξ, the absolute values $a^{(\nu)}$ of $f(\alpha^{(\nu)})$ could not sink below every positive value, which is precisely what we have proved for them.

§ 40. Continuity of the roots.

We close this section with the proof of the theorem:

The roots of an algebraic equation are continuous functions of the coefficients.

We have first to explain the meaning of this theorem.

Let

$$f(x) = x^n + a_1 x^{n-1} + a_2 x^{n-2} + \cdots \tag{1}$$

be an integral rational function of x of degree n. By what has been proved in the preceding paragraphs, $f(x)$ can be decomposed into n linear factors, some of which may be equal to one another. Combining equal factors and denoting the roots by $\alpha, \beta, \gamma, \ldots$, we put

$$f(x) = (x-\alpha)^a (x-\beta)^b (x-\gamma)^c \cdots, \tag{2}$$

where $a, b, c, \ldots$ are positive whole numbers whose sum is equal to n.

The roots $\alpha, \beta, \gamma, \ldots$ will change with the coefficients $a_1, a_2, a_3, \ldots$; the degree of their multiplicity can also become another one.

We denote the changes of $a_1, a_2, \ldots$ by $\varepsilon_1, \varepsilon_2, \ldots$ and put

$$\varphi(x) = \varepsilon_1 x^{n-1} + \varepsilon_2 x^{n-2} + \cdots, \tag{3}$$

$$f(x) + \varphi(x) = f_1(x). \tag{4}$$

We surround the points $\alpha, \beta, \gamma, \ldots$ by regions of arbitrary smallness, but in such a way that these regions exclude one another; for instance, we enclose the points $\alpha, \beta, \gamma, \ldots$ by circular peripheries with radii $\rho, \rho', \rho'', \ldots$, and denote these regions by $(\rho), (\rho'), (\rho''), \ldots$.

If the absolute values of $\varepsilon_1, \varepsilon_2, \ldots$ lie below sufficiently small values, then we can prove first of the function $f_1(x)$

that it has no roots outside the regions $(\rho), (\rho'), (\rho''), \ldots$,

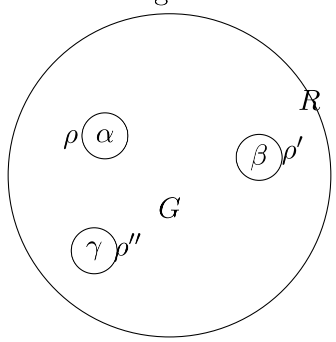
and secondly,

that the number of roots of $f_1(x)$ within (ρ) is exactly a, within (ρ') exactly b, within (ρ'') exactly c, and so on.

In the last part of the theorem, however, it must be observed that, if $f_1(x)$ has multiple roots, these must be counted according to their multiplicity.

We construct, according to §31, IV, in the x-plane a circle with radius R and with the origin as centre, which encloses the regions $(\rho), (\rho'), (\rho''), \ldots$, so that outside this circle no more roots of $f_1(x)$ lie, and we denote by G the region lying within this circle, but outside $(\rho), (\rho'), (\rho''), \ldots$. It must also be remarked that R can be assumed independently of $\varepsilon_1, \varepsilon_2, \ldots$, so long as a definite upper bound is fixed for the absolute values of these quantities.

Fig. 4.



If now x is a point of the region G, then the absolute value of $x - \alpha$ is greater than ρ, and consequently, by (2),

$$|f(x)| > \rho^a \rho'^b \rho''^c \cdots . \tag{5}$$

If now α_1 is a root of $f_1(x)$, then from (4) it follows that

$$f(\alpha_1) = -\varphi(\alpha_1), \tag{6}$$

and α_1 therefore cannot lie in the region G if one takes care that everywhere within G

$$|\varphi(x)| < \rho^a \rho'^b \rho''^c \cdots , \tag{7}$$

which is always possible by sufficient diminution of the upper bound of $\varepsilon_1, \varepsilon_2, \ldots$. Thus the first part of our assertion is proved, namely that, when the inequality (7) is satisfied, no roots of $f_1(x)$ lie outside the regions $(\rho), (\rho'), (\rho''), \ldots$.

To prove the second part, we put

$$f(x) = \psi(x)(x - \alpha)^a, \tag{8}$$
$$\psi(x) = (x - \beta)^b (x - \gamma)^c \cdots . \tag{9}$$

We now choose a positive number A, which shall however be independent of ρ, in such a way that, as long as x lies in the interior or on the periphery of the region (ρ),

$$|\psi(x)| > A \qquad (\text{for } |x - \alpha| \leq \rho) \tag{10}$$

remains true. Such a number is obtained, for example, if l is taken equal to half the least of the distances $(\alpha, \beta), (\alpha, \gamma), \ldots$ and one puts

$$A = l^{b+c+\cdots};$$

then the requirement expressed in (10) is fulfilled at least as long as ρ is smaller than l. If only one point α is present, then $\psi(x) = 1$ is to be put, and for A any proper fraction can be chosen.

Let now $\alpha_1, \alpha_2, \ldots$ be the roots of $f_1(x)$ within (ρ), and let $a_1, a_2, \ldots$ be the degrees of their multiplicity; and let $\beta_1, \beta_2, \ldots$, $b_1, b_2, \ldots$, $\gamma_1, \gamma_2, \ldots$, $c_1, c_2, \ldots$ have the same meaning for the regions $(\rho'), (\rho''), \ldots$. Then

$$n = a_1 + a_2 + \cdots + b_1 + b_2 + \cdots + c_1 + c_2 + \cdots , \tag{11}$$

since the total number of all roots must be equal to n. The function $f_1(x)$ can then be represented as

$$f_1(x) = \psi_1(x)(x - \alpha_1)^{a_1}(x - \alpha_2)^{a_2} \cdots , \tag{12}$$

where

$$\psi_1(x) = (x - \beta_1)^{b_1}(x - \beta_2)^{b_2} \cdots (x - \gamma_1)^{c_1}(x - \gamma_2)^{c_2} \cdots . \tag{13}$$

Now we can determine a positive number B independent of $\rho, \rho', \rho'', \ldots$ such that, as long as x remains within or on the boundary of (ρ),

$$|\psi_1(x)| < B \qquad (\text{for } |x - \alpha| \leq \rho). \tag{14}$$

For example, one can choose a magnitude L which is greater than twice the distance of the point α from one of the points $\beta, \gamma, \ldots$ and in every case greater than 1, and then assume

$$B > L^n.$$

Now from (4) it follows that

$$|f(x)| < |f_1(x)| + |\varphi(x)|, \tag{15}$$

hence by (8) and (12)

$$|\psi(x)|\,|x-\alpha|^a < |\psi_1(x)|\,|x-\alpha_1|^{a_1}\,|x-\alpha_2|^{a_2}\cdots + |\varphi(x)|. \tag{16}$$

If we now assume x on the periphery of ρ, and therefore put

$$|x-\alpha| = \rho,$$

then

$$|x-\alpha_1| < 2\rho, qquad |x-\alpha_2| < 2\rho, \ldots,$$

and consequently, by (10) and (16),

$$\rho^a A < (2\rho)^{a_1+a_2+\cdots} B + |\varphi(x)|. \tag{17}$$

We now choose the upper bounds for the coefficients $\varepsilon_1, \varepsilon_2, \ldots$ of φ so small that

$$|\varphi(x)| < \rho^a A'$$

where A' denotes a number which is smaller than A; then from (17) it follows that

$$A - A' < 2^a \rho^{-a+a_1+a_2+\cdots} B. \tag{18}$$

But, for sufficiently small ρ, this would no longer be possible if a were greater than the sum $a_1 + a_2 + \cdots$. Hence

$$a \le a_1 + a_2 + \cdots\,. \tag{19}$$

In the same way one proves that

$$b \le b_1 + b_2 + \cdots\,, \qquad c \le c_1 + c_2 + \cdots\,, \qquad \ldots. \tag{20}$$

But since the sums of the left sides as well as of the right sides in the inequalities (19), (20) must be equal to n, only equality signs can hold, and therefore

$$a = a_1 + a_2 + \cdots\,, \quad b = b_1 + b_2 + \cdots\,, \quad c = c_1 + c_2 + \cdots\,, \quad \ldots,$$

whereby the second part of our theorem is also proved.

We shall add to the theorem just proved the following remark concerning the case of multiple roots.

The necessary and sufficient condition that $f(x) = 0$ should have multiple roots at all is that $f(x)$ and $f'(x)$ should have a common factor. Therefore, if the algorithm of the greatest common divisor is applied to these two functions according to the principles developed in the first section, the condition for common factors is obtained by setting the last constant remainder equal to zero, in the form of an equation between the coefficients,

$$F(a_1, a_2, \ldots, a_n) = 0,$$

where F is an integral rational function of the arguments $a_1, a_2, \ldots, a_n$, which is called the discriminant of f, and whose properties and mode of formation will be treated more fully in the next section. In any case, since equations of degree n without multiple roots do exist at all, F cannot vanish identically.

Consider now

$$F(a_1 + \varepsilon_1, a_2 + \varepsilon_2, \ldots, a_n + \varepsilon_n).$$

This function too, if $a_1, a_2, \ldots, a_n$ are definite and $\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n$ are indeterminate numbers, will not vanish identically; and one can dispose of the ε's so that its value comes out different from zero, even when any arbitrary upper bound has been fixed for the absolute values of the ε's.[10]

[10] This will only later be rigorously justified. For the time being we presuppose it in these preliminary remarks, on which no further conclusions will be founded.

One therefore sees from these considerations how an a-fold root α of $f(x)$, under continuous variation of the coefficients, radiates apart into a simple roots, so that from this point of view too one is justified in considering the a-fold root as having arisen from the coincidence of a simple roots.

If we do not assume the coefficient of the highest power of x equal to 1, and therefore examine the equation in the form

$$a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_{n-1}x + a_n = 0, \tag{21}$$

then, if $a_n = 0$, one or more of the roots are equal to zero, and equation (21) can be divided by x or by a higher power of x, whereby an equation of correspondingly lower degree arises in which the last term is different from zero. We shall therefore assume from the outset that a_n is different from zero, and now put in (21)

$$x = \frac{1}{y}. \tag{22}$$

By multiplication by y^n and division by a_n, (21) then passes into

$$y^n + \frac{a_{n-1}}{a_n}y^{n-1} + \cdots + \frac{a_1}{a_n}y + \frac{a_0}{a_n} = 0, \tag{23}$$

and we can apply to the roots of this equation the theorem just stated, by assuming $a_0 = 0$, so that one of the roots of (23) vanishes. We thus obtain the theorem:

One can take a_0 in the equation (21) so small that one of its roots becomes large beyond all bounds, while the others differ arbitrarily little from the roots of the equation of degree $n - 1$

$$a_1x^{n-1} + a_2x^{n-2} + \cdots + a_{n-1}x + a_n = 0.$$

This is also expressed by saying that, as a_0 vanishes, one of the roots of (21) becomes infinite.

If a_1 also vanishes, or a_1 and a_2, and so on, two or three roots become infinite.

This justifies the expression sometimes used, that an equation of degree n passes, by the becoming-infinite of one root, into an equation of degree $n - 1$.

Fourth Section. Symmetric functions.

§ 41. Concept of symmetric functions. Symmetric fundamental functions.

In this section we consider whole functions, of arbitrary degree, in an arbitrary number n of variables

$$\alpha_1, \alpha_2, \ldots, \alpha_n.$$

Such a function is called symmetric if it remains unchanged when the variables $\alpha_1, \alpha_2, \ldots, \alpha_n$ are subjected to an arbitrary permutation; and it is such symmetric functions whose properties and laws of formation we now have to learn more precisely.

In order that a function $\Phi(\alpha_1, \alpha_2, \ldots, \alpha_n)$ be symmetric, it is sufficient that it not be changed when any two of the arguments $\alpha_1, \alpha_2, \ldots, \alpha_n$ are interchanged, since by § 18 all permutations can be formed by a succession of transpositions.

The function Φ will in general not be homogeneous, but will contain terms of different dimensions. If, however, one collects all terms of the same dimension, every Φ can be represented as a sum of homogeneous functions of different degrees; and, if Φ is to be symmetric, then each homogeneous constituent of a definite degree must by itself be symmetric, since the dimensions of the terms are not changed by the permutations of the variables.

Accordingly we may confine ourselves to the consideration of homogeneous symmetric functions, from which all the others can be put together.

After this it is easy to give the general form of a symmetric function. We obtain it if, in one term of such a function,

$$\alpha_1^{\nu_1}\alpha_2^{\nu_2}\cdots\alpha_n^{\nu_n},$$

as in the formation of determinants, we permute the lower indices in all possible ways and take the sum of all terms so formed. Such a function may be called an elementary constituent of a symmetric function. If we take several such elementary constituents, multiply them by arbitrary factors independent of the α's, and add them, then we obtain the most general symmetric function. The number of terms of one of these elementary constituents is $n!$ when the exponents $\nu_1, \nu_2, \ldots, \nu_n$ are all different; but if one and the same exponent ν occurs several times, the permutations which give no different terms are to be omitted. For example, for three variables, if ν_1, ν_2, ν_3 are different, an elementary constituent is

$$\alpha_1^{\nu_1}\alpha_2^{\nu_2}\alpha_3^{\nu_3} + \alpha_1^{\nu_1}\alpha_2^{\nu_3}\alpha_3^{\nu_2} + \alpha_1^{\nu_2}\alpha_2^{\nu_1}\alpha_3^{\nu_3} + \alpha_1^{\nu_2}\alpha_2^{\nu_3}\alpha_3^{\nu_1} + \alpha_1^{\nu_3}\alpha_2^{\nu_1}\alpha_3^{\nu_2} + \alpha_1^{\nu_3}\alpha_2^{\nu_2}\alpha_3^{\nu_1},$$

whereas, if $\nu_2 = \nu_3$,

$$\alpha_1^{\nu_1}\alpha_2^{\nu_2}\alpha_3^{\nu_2} + \alpha_1^{\nu_2}\alpha_2^{\nu_1}\alpha_3^{\nu_2} + \alpha_1^{\nu_2}\alpha_2^{\nu_2}\alpha_3^{\nu_1}.$$

The simplest example of a symmetric function is the sum of the variables

$$\alpha_1 + \alpha_2 + \alpha_3 + \cdots + \alpha_n.$$

Likewise the product

$$\alpha_1\alpha_2\cdots\alpha_n$$

belongs here.

These two are the extreme cases of a series of symmetric functions which we call the symmetric fundamental functions and obtain as follows.

The product

$$f(x) = (x-\alpha_1)(x-\alpha_2)\cdots(x-\alpha_n) \tag{1}$$

is, whatever x may be, provided only that x is independent of the α's, a symmetric function of $\alpha_1, \alpha_2, \ldots, \alpha_n$. If therefore we carry out the multiplication of the several factors and arrange according to powers of x, then the coefficients of the separate powers of x are likewise symmetric functions. For they change no more under interchange of the α's than does the function $f(x)$ (§ 1). We put

$$f(x) = x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_{n-1}x + a_n. \tag{2}$$

The functions $a_1, a_2, \ldots, a_n$ are the coefficients of the algebraic equation whose roots are $\alpha_1, \alpha_2, \ldots, \alpha_n$. They are called the symmetric fundamental functions. We already formed these symmetric fundamental functions in § 7; as we recall, a_1 is the sum of the α's taken with negative sign, a_2 the sum of the products two at a time, a_3 the sum of the products three at a time taken with negative sign, and so on; finally a_n is the product of all the α's with positive or negative sign according as n is even or odd. We write, for short,

$$\begin{aligned} a_1 &= -\sum \alpha_1, \\ a_2 &= +\sum \alpha_1\alpha_2, \\ a_3 &= -\sum \alpha_1\alpha_2\alpha_3, \\ &\vdots \\ a_n &= \pm\alpha_1\alpha_2\cdots\alpha_n. \end{aligned} \tag{3}$$

The goal of our considerations is the proof of the principal theorem that all symmetric functions of the α's can be expressed rationally by the fundamental functions.

§ 42. The power sums.

We occupy ourselves first with another special kind of symmetric functions, the power sums. Namely, if ν denotes any positive integral exponent, then

$$s_\nu = \alpha_1^\nu + \alpha_2^\nu + \cdots + \alpha_n^\nu \tag{1}$$

obviously belongs to the symmetric functions, and s_ν is called the νth power sum. We shall derive formulae according to which the power sums are expressible by the fundamental functions.

In what follows, whenever $\varphi(x)$ is any function of x, we always denote by

$$S[\varphi(\alpha)]$$

the sum obtained by replacing x in $\varphi(x)$ successively by each of the variables $\alpha_1, \alpha_2, \ldots, \alpha_n$ and adding the functions so formed, thus

$$S[\varphi(\alpha)] = \varphi(\alpha_1) + \varphi(\alpha_2) + \cdots + \varphi(\alpha_n).$$

Accordingly, for example,

$$S[\alpha^\nu] = s_\nu$$

is the νth power sum.

We now divide $f(x)$, by § 4, by an arbitrary linear function $x - \alpha$ and obtain

$$\frac{f(x) - f(\alpha)}{x - \alpha} = x^{n-1} + f_1(\alpha)x^{n-2} + f_2(\alpha)x^{n-3} + \cdots + f_{n-1}(\alpha), \tag{2}$$

where, by § 4, (8),

$$\begin{aligned} f_1(\alpha) &= \alpha + a_1, \\ f_2(\alpha) &= \alpha^2 + a_1\alpha + a_2, \\ &\vdots \\ f_{n-1}(\alpha) &= \alpha^{n-1} + a_1\alpha^{n-2} + \cdots + a_{n-1}. \end{aligned} \tag{3}$$

We put in (2), for α, each of the quantities $\alpha_1, \alpha_2, \ldots, \alpha_n$, whereby $f(\alpha) = 0$, and form the sum S. The left hand side then gives, by § 14, (11),

$$nx^{n-1} + (n-1)a_1x^{n-2} + (n-2)a_2x^{n-3} + \cdots + a_{n-1}, \tag{4}$$

whereas the right hand side becomes

$$nx^{n-1} + x^{n-2}S[f_1(\alpha)] + x^{n-3}S[f_2(\alpha)] + \cdots + S[f_{n-1}(\alpha)]. \tag{5}$$

The comparison of coefficients of like powers of x in (4) and (5) gives

$$\begin{aligned} S[f_1(\alpha)] &= (n-1)a_1, \\ S[f_2(\alpha)] &= (n-2)a_2, \\ &\vdots \\ S[f_{n-1}(\alpha)] &= a_{n-1}. \end{aligned} \tag{6}$$

But by (1) and (3)

$$\begin{aligned} S[f_1(\alpha)] &= s_1 + na_1, \\ S[f_2(\alpha)] &= s_2 + a_1s_1 + na_2, \\ &\vdots \\ S[f_{n-1}(\alpha)] &= s_{n-1} + a_1s_{n-2} + a_2s_{n-3} + \cdots + na_{n-1}. \end{aligned}$$

Consequently from (6) one obtains the following system of formulae due to Newton:

$$\begin{aligned}
0 &= s_1 + a_1,\\
0 &= s_2 + a_1 s_1 + 2a_2,\\
0 &= s_3 + a_1 s_2 + a_2 s_1 + 3a_3,\\
&\vdots\\
0 &= s_{n-1} + a_1 s_{n-2} + a_2 s_{n-3} + \cdots + (n-1)a_{n-1}.
\end{aligned} \tag{7}$$

This system of formulae can, however, be continued still further; for, since

$$S[f(\alpha)] = 0, \qquad S[\alpha f(\alpha)] = 0, \qquad S[\alpha^2 f(\alpha)] = 0, \ldots,$$

it follows that

$$\begin{aligned}
0 &= s_n + a_1 s_{n-1} + a_2 s_{n-2} + \cdots + a_n,\\
0 &= s_{n+1} + a_1 s_n + a_2 s_{n-1} + \cdots + a_n s_1,\\
0 &= s_{n+2} + a_1 s_{n+1} + a_2 s_n + \cdots + a_n s_2, \quad \ldots
\end{aligned} \tag{8}$$

By the formulae (7), (8) the problem is now solved of representing, one after another, the functions $s_1, s_2, \ldots$ up to any height as whole rational functions of the symmetric fundamental functions; for example

$$\begin{aligned}
s_1 &= -a_1,\\
s_2 &= a_1^2 - 2a_2,\\
s_3 &= -a_1^3 + 3a_1 a_2 - 3a_3,\\
s_4 &= a_1^4 - 4a_1^2 a_2 + 4a_1 a_3 + 2a_2^2 - 4a_4.
\end{aligned} \tag{9}$$

The manner of formation of these expressions shows that the coefficients in these representations are whole numbers.

Conversely, by means of the formulae (7) and (8), one can express the symmetric fundamental functions $a_1, a_2, a_3, \ldots$ rationally by the power sums $s_1, s_2, s_3, \ldots$. This representation is, however, far less simple in so far as the coefficients are not whole but fractional numbers, for example

$$a_1 = -s_1, \qquad 2a_2 = s_1^2 - s_2, \qquad 6a_3 = -s_1^3 + 3s_1 s_2 - 2s_3. \tag{10}$$

One can also continue the system of formulae (8) in the opposite direction, if one forms the equations

$$S[\alpha^{-1} f(\alpha)] = 0, \qquad S[\alpha^{-2} f(\alpha)] = 0, \ldots.$$

In this way one obtains a means of expressing the power sums s_ν also for negative exponents ν by the fundamental functions. These sums of negative powers likewise belong to the symmetric functions, although no longer to the whole, but to the fractional, ones. They pass into whole functions only after multiplication by powers of the product $\alpha_1 \alpha_2 \cdots \alpha_n$. For completeness we put down here the first two of these formulae:

$$\begin{aligned}
0 &= a_{n-1} + a_n s_{-1},\\
0 &= 2a_{n-2} + a_{n-1} s_{-1} + a_n s_{-2}.
\end{aligned} \tag{11}$$

§ 43. Proof of the principal theorem for two variables.

We now pass to the proof of the fundamental theorem of the theory of symmetric functions, that they are all rationally expressible by the symmetric fundamental functions. Since we use complete induction as the means of proof, we first derive the theorem under the assumption that only two independent variables α, β are given, but in a way which at the same time will bring out the guiding thought for the general proof.

We denote the symmetric fundamental functions by

$$a = -(\alpha + \beta), \qquad b = \alpha\beta \tag{1}$$

and accordingly put

$$f(x) = (x - \alpha)(x - \beta) = x^2 + ax + b. \tag{2}$$

Let $S(\alpha, \beta)$ be any whole rational and symmetric function of α and β. We can substitute for β from (1) the value $-(\alpha + a)$, and obtain, if we arrange according to powers of α,

$$S(\alpha, \beta) = S(\alpha, -\alpha - a) = A_0\alpha^m + A_1\alpha^{m-1} + \cdots + A_m, \tag{3}$$

where the coefficients $A_0, A_1, \ldots, A_m$ depend only on a and on the coefficients still occurring in S. We note expressly, however, that if in $S(\alpha, \beta)$ no fractional numerical coefficients occur, then no fractions occur in the coefficients $A_0, A_1, \ldots, A_m$ either.

We now put

$$\Phi(x) = A_0x^m + A_1x^{m-1} + \cdots + A_{m-1}x + A_m \tag{4}$$

and divide $\Phi(x)$ by $f(x)$, by § 3. We obtain a quotient Q and a remainder which, with respect to x, is at most of the first degree; thus

$$\Phi(x) = Qf(x) + A + Bx. \tag{5}$$

Here A and B are now whole functions of a and b, and they contain no fractional numerical coefficients if none occur in S. If now we put $x = \alpha$, it follows from (5) and (3), since $f(\alpha)$ vanishes, that

$$S(\alpha, \beta) = A + B\alpha. \tag{6}$$

But since $S(\alpha, \beta)$ and likewise A, B are symmetric, it follows by interchanging α and β that

$$S(\alpha, \beta) = A + B\beta. \tag{7}$$

From this one concludes, since α and β are independent variables, that $B = 0$, and consequently

$$S(\alpha, \beta) = A, \tag{8}$$

which proves the fundamental theorem in this case.

§ 44. General proof of the principal theorem.

We now suppose that the fundamental theorem has been proved for symmetric functions of $n - 1$ variables, and we derive it, by a procedure entirely analogous to that applied in § 43, for n variables.

Let again

$$S = S(\alpha_1, \alpha_2, \ldots, \alpha_n) \tag{1}$$

be a whole symmetric function of the n variables $\alpha_1, \alpha_2, \ldots, \alpha_n$. If we arrange it according to powers of α_1, and accordingly put

$$S = S_0\alpha_1^\mu + S_1\alpha_1^{\mu-1} + \cdots + S_{\mu-1}\alpha_1 + S_\mu, \tag{2}$$

then the coefficients $S_0, S_1, \ldots, S_\mu$ are whole symmetric functions of the $n - 1$ variables $\alpha_2, \ldots, \alpha_n$.

If we denote the symmetric fundamental functions of these latter variables by

$$a_1', a_2', \ldots, a_{n-1}',$$

then, by our supposition, the coefficients $S_0, S_1, \dots, S_\mu$ can be expressed rationally by these. But by § 42, (2) and (3),

$$\begin{aligned} a_1' &= f_1(\alpha_1) = \alpha_1 + a_1, \\ a_2' &= f_2(\alpha_1) = \alpha_1^2 + a_1\alpha_1 + a_2, \\ a_3' &= f_3(\alpha_1) = \alpha_1^3 + a_1\alpha_1^2 + a_2\alpha_1 + a_3, \end{aligned}$$

and so on. Thus the coefficients $S_0, S_1, \dots, S_\mu$ can be expressed wholly and rationally by

$$\alpha_1, a_1, a_2, \dots, a_n.$$

If, therefore, after these expressions have been introduced, we arrange (2) anew according to powers of α_1, we obtain

$$S = A_0\alpha_1^m + A_1\alpha_1^{m-1} + \cdots + A_{m-1}\alpha_1 + A_m, \tag{3}$$

where in general m is an exponent different from μ, and where the coefficients $A_0, A_1, \dots, A_m$ depend wholly and rationally on $a_1, a_2, \dots, a_n$. We put again

$$\Phi(x) = A_0x^m + A_1x^{m-1} + \cdots + A_{m-1}x + A_m \tag{4}$$

and divide $\Phi(x)$ by

$$\begin{aligned} f(x) &= x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_n \\ &= (x-\alpha_1)(x-\alpha_2)\cdots(x-\alpha_n). \end{aligned}$$

There results a quotient and a remainder which, with respect to x, is at most of degree $n-1$. We therefore put

$$\Phi(x) = Qf(x) + \psi(x), \tag{5}$$

$$\psi(x) = C_0x^{n-1} + C_1x^{n-2} + \cdots + C_{n-2}x + C_{n-1}, \tag{6}$$

and here $C_0, C_1, \dots, C_{n-1}$ are whole rational functions of $a_1, a_2, \dots, a_n$, in which, if S in its original form contains no fractional coefficients, no fractions occur either.

But now, since $f(\alpha_1)$ vanishes and $S = \Phi(\alpha_1)$, it follows from (5) that

$$S = \psi(\alpha_1). \tag{7}$$

Since S is symmetric, α_1 may here be replaced by $\alpha_2, \alpha_3, \dots, \alpha_n$. This gives the following n equations:

$$\begin{aligned} C_0\alpha_1^{n-1} + C_1\alpha_1^{n-2} + \cdots + C_{n-2}\alpha_1 + (C_{n-1} - S) &= 0, \\ C_0\alpha_2^{n-1} + C_1\alpha_2^{n-2} + \cdots + C_{n-2}\alpha_2 + (C_{n-1} - S) &= 0, \\ &\vdots \\ C_0\alpha_n^{n-1} + C_1\alpha_n^{n-2} + \cdots + C_{n-2}\alpha_n + (C_{n-1} - S) &= 0. \end{aligned} \tag{8}$$

If we regard this system as a system of homogeneous linear equations with the n unknowns

$$C_0, C_1, \dots, C_{n-2}, C_{n-1} - S,$$

then its determinant is

$$\begin{vmatrix} \alpha_1^{n-1} & \alpha_1^{n-2} & \cdots & \alpha_1 & 1 \\ \alpha_2^{n-1} & \alpha_2^{n-2} & \cdots & \alpha_2 & 1 \\ \vdots & \vdots & & \vdots & \vdots \\ \alpha_n^{n-1} & \alpha_n^{n-2} & \cdots & \alpha_n & 1 \end{vmatrix},$$

which, by § 22, formula (12), is equal to the product of all differences

$$\alpha_1 - \alpha_2, \quad \alpha_1 - \alpha_3, \dots$$

and is therefore different from zero. Consequently, by Theorem II in § 24,

$$C_0 = 0, \quad C_1 = 0, \ldots, \quad C_{n-2} = 0,$$

and

$$S = C_{n-1}, \tag{9}$$

which contains the fundamental theorem to be proved.

One reaches the same result also by Theorem II, § 30; for since the function of degree $n-1$ in x,

$$C_0 x^{n-1} + C_1 x^{n-2} + \cdots + C_{n-2} x + C_{n-1} - S,$$

vanishes for $x = \alpha_1, \alpha_2, \ldots, \alpha_n$, that is, for more than $n-1$ values, all its coefficients must be zero.

The proof which we have here given for the fundamental theorem in connection with Cauchy[11] at the same time offers a means, in special cases, actually to compute the expression of a symmetric function by the fundamental functions. The other proofs known for the theorem offer the same possibility. We shall communicate here a second proof, which leads to a method of computation that is often simpler.[12]

§ 45. Second proof of the theorem on symmetric functions.

Let S be a whole symmetric function of the variables $\alpha_1, \alpha_2, \ldots, \alpha_n$. The individual terms of this function are all of the form

$$M \alpha_1^{\nu_1} \alpha_2^{\nu_2} \cdots \alpha_n^{\nu_n},$$

where M is a coefficient independent of the α's. These terms shall now be ordered in a definite manner. After the order of the variables $\alpha_1, \alpha_2, \ldots, \alpha_n$ has been fixed, of two terms

$$A = M \alpha_1^{\nu_1} \alpha_2^{\nu_2} \cdots \alpha_n^{\nu_n}, \qquad A' = M' \alpha_1^{\nu_1'} \alpha_2^{\nu_2'} \cdots \alpha_n^{\nu_n'},$$

A shall be called the higher if the first of the differences

$$\nu_1 - \nu_1', \quad \nu_2 - \nu_2', \ldots, \nu_n - \nu_n'$$

which is different from zero has a positive value; thus if either $\nu_1 > \nu_1'$, or $\nu_1 = \nu_1', \nu_2 > \nu_2'$, or $\nu_1 = \nu_1', \nu_2 = \nu_2', \nu_3 > \nu_3'$, and so on.

Since we suppose that all terms in which all exponents $\nu_1, \nu_2, \ldots, \nu_n$ agree have been united into one term, it is thereby decided, for any two terms, which is the higher; and if A is higher than A', and A' higher than A'', then A is also higher than A''.

Let now, according to this ordering,

$$A = M \alpha_1^{\nu_1} \alpha_2^{\nu_2} \cdots \alpha_n^{\nu_n}$$

be the highest term of our function S. It follows that

$$\nu_1 \geq \nu_2$$

must hold; for if $\nu_1 < \nu_2$, then the term

$$M \alpha_1^{\nu_2} \alpha_2^{\nu_1} \cdots \alpha_n^{\nu_n},$$

which must also occur in S because of symmetry, would stand higher in the order than A, contrary to the supposition. In the same way it follows that $\nu_2 \geq \nu_3$ must hold; for if $\nu_2 < \nu_3$, then

$$M \alpha_1^{\nu_1} \alpha_2^{\nu_3} \alpha_3^{\nu_2} \cdots \alpha_n^{\nu_n}$$

[11] Cauchy, *Exercices de mathématiques*, fourth year.

[12] Waring, *Meditationes algebraicae*. Gauss, *Demonstratio nova altera theorematis omnem functionem algebraicam rationalem integram unius variabilis in factores reales primi vel secundi gradus resolvi posse.* Werke, vol. III, p. 36.

would stand higher than A, and so forth. We therefore conclude that the exponents in A,

$$\nu_1, \nu_2, \nu_3, \ldots, \nu_n,$$

form a decreasing, or at least never increasing, sequence of numbers; that is, the differences

$$\nu_1 - \nu_2, \quad \nu_2 - \nu_3, \ldots, \nu_{n-1} - \nu_n$$

are all positive, or at least not negative.

We conclude secondly that in no term of the function S can an exponent higher than ν_1 occur; for otherwise there would also occur a term in which α_1 had this higher exponent, and that would be of higher order than A, contrary to the supposition. Thus the terms that can occur at all in a symmetric function are, apart from the coefficients, completely determined by the highest term; and the number of possible terms is finite when the highest term is given.

We can denote the order of the symmetric function by the exponent sequence of its highest term,

$$(\nu_1, \nu_2, \ldots, \nu_n).$$

Apart from the case in which all exponents are zero, so that the function is a constant, the lowest possible order is

$$(1, 0, \ldots, 0),$$

to which corresponds the single symmetric function

$$M(\alpha_1 + \alpha_2 + \cdots + \alpha_n) = -Ma_1.$$

The next higher order is

$$(1, 1, 0, \ldots, 0),$$

to which corresponds the symmetric function

$$Ma_1 + M'a_2,$$

and so in the first n orders we obtain the fundamental functions themselves and their linear combinations, and no others.

The next following order is characterized by

$$(2, 0, 0, \ldots, 0),$$

and contains the linear combinations of the fundamental functions with the sum of the squares.

In all these cases the representability of the symmetric functions has already been proved; and we shall now carry out the general proof by deriving the representation of the function S, of order $(\nu_1, \nu_2, \ldots, \nu_n)$, by the fundamental functions under the supposition that it is already known for functions of lower order, thus by the procedure of complete induction.

We note for this purpose that, by multiplying two symmetric functions S and S', whose highest terms are

$$A = M\alpha_1^{\nu_1}\alpha_2^{\nu_2}\cdots\alpha_n^{\nu_n}, \qquad A' = M'\alpha_1^{\nu'_1}\alpha_2^{\nu'_2}\cdots\alpha_n^{\nu'_n},$$

there arises a symmetric function whose highest term is the product

$$AA' = MM'\alpha_1^{\nu_1+\nu'_1}\alpha_2^{\nu_2+\nu'_2}\cdots\alpha_n^{\nu_n+\nu'_n}.$$

For suppose that in SS' there were a higher term than AA', arising from the multiplication of the two terms

$$B = N\alpha_1^{\mu_1}\alpha_2^{\mu_2}\cdots\alpha_n^{\mu_n}, \qquad B' = N'\alpha_1^{\mu'_1}\alpha_2^{\mu'_2}\cdots\alpha_n^{\mu'_n},$$

so that

$$BB' = NN'\alpha_1^{\mu_1+\mu'_1}\alpha_2^{\mu_2+\mu'_2}\cdots\alpha_n^{\mu_n+\mu'_n},$$

and at the same time not $B = A$ and $B' = A'$. Then the first of the differences

$$\mu_1 - \nu_1 + \mu_1' - \nu_1', \quad \mu_2 - \nu_2 + \mu_2' - \nu_2', \ldots, \quad \mu_n - \nu_n + \mu_n' - \nu_n'$$

which is not zero would be positive; this is impossible, since the first non-vanishing one among the differences

$$\mu_1 - \nu_1, \quad \mu_2 - \nu_2, \ldots, \mu_n - \nu_n$$

and among the differences

$$\mu_1' - \nu_1', \quad \mu_2' - \nu_2', \ldots, \mu_n' - \nu_n'$$

is, by supposition, negative. By repeating this conclusion one obtains the more general theorem that the highest term of a product of several factors is obtained by multiplying the highest terms of the individual factors.

The highest terms of the symmetric fundamental functions

$$a_1, a_2, a_3, \ldots, a_n$$

are now, apart from the sign,

$$\alpha_1, \quad \alpha_1\alpha_2, \quad \alpha_1\alpha_2\alpha_3, \ldots, \alpha_1\alpha_2\alpha_3 \cdots \alpha_n;$$

and if therefore we form the product

$$P = \pm M a_1^{\nu_1 - \nu_2} a_2^{\nu_2 - \nu_3} \cdots a_{n-1}^{\nu_{n-1} - \nu_n} a_n^{\nu_n},$$

we obtain a symmetric function whose highest term is likewise A, as in S.

The difference

$$S - P = S'$$

is therefore again a symmetric function whose highest term is lower than that of S, and which, by the supposition, we can represent rationally by the fundamental functions. Since P is represented in the same way, the goal is reached. Here again it is clear that through the representation by means of the fundamental functions no fractions are introduced if originally none are contained in S.

This method of computing symmetric functions at the same time gives information about the degree of the whole rational function of $a_1, a_2, \ldots, a_n$ by which a definite symmetric function of the $\alpha_1, \alpha_2, \ldots, \alpha_n$ is expressed. Namely, if $(\nu_1, \nu_2, \ldots, \nu_n)$ is the order of the symmetric function, measured by the exponent sequence of the first term, then ν_1 is the degree of P with respect to all $a_1, a_2, \ldots, a_n$, and in the developed expression for S no term of higher degree can occur. Nor can the term P be destroyed by any of the following terms, since P is completely determined by the order of S, and the order of S', and of all following symmetric functions, is lower than the order of S.

If therefore, in place of $a_1, a_2, \ldots, a_n$, we put

$$\frac{a_1}{a_0}, \quad \frac{a_2}{a_0}, \ldots, \frac{a_n}{a_0},$$

then $a_0^{\nu_1} S$ is a whole homogeneous function of degree ν_1 of $a_0, a_1, \ldots, a_n$, and one cannot transform S into a whole function of the a's by multiplication with a lower power of a_0.

§ 46. Discriminants.

The theorems obtained above teach how to express symmetric functions of the roots of an equation rationally by the coefficients of the equation. The results were first derived under the supposition that the roots, or, what by the third section is equivalent to this, the coefficients were independent variables; but they cannot become false when definite numerical values are substituted for them, since at least for whole symmetric functions no denominators occur which could vanish.

We first make some applications to special symmetric functions which are important for what follows. Already in § 19 it was remarked that the product of differences

$$P = (\alpha_1 - \alpha_2)(\alpha_1 - \alpha_3)\cdots(\alpha_1 - \alpha_n)(\alpha_2 - \alpha_3)\cdots(\alpha_2 - \alpha_n)\cdots(\alpha_{n-1} - \alpha_n) \tag{1}$$

changes only its sign when two elements are interchanged. The square of this product will therefore remain unchanged under all interchanges of two of the variables α, and hence also under all permutations; that is, it is a symmetric function of the variables. The order of this symmetric function is, as one sees from (1),

$$(2n-2, 2n-4, \ldots, 2, 0).$$

If therefore we take the function $f(x)$, whose roots are $\alpha_1, \alpha_2, \ldots, \alpha_n$, in the form

$$f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_{n-1}x + a_n,$$

so that the highest power of x has not necessarily the coefficient 1, then P^2 is a whole rational function of

$$\frac{a_1}{a_0}, \quad \frac{a_2}{a_0}, \ldots, \frac{a_n}{a_0}, \tag{2}$$

which, by multiplication by a_0^{2n-2}, passes into a homogeneous function of $a_0, a_1, \ldots, a_n$. We call this homogeneous function, which is therefore defined by

$$a_0^{2n-2}P^2 = D, \tag{3}$$

the discriminant of the function $f(x)$. If we put $a_0 = 1$, we obtain the non-homogeneous form of the discriminant.[13]

The vanishing of the discriminant is the necessary and sufficient condition that two of the values $\alpha_1, \alpha_2, \ldots, \alpha_n$ become equal.

We can set up a second expression for the discriminant. By § 29, (4),

$$\begin{aligned}
f'(\alpha_1) &= a_0(\alpha_1 - \alpha_2)(\alpha_1 - \alpha_3)\cdots(\alpha_1 - \alpha_n),\\
f'(\alpha_2) &= a_0(\alpha_2 - \alpha_1)(\alpha_2 - \alpha_3)\cdots(\alpha_2 - \alpha_n),\\
&\vdots\\
f'(\alpha_n) &= a_0(\alpha_n - \alpha_1)(\alpha_n - \alpha_2)\cdots(\alpha_n - \alpha_{n-1}),
\end{aligned}$$

and consequently, since here every factor $(\alpha_h - \alpha_k)$ occurs twice with opposite signs,

$$D = (-1)^{\frac{n(n-1)}{2}} a_0^{n-2} f'(\alpha_1)f'(\alpha_2)\cdots f'(\alpha_n). \tag{5}$$

For actually representing the discriminant we first have a general means. By § 22,

$$(-1)^{\frac{n(n-1)}{2}} P = \begin{vmatrix} 1 & \alpha_1 & \alpha_1^2 & \cdots & \alpha_1^{n-1} \\ 1 & \alpha_2 & \alpha_2^2 & \cdots & \alpha_2^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & \alpha_n & \alpha_n^2 & \cdots & \alpha_n^{n-1} \end{vmatrix}, \tag{6}$$

and if, by § 27, we represent the square of this determinant by multiplication according to vertical rows as a new determinant, and put

$$s_m = \alpha_1^m + \alpha_2^m + \cdots + \alpha_n^m,$$

[13]The discriminant is, apart from the factor $(-1)^{\frac{n(n-1)}{2}} a_0^{2n-2}$, the same as what Gauss calls the determinant of the function (loc. cit., art. 6).

then

$$D = a_0^{2n-2} \begin{vmatrix} s_0 & s_1 & s_2 & \cdots & s_{n-1} \\ s_1 & s_2 & s_3 & \cdots & s_n \\ \vdots & \vdots & \vdots & & \vdots \\ s_{n-1} & s_n & s_{n+1} & \cdots & s_{2n-2} \end{vmatrix}. \tag{7}$$

The quantities s_m, however, are the power sums which we have already expressed by the coefficients in § 42; only in place of $a_1, a_2, \ldots, a_n$ the quotients (2) are to be put.

Thus we find, for example, for $n = 2$

$$D = a_0^2(s_0 s_2 - s_1^2) = a_1^2 - 4a_0 a_2, \tag{8}$$

and for $n = 3$

$$D = a_0^4(s_0 s_2 s_4 - s_0 s_3^2 + 2s_1 s_2 s_3 - s_1^2 s_4 - s_2^3). \tag{9}$$

§ 46. Discriminants

Here, according to § 42, one is to set

$$s_0 = 3, \qquad a_0 s_1 = -a_1,$$
$$a_0^2 s_2 = a_1^2 - 2a_0 a_2,$$
$$a_0^3 s_3 = -a_1^3 + 3a_0 a_1 a_2 - 3a_0^2 a_3,$$
$$a_0^4 s_4 = a_1^4 - 4a_0 a_1^2 a_2 + 4a_0^2 a_1 a_3 + 2a_0^2 a_2^2,$$

from which one obtains

$$D = a_1^2 a_2^2 + 18 a_0 a_1 a_2 a_3 - 4a_0 a_2^3 - 4a_1^3 a_3 - 27 a_0^2 a_3^2. \tag{10}$$

§ 47. Discriminants of forms of the third and fourth orders

For functions of the third and fourth orders, the solutions of the equations of these two degrees (§ 35, 37) give us a simple means for computing the discriminants.

First, for the equation of the third order, the three roots of

$$x^3 + ax + b = 0 \tag{1}$$

were represented in the form

$$\begin{aligned} \alpha &= u + v, \\ \beta &= \varepsilon u + \varepsilon^2 v, \\ \gamma &= \varepsilon^2 u + \varepsilon v, \end{aligned} \tag{2}$$

where

$$\varepsilon = \frac{-1 + \sqrt{-3}}{2}, \qquad \varepsilon^2 = \frac{-1 - \sqrt{-3}}{2} \tag{3}$$

are the complex third roots of unity, and u, v are determined by § 35, (8). From (3) one obtains

$$\varepsilon - \varepsilon^2 = \sqrt{-3}, \qquad 1 - \varepsilon = \varepsilon^2 \sqrt{-3}, \qquad 1 - \varepsilon^2 = -\varepsilon\sqrt{-3}, \tag{4}$$

and from (2)

$$\begin{aligned} \alpha - \beta &= \sqrt{-3}\,(\varepsilon^2 u - \varepsilon v), \\ \alpha - \gamma &= -\sqrt{-3}\,(\varepsilon u - \varepsilon^2 v), \\ \beta - \gamma &= \sqrt{-3}\,(u - v). \end{aligned}$$

Hence, by multiplication,

$$(\alpha-\beta)(\alpha-\gamma)(\beta-\gamma) = 3\sqrt{-3}\,(u^3 - v^3). \tag{5}$$

Squaring, one obtains for the discriminant of the cubic equation (1), according to § 35, (6),

$$D = -(27b^2 + 4a^3). \tag{6}$$

From this one obtains the discriminant of the general function

$$a_0x^3 + a_1x^2 + a_2x + a_3 \tag{7}$$

by putting for a and b the expressions of § 34, (4), replacing a_1, a_2, a_3 there by

$$\frac{a_1}{a_0}, \quad \frac{a_2}{a_0}, \quad \frac{a_3}{a_0},$$

and multiplying by a_0^4 [§ 46, (3)]. This gives

$$a = \frac{3a_0a_2 - a_1^2}{3a_0^2}, \qquad b = \frac{2a_1^3 - 9a_0a_1a_2 + 27a_0^2a_3}{27a_0^3},$$

and consequently

$$D = -\frac{1}{27a_0^2}\left\{(2a_1^3 - 9a_0a_1a_2 + 27a_0^2a_3)^2 + 4(3a_0a_2 - a_1^2)^3\right\}. \tag{8}$$

If one carries out the square and the cube, one returns to formula (10) of the preceding paragraph; it is noteworthy, however, that formula (8) apparently contains the denominator $27a_0^2$, which cancels when the expansion is carried out.

In exactly the same way we may proceed for biquadratic equations. If $\alpha, \beta, \gamma, \delta$ are the roots of the biquadratic equation

$$x^4 + ax^2 + bx + c = 0, \tag{9}$$

then, by § 37, (5), we obtain

$$\begin{aligned} \alpha-\beta &= v+w, & \gamma-\delta &= v-w,\\ \alpha-\gamma &= w+u, & \delta-\beta &= w-u,\\ \alpha-\delta &= u+v, & \beta-\gamma &= -u+v. \end{aligned} \tag{10}$$

Accordingly the discriminant of equation (9) is

$$D = (v^2-w^2)^2(w^2-u^2)^2(u^2-v^2)^2,$$

that is, D is equal to the discriminant of the cubic resolvent § 37, (4),

$$z^3 + 2az^2 + (a^2 - 4c)z - b^2 = 0, \tag{11}$$

whose roots are u^2, v^2, w^2.

If, however, this discriminant is formed by formula (10), § 46, with

$$a_0 = 1, \quad a_1 = 2a, \quad a_2 = a^2 - 4c, \quad a_3 = -b^2,$$

one obtains

$$D = -4a^3b^2 + 144acb^2 + 16a^4c - 128a^2c^2 + 256c^3 - 27b^4. \tag{12}$$

If, on the other hand, formula (8) is used to form the discriminant of the cubic equation (11), one obtains

$$27D = 4(a^2 + 12c)^3 - (2a^3 - 72ac + 27b^2)^2. \tag{13}$$

From this one derives the discriminant of the general fourth-degree function

$$f(x) = a_0x^4 + a_1x^3 + a_2x^2 + a_3x + a_4 \tag{14}$$

by expressing a, b, c, according to § 34, (4), by

$$-\frac{3a_1^2}{8} + a_2, \qquad \frac{a_1^3}{8} - \frac{a_1a_2}{2} + a_3, \qquad -\frac{3a_1^4}{256} + \frac{a_1^2a_2}{16} - \frac{a_1a_3}{4} + a_4,$$

then replacing a_1, a_2, a_3, a_4 by

$$\frac{a_1}{a_0}, \quad \frac{a_2}{a_0}, \quad \frac{a_3}{a_0}, \quad \frac{a_4}{a_0},$$

and multiplying by a_0^6. After carrying out this simple calculation, set, for abbreviation,

$$A = a_2^2 - 3a_1a_3 + 12a_0a_4, \tag{15}$$

$$B = 27a_1^2a_4 + 27a_0a_3^2 + 2a_2^3 - 72a_0a_2a_4 - 9a_1a_2a_3. \tag{16}$$

Then (13) gives

$$27D = 4A^3 - B^2. \tag{17}$$

The quantities A and B, which will occupy us further in the next section, are called the first and second invariants of the biquadratic function f. Thus, by (17), one may express the discriminant rationally through these quantities, though not in such a way that the coefficients become integral. If the calculation prescribed in (17) is carried out in order to represent D itself through the coefficients a_0, a_1, a_2, a_3, a_4, then the factor 27 must still cancel. We shall not write down here the long expression, whose calculation offers no difficulty.

§ 48. Resultants

Another application of the theory of symmetric functions, which also contains the formation of discriminants as a special case, is the formation of resultants.

Let

$$f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_n, \tag{1}$$

$$\varphi(x) = b_0x^m + b_1x^{m-1} + b_2x^{m-2} + \cdots + b_m \tag{2}$$

be two integral rational functions of degrees n and m, and let $\alpha_1, \alpha_2, \ldots, \alpha_n$ be the roots of the first, so that one may write

$$f(x) = a_0(x - \alpha_1)(x - \alpha_2)\cdots(x - \alpha_n). \tag{3}$$

Now the product

$$\varphi(\alpha_1)\varphi(\alpha_2)\cdots\varphi(\alpha_n)$$

is a symmetric function of the roots of $f(x)$ and hence can be expressed integrally and rationally through

$$\frac{a_1}{a_0}, \quad \frac{a_2}{a_0}, \ldots, \frac{a_n}{a_0}.$$

The order of this symmetric function is $(m, m \ldots m)$ (§ 45), and therefore

$$R = a_0^m\varphi(\alpha_1)\varphi(\alpha_2)\cdots\varphi(\alpha_n) \tag{4}$$

is an integral rational and homogeneous function of order m in $a_0, a_1, \ldots, a_n$. It is also, as its expression shows, an integral homogeneous function of order n in $b_0, b_1, \ldots, b_m$. It is called the resultant of the two functions $f(x)$ and $\varphi(x)$. This designation is justified by a second representation.

Let the roots of $\varphi(x)$ be denoted by $\beta_1, \beta_2, \ldots, \beta_m$, so that

$$\varphi(x) = b_0(x - \beta_1)(x - \beta_2) \cdots (x - \beta_m). \tag{5}$$

Then, by (4),

$$R = a_0^m b_0^n \prod_{i,k} (\alpha_i - \beta_k), \tag{6}$$

where the product sign $\prod$ refers to all indices $i = 1, 2, \ldots, n$, $k = 1, 2, \ldots, m$.

This expression now shows that R, apart from the factor $(-1)^{mn}$, depends on $\varphi(x)$ in the same way as on $f(x)$; hence one can also set

$$(-1)^{mn} R = b_0^n f(\beta_1) f(\beta_2) \cdots f(\beta_m). \tag{7}$$

Accordingly, discriminants are a special case of resultants, obtained by taking $\varphi(x)$ to be the derivative $f'(x)$.

The vanishing of the resultant of $f(x)$ and $\varphi(x)$ is the necessary and sufficient condition that the two equations

$$f(x) = 0 \quad \text{and} \quad \varphi(x) = 0$$

have a common root.

The equation obtained by setting the resultant equal to zero is also called the final equation from $f(x) = 0$ and $\varphi(x) = 0$, and the formation of the final equation is the elimination of x.

For the computation of resultants one may use the algorithm of the greatest common divisor (§ 6). Namely, if $m \leq n$, then one can determine an integral function Q and likewise a function $\psi(x)$, of degree lower than m, such that

$$f(x) = Q\varphi(x) + \psi(x).$$

Thus, putting x equal to one of the roots β_i,

$$f(\beta_i) = \psi(\beta_i),$$

and hence

$$b_0^n \prod_i f(\beta_i) = b_0^n \prod_i \psi(\beta_i). \tag{8}$$

If m' is the degree of $\psi(x)$, then

$$b_0^{m'} \prod_i \psi(\beta_i) = R' \tag{9}$$

is the resultant of $\varphi(x)$ and $\psi(x)$, and therefore

$$R = b_0^{n-m'} R'. \tag{10}$$

The formation of the resultant R is thereby reduced to the formation of R'. One can now divide $\varphi(x)$ again by $\psi(x)$, and so reduce the formation of R' to the formation of a resultant of functions of ever lower degree, until one finally reaches a constant, that is, a function of the coefficients a, b, which must be identical with the resultant sought.

The computation of resultants is usually quite lengthy. We shall record only the simplest example, the resultant of two quadratic functions:

$$f(x) = a_0 x^2 + a_1 x + a_2, \qquad \varphi(x) = b_0 x^2 + b_1 x + b_2.$$

Here one may directly compute the product

$$R = a_0^2 (b_0 \alpha_1^2 + b_1 \alpha_1 + b_2)(b_0 \alpha_2^2 + b_1 \alpha_2 + b_2),$$

and obtains, after setting

$$-a_0(\alpha_1 + \alpha_2) = a_1, \qquad a_0\alpha_1\alpha_2 = a_2,$$

$$R = a_0^2 b_2^2 + a_2^2 b_0^2 - b_0 b_1 a_1 a_2 - a_0 a_1 b_1 b_2 + b_0 b_2 a_1^2 + a_0 a_2 b_1^2 - 2a_0 b_0 a_2 b_2, \tag{11}$$

or, equivalently,

$$R = (a_0 b_2 - a_2 b_0)^2 - (a_0 b_1 - a_1 b_0)(a_1 b_2 - a_2 b_1). \tag{12}$$

The individual terms of the resultant of the two functions $f(x)$ and $\varphi(x)$ of degrees n and m have, apart from a numerical integral factor, the form

$$a_0^{\nu_0} a_1^{\nu_1} \cdots a_n^{\nu_n} b_0^{\mu_0} b_1^{\mu_1} \cdots b_m^{\mu_m}, \tag{13}$$

where, as follows from the degrees determined above,

$$\nu_0 + \nu_1 + \cdots + \nu_n = m, \qquad \mu_0 + \mu_1 + \cdots + \mu_m = n. \tag{14}$$

But we can state still another relation among these exponents. Since, by (6), R is a homogeneous function of degree nm in the $n+m$ variables α, β, and since further a_0 is of degree zero, a_1 of degree one, a_2 of degree two, and so on in the α, and in general a_k and b_k are of degree k in these variables, it follows that

$$\nu_1 + 2\nu_2 + 3\nu_3 + \cdots + n\nu_n + \mu_1 + 2\mu_2 + 3\mu_3 + \cdots + m\mu_m = nm. \tag{15}$$

This is a relation from which an important conclusion follows, one that we shall discuss somewhat more fully in the following paragraph.

§ 49. Elimination. Bezout's theorem

Suppose that in the two functions

$$\begin{aligned} f(x) &= a_0 x^n + a_1 x^{n-1} + \cdots + a_n, \\ \varphi(x) &= b_0 x^m + b_1 x^{m-1} + \cdots + b_m \end{aligned} \tag{1}$$

the coefficients a, b are themselves again integral rational functions of a quantity y. Then the resultant will also become an integral rational function of y, which we shall denote by $F(y)$. The formation of this equation is the elimination of x from the two equations (1).

The equation

$$F(y) = 0 \tag{2}$$

has as roots all those values of y for which the two equations

$$f(x) = 0, \qquad \varphi(x) = 0 \tag{3}$$

have a common root. If condition (2) is satisfied, one finds the values of x satisfying the two equations (3) by seeking the greatest common divisor of $f(x)$ and $\varphi(x)$. This common divisor is of the first degree and hence determines x rationally when there is only one such common root; otherwise it is of correspondingly higher degree, and the determination of x still requires the solution of an equation of higher degree.

If only one common root x belongs to each root of the resultant, then the number of value-pairs x, y satisfying equations (3) is equal to the degree of the resultant with respect to y. Thus, if the coefficients a are of degree ν, and the coefficients b of degree μ, with respect to y, then by the degree determination of the preceding paragraph the degree of the resultant is

$$n\nu + m\mu; \tag{5}$$

this therefore is the number of common root-pairs. This number can, however, still be modified by the possibility that the coefficients are so constituted that the highest power of y disappears from the final equation. One would then be able to save the agreement of the number with (5) only by arbitrarily adding infinitely distant roots. Multiple roots also raise difficulties. One partly avoids these inconveniences by taking as the basis the homogeneous form of the equations, a form in which the elimination problem arises especially naturally in geometry.

Namely, if the function $f(x)$ has arisen from a homogeneous function of order n in the three variables x, y, z (a ternary form) by arranging it according to powers of x, then the coefficients $a_0, a_1, a_2, \ldots, a_n$ are homogeneous forms in the two variables y, z, of the degree indicated by the index; thus a_0 is a constant, a_1 a linear form, a_2 a quadratic form, and so forth. The same applies to the coefficients $b_0, b_1, \ldots, b_m$ of the function $\varphi(x)$.

If x, y, z denote triangular coordinates in the plane, then $f(x) = 0$, $\varphi(x) = 0$ are the equations of curves of orders n and m, and the common roots of these two equations, that is the ratios $x : y : z$, denote the points of intersection of the two curves. The final equation obtained by eliminating x is a homogeneous equation whose degree follows from (15), § 48 as mn.

A diminution of the degree cannot occur here; the final equation is always a homogeneous equation of degree nm in the two unknowns y, z. The only exception to which one must attend is the case in which the resultant vanishes identically (for all y, z), so that infinitely many common roots are present. Geometrically, this case means that the two curves have a common curve-part and not merely individual points in common.

If the equations $f = 0$, $\varphi = 0$ have only a finite number of common roots, then in the expressions

$$y_1 = y + \beta x, \qquad z_1 = z + \gamma x$$

we can choose the constants β, γ so that, if to one value of the ratio $y : z$ there correspond several different values of x satisfying equations (1), then the corresponding values of $y_1 : z_1$ come out different from one another.

We may then regard f and φ also as homogeneous functions of degrees m and n in the variables x, y_1, z_1, and the resultant then becomes a homogeneous function of degree mn, $R(y_1, z_1)$, in y_1 and z_1. If in it we put

$$y_1 = b' + b\lambda, \qquad z_1 = c' + c\lambda,$$

then we obtain an equation of degree mn in λ, and, provided that $R(y_1, z_1)$ does not vanish identically, we may choose the constants b, b', c, c' so that the coefficient of the highest power of λ, namely $R(b, c)$, is non-zero. Then to each root of the resultant there corresponds only one system of values of the ratio $x : y : z$, and the number of common roots of f and φ is at most equal to mn. To be able to state the theorem that the number of common roots is always equal to mn, one must still make a convention according to which certain value-pairs are to be counted more than once.

The theorem which, in geometrical garb, says that two curves of orders m and n meet in mn points, is called Bezout's theorem.

§ 50. Elimination from several equations

The principles of elimination that we have applied in the preceding discussion to two equations may be extended without difficulty to several equations with a corresponding number of unknowns.

First take the equations

$$f(x) = 0, \qquad \varphi(x) = 0, \qquad \psi(x) = 0, \tag{1}$$

of degrees m, n, p, whose coefficients a_i, b_i, c_i are for the present to be regarded as independent variables.

We apply the algorithm of the greatest common divisor to two of them, say to $f(x)$ and $\varphi(x)$, and obtain a last remainder which is a rational function of the a and b and, after clearing denominators, agrees with the resultant R of $f(x)$ and $\varphi(x)$, and a next-to-last remainder which is a linear function of x. The condition for the existence of a common root of $f = 0$ and $\varphi = 0$ is the vanishing of the resultant; under this condition the next-to-last remainder gives a rational expression for the common root in terms of the coefficients.

If one substitutes this expression in $\psi(x)$, then, after all denominators have been removed, one obtains an integral rational function of the coefficients a, b, c, which we shall denote by S; its vanishing, together with $R = 0$, contains the necessary and sufficient condition that the three equations (1) have a common root.

Now replace the coefficients a, b, c of the functions (1) by integral rational functions of a second unknown y, whose coefficients will be denoted by α, β, γ and for the present may be regarded as independent variables. Then R and S become integral rational functions of α, β, γ and y, and if, as equations in y, we consider

$$R = 0, \qquad S = 0, \tag{2}$$

then, according to the rules of the preceding paragraph, we may form their resultant T. If it does not vanish identically, this will be an integral rational function of the coefficients α, β, γ, and the equation

$$T = 0 \tag{3}$$

is the necessary and sufficient condition that the system of equations (1) can be satisfied by one or more value-pairs x, y. T is called the resultant of the system (1). Thus one may form from the three equations (1), by eliminating the unknowns x and y, a final equation for z.

If the α, β, γ are now replaced again by integral rational functions of a third unknown z, then T passes into an equation in z. The functions f, φ, ψ become functions of the three unknowns x, y, z. The degree of equation (3) gives the number of systems of values x, y, z for which the three equations (1) are simultaneously satisfied.

If, however, the coefficients are so constituted that T vanishes identically for all values of z, then there are infinitely many systems of values x, y, z simultaneously satisfying equations (1). If one regards x, y, z as Cartesian coordinates, then equations (1) each represent a surface. The degree of the final equation $T = 0$ with respect to z gives the number of intersection points of these three surfaces. If, however, T is identically zero, the three surfaces have either a common surface part or a common curve.

The degree of the final equation can be determined in the following way, which requires no lengthy considerations about the degrees of the functions R, S, T. To make ourselves independent of exceptional cases, we introduce homogeneous variables by adjoining a fourth variable u, and thus consider three homogeneous quaternary equations

$$F(x, y, z, u) = 0, \qquad \Phi(x, y, z, u) = 0, \qquad \Psi(x, y, z, u) = 0 \tag{4}$$

of orders m, n, p.

Assume the special case in which each of these three functions splits into linear factors. It is then clear that, if the number of common solutions is not infinite, their number must be equal to the product mnp; for we may set one linear factor of F, one of Φ, and one of Ψ simultaneously equal to zero, and thereby obtain mnp systems of three linear equations, which, if independent of one another, yield the same number of systems of values for the ratios $x : y : z : u$.

At first sight, it is questionable to conclude that in the general case the degree of the final equation of (4) is just as large. But this concern can be entirely removed by the following simple consideration, due to a remark of F. Klein.

Imagine the final equation from equations (4) first to have been formed under the assumption that the coefficients in equations (4) are independent variables. Then the resultant obtained after eliminating x, y is certainly not identically zero, since even for special systems of coefficients, for example when F, Φ, Ψ split into linear factors, it does not vanish identically. We therefore obtain a homogeneous final equation in the two unknowns z, u of some degree, which is to be determined.

If the coefficients are specialized in any way, it is possible that the final equation becomes identically satisfied; but if this case does not occur, then its order cannot change, since all its terms have the same order. Since under the specialization assumed above, expressed by the splitting of the functions F, Φ, Ψ, the identical vanishing does not occur but instead a final equation of order mnp appears, this must also be the degree of the final equation in the general case.

Thus Bezout's theorem is proved for the case of three homogeneous quaternary equations.

That we have restricted ourselves here to three equations was done solely in the interest of simplicity. The extension required for the case of n equations in $n+1$ homogeneous unknowns follows of itself and may be left to the reader.

For our general problem we shall only add that these considerations show that the problem of algebra, insofar as it concerns the solution of algebraic equations of any kind, is thereby reduced to the treatment of a chain of equations each with one unknown.

§ 51. Decomposable and indecomposable functions

Among integral rational functions of several variables one must distinguish decomposable and indecomposable functions.

An integral rational function of any variables is called decomposable if it can be represented as a product of at least two integral rational functions of the same variables; if it cannot be represented as such a product, it is called indecomposable.

If a function is decomposable, then the degree of each factor is lower than the degree of the function itself. A linear function is therefore always indecomposable, and every integral rational function can be decomposed into a finite number of indecomposable functions.

We say that an integral function W is divisible by another w if a third integral function w' exists such that

$$W = ww'.$$

It follows that if U is divisible by w and V is any integral function, then the product UV is also divisible by w; and if $U, U', U'', \ldots$ are divisible by w, while $V, V', V'', \ldots$ are arbitrary integral rational functions, then $UV + U'V' + U''V'' + \cdots$ is also divisible by w.

If W is divisible by w, one also says that w divides W.

Two integral rational functions U, V that are not divisible by one and the same integral rational function are called relatively prime, or prime to one another.

We prove the following theorem:

I. If U, V, v are integral rational functions of any variables, if U and v are relatively prime, and if UV is divisible by v, then V is divisible by v.

This theorem corresponds exactly to a familiar fundamental theorem from the theory of integers: if a product of two integers is divisible by a third integer that is prime to one factor, then the other factor must be divisible by that integer.

We prove it by complete induction. If U, V, v depend on only one variable x, the theorem is correct; for by § 6 (Theorem I), when U and v are relatively prime, one can determine two other integral functions P and p of x such that

$$PU + pv = 1.$$

Multiplication by V gives

$$PUV + pVv = V,$$

and from this it is seen that if UV is divisible by v, then V must also be divisible by v.

We now assume that the theorem to be proved has been established for functions of n or fewer variables, and derive from this its validity for functions of $n+1$ variables.

For this it is necessary to draw from the theorem I, assumed to be true, some consequences whose final result will give the validity of the theorem for the next higher number of variables.

If v is an indecomposable function, then another function U of the same variables is either relatively prime to v or divisible by v. It follows by I that a product UV of two functions can be divisible by v only when one of the two factors is divisible by v. The same holds for a product of several functions, and thus from I follows the theorem:

II. A product of several integral rational functions is divisible by an indecomposable function v only when at least one of the factors of the product is divisible by v.

If an integral rational function U is decomposed into indecomposable factors in two ways,

$$U = vv'v'' \cdots = ww'w'' \cdots ,$$

then by II at least one of the factors $v, v', v'', \ldots$ must be divisible by w, say v. But since v too is indecomposable, v can differ from w only by a constant factor.

Thus, if c is this constant factor,

$$cv'v'' \cdots = w'w'' \cdots ,$$

from which it follows that one of the functions $v', v'', \ldots$, say v', is divisible by w' and hence differs from w' only by a constant factor.

We therefore obtain as a second consequence of Theorem I:

III. An integral rational function can, apart from constant factors, be decomposed into indecomposable factors in only one way.

From this arises the concept of the greatest common divisor of two or more integral rational functions $U, V, \ldots$. By this one understands the product of all indecomposable factors that occur in the decompositions of each of the functions $U, V, \ldots$, or the function of highest possible degree that divides all the functions $U, V, \ldots$. By III this function is completely determined, up to a constant factor, for every system of functions $U, V, \ldots$.

Several functions $U, V, W, \ldots$ are called relatively prime if no two of them have a common divisor.

All these definitions and theorems are exactly analogous to well-known theorems of elementary number theory, where they arise as consequences of the algorithm of the greatest common divisor; only here the indecomposable functions take the role of prime numbers.

We record one more such theorem:

IV. If u, v are relatively prime and if Uu, Uv are divisible by w, then U is divisible by w.

For if one decomposes the integral rational functions U, u, v, w into their indecomposable factors, then some factor of w, since it cannot occur in both u and v, must occur in U. Removing it from U and w, one argues in the same way for the next factor of w, and so on.

Now consider integral rational functions of one variable t,

$$f(t) = u_0 t^m + u_1 t^{m-1} + \cdots + u_{m-1} t + u_m,$$

whose coefficients are integral rational functions of n variables x, independent of t.

If the coefficients $u_0, u_1, \ldots, u_m$ have no common divisor, then $f(t)$ is called primitive; otherwise the greatest common divisor of the coefficients $u_0, u_1, \ldots, u_m$ is called the divisor of the function $f(t)$ (compare § 2).

Still assuming the validity of I, we conclude:

V. The product of two primitive functions $f(t)$ and $\varphi(t)$ is again a primitive function; and the divisor of a product of two imprimitive functions is equal to the product of the divisors of the two factors.

The proof is exactly as for the corresponding theorem in § 2.

Let

$$\begin{aligned} f(t) &= u_0 t^m + u_1 t^{m-1} + \cdots + u_m, \\ \varphi(t) &= v_0 t^\mu + v_1 t^{\mu-1} + \cdots + v_\mu \end{aligned} \tag{1}$$

be two primitive functions, and let

$$F(t) = U_0 t^{m+\mu} + U_1 t^{m+\mu-1} + \cdots + U_{m+\mu} \tag{2}$$

be their product. Then, if r and s denote any two indices from the rows $0, 1, \ldots, m$ and $0, 1, 2, \ldots, \mu$ (§ 2),

$$U_{r+s} = u_r v_s + u_{r-1} v_{s+1} + \cdots + u_{r+1} v_{s-1} + \cdots . \tag{3}$$

Now let w be any indecomposable function that does not divide all the u_r and also does not divide all the v_s. We choose r and s in (3) so that u_r is the first u not divisible by w, so that $u_{r-1}, u_{r-2}, \ldots$ are divisible by w, and similarly v_s is the first v not divisible by w. Then U_{r+s} cannot be divisible by w, since all terms except the first are divisible by w; that is, $F(t)$ is primitive.

It follows immediately, if p and q are arbitrary integral rational functions of the x, that pq is the divisor of the product of the two imprimitive functions $pf(t)$ and $q\varphi(t)$, which proves the second part of theorem V. It follows further:

VI. If an integral rational function $F(t)$ of the $n+1$ variables x and t is decomposable into two factors that are integral with respect to t and at least rational with respect to the x, then it is also decomposable into two functions that are integral and rational in x and t.

For by hypothesis there is an integral rational function w of the x alone, and two integral rational functions of x and t, $f_1(t), \varphi_1(t)$, such that

$$wF(t) = f_1(t)\varphi_1(t).$$

By (V), w must divide the product of the divisors of $f_1(t)$ and $\varphi_1(t)$, and hence we also obtain

$$F(t) = f(t)\varphi(t),$$

where $f(t), \varphi(t)$ are again integral rational functions of x and t, and with respect to t have the same degrees as $f_1(t)$ and $\varphi_1(t)$, as was to be proved.

From this we conclude further that two functions $F(t), f(t)$ which, considered as integral rational functions of the $n+1$ variables x and t, are relatively prime in the sense defined above, must also prove to be relatively prime when considered as functions of t alone and treated by the algorithm of the greatest common divisor (§ 6). For if they had a common divisor that was integral with respect to t but fractional with respect to the x, then one could determine an integral function T of x and t and an integral function P of the x alone such that

$$PF(t) = TF_1(t), \qquad Pf(t) = Tf_1(t),$$

where F_1, f_1 are integral functions without a common divisor. By IV, P would have to occur in the divisor of T, and $F(t)$ and $f(t)$ could not be relatively prime. Therefore by § 6 one can determine two integral rational functions Q, q of x and t, and an integral rational function X of the x alone, such that the identity

$$QF(t) + qf(t) = X \tag{4}$$

holds.

Now let $\Phi(t)$ be another integral rational function of x and t. From (4), by multiplication with $\Phi(t)$, one obtains

$$Q\Phi(t)F(t) + q\Phi(t)f(t) = X\Phi(t),$$

and if $\Phi(t)F(t)$ is divisible by $f(t)$, then $X\Phi(t)$ is consequently also divisible by $f(t)$.

Thus, if $\varphi(t)$ and $\psi(t)$ again denote two integral functions of x and t, we may put

$$\begin{aligned} X\Phi(t) &= \varphi(t)f(t), \\ F(t)\Phi(t) &= \psi(t)f(t). \end{aligned} \tag{5}$$

Multiplying the second of these equations by X and substituting from the first the expression $\varphi(t)f(t)$ for $X\Phi(t)$, one may cancel $f(t)$ and obtains

$$\varphi(t)F(t) = X\psi(t).$$

By IV, X must therefore occur both in the divisor of $\varphi(t)f(t)$ and in the divisor of $\varphi(t)F(t)$. Since the functions $f(t)$ and $F(t)$, and hence also their divisors, are relatively prime, X must occur in the divisor of $\varphi(t)$. Thus, putting

$$\varphi(t) = X\varphi_1(t),$$

one obtains from (5)

$$\Phi(t) = \varphi_1(t)f(t),$$

that is, $\Phi(t)$ is divisible by $f(t)$. Hence:

VII. If $F(t)$ and $f(t)$ are relatively prime and $\Phi(t)F(t)$ is divisible by $f(t)$, then $\Phi(t)$ is divisible by $f(t)$.

But this is nothing other than Theorem I for functions of $n+1$ variables, and so I is proved in general.

§ 52. Tschirnhausen transformation

We now make another important application of the theory of symmetric functions and the formation of resultants, namely to the transformation of algebraic equations due to Tschirnhausen.[14]

Let

$$f(x) = x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_{n-1}x + a_n \tag{1}$$

be any function of the nth degree, and let

$$y = \frac{\chi(x)}{\psi(x)} \tag{2}$$

be any rational function, integral or fractional, of x, subject only to the assumption that $f(x)$ and $\psi(x)$ are to have no common divisor. By the concluding theorem of § 6 one can then determine functions $F(x)$ and $\varphi(x)$, the latter of degree at most $n-1$, such that

$$\chi(x) = F(x)f(x) + \varphi(x)\psi(x), \tag{3}$$

and consequently, as soon as x is put equal to a root of the equation

$$f(x) = 0, \tag{4}$$

one has

$$\frac{\chi(x)}{\psi(x)} = \varphi(x).$$

[14] Acta eruditorum, Leipzig 1683.

Since we shall consider the values of y only for those values of x that are roots of (4), we lose no generality if from the outset we put

$$y = \varphi(x),$$

therefore equal to an integral function of degree $n-1$. Thus let

$$y = \varphi(x) = \alpha_0 + \alpha_1 x + \alpha_2 x^2 + \cdots + \alpha_{n-1} x^{n-1}, \tag{5}$$

where the coefficients α remain for the moment wholly undetermined.

Let the roots of equation (4) be denoted by

$$x_1, x_2, \ldots, x_n. \tag{6}$$

Then from (5) there arise the corresponding values of y:

$$y_1 = \varphi(x_1), \qquad y_2 = \varphi(x_2), \ldots, y_n = \varphi(x_n), \tag{7}$$

and the product

$$\Phi(y) = (y - y_1)(y - y_2) \cdots (y - y_n) \tag{8}$$

is a symmetric function of the quantities (6), and can therefore be expressed rationally through the coefficients of $f(x)$.

Represented in this way, let $\Phi(y)$ have the expression

$$\Phi(y) = y^n + b_1 y^{n-1} + b_2 y^{n-2} + \cdots + b_{n-1} y + b_n. \tag{9}$$

The coefficients b_ν are determined by the y_i, according to § 7, by means of the formulas

$$b_1 = -\sum y_1, \qquad b_2 = \sum y_1 y_2, \qquad b_3 = -\sum y_1 y_2 y_3, \ldots,$$

from which it follows that b_ν is an integral rational and homogeneous function of degree ν in the variables $\alpha_0, \alpha_1, \ldots, \alpha_{n-1}$.

$\Phi(y)$ is nothing other than the resultant of the two equations

$$f(x) = 0, \qquad \varphi(x) - y = 0, \tag{10}$$

and conversely, by § 49, if y becomes equal to one of the values $y_1, y_2, \ldots, y_n$, one can also express the common root of the two equations (10) rationally through the coefficients of (10), that is, rationally through a_i, α_i, y, provided that only one such common root is present.

This will happen when the n values $y_1, y_2, \ldots, y_n$ are distinct. Then one obtains an expression

$$x = \beta_0 + \beta_1 y + \beta_2 y^2 + \cdots + \beta_{n-1} y^{n-1}$$

that is valid when one of the values (6) is put for x and the corresponding value (7) for y. The quantities β_i are rationally composed from the a_i and α_i.

If, however, several of the values $y_1, y_2, \ldots, y_n$ coincide, then determining the corresponding values of x still requires solving an equation of the corresponding degree. But this degree is always less than n, since the function $\varphi(x)$ can take the same value y for at most $n-1$ values of x. Thus, when the equation

$$\Phi(y) = 0 \tag{11}$$

has been completely solved, the equation (1) has thereby also been completely solved if (11) has n distinct roots, or at least reduced to the solution of an equation of lower degree if (11) has equal roots.

Accordingly we regard (11) as a transformation of equation (1), and the formation of (11) from (1) is called the Tschirnhausen transformation of equation (1).

The purpose of such a transformation is to determine the coefficients α_i so that $\Phi(y)$ obtains a simpler form than $f(x)$.

The coefficients b_ν can be formed when the power sums of the y_i are known (§ 42).

Let the sums of the mth powers of the x_i be denoted by s_m, and those of the y_i by σ_m, so that

$$s_m = Sx_i^m, \qquad \sigma_m = Sy_i^m.$$

Then σ_m can be formed from the s_m in the following way. First expand y^m by the polynomial theorem (§ 12), putting

$$y^m = \sum \frac{\Pi(m)}{\Pi(\nu_0)\Pi(\nu_1)\cdots\Pi(\nu_{n-1})} \alpha_0^{\nu_0}\alpha_1^{\nu_1}\cdots\alpha_{n-1}^{\nu_{n-1}} x^{\nu_1+2\nu_2+\cdots+(n-1)\nu_{n-1}}, \tag{12}$$

where the sum extends over all non-negative integral values of $\nu_0, \nu_1, \dots, \nu_{n-1}$ satisfying the condition

$$\nu_0 + \nu_1 + \cdots + \nu_{n-1} = m. \tag{13}$$

If in (12) one then puts $x_1, x_2, \dots, x_n$ for x and forms the sum of the values thereby obtained for $y_1^m, y_2^m, \dots, y_n^m$, it follows that

$$\sigma_m = \sum \frac{\Pi(m)}{\Pi(\nu_0)\Pi(\nu_1)\cdots\Pi(\nu_{n-1})} s_{\nu_1+2\nu_2+\cdots+(n-1)\nu_{n-1}} \alpha_0^{\nu_0}\alpha_1^{\nu_1}\cdots\alpha_{n-1}^{\nu_{n-1}}. \tag{14}$$

Thus σ_m is represented as an integral homogeneous function of order m in $\alpha_0, \alpha_1, \dots, \alpha_{n-1}$. A somewhat simplified notation is obtained from the second representation of forms (§ 15). Let $\mu_1, \mu_2, \dots, \mu_m$ run independently through the sequence of numbers $0, 1, \dots, n-1$. If in one combination the index 0 occurs ν_0 times, the index 1 occurs ν_1 times, the index 2 occurs ν_2 times, and so on, the index $n-1$ occurs ν_{n-1} times, then

$$\mu_1 + \mu_2 + \cdots + \mu_m = \nu_1 + 2\nu_2 + \cdots + (n-1)\nu_{n-1},$$

and accordingly the expression for σ_m can be represented as

$$\sigma_m = \sum_\mu s_{\mu_1+\mu_2+\cdots+\mu_m} \alpha_{\mu_1}\alpha_{\mu_2}\cdots\alpha_{\mu_m}. \tag{15}$$

For example,

$$\begin{aligned} \sigma_1 &= \alpha_0 s_0 + \alpha_1 s_1 + \cdots + \alpha_{n-1} s_{n-1}, \\ \sigma_2 &= \sum_{i,k} s_{i+k}\alpha_i\alpha_k, \\ \sigma_3 &= \sum_{h,i,k} s_{h+i+k}\alpha_h\alpha_i\alpha_k, \end{aligned} \tag{16}$$

where h, i, k run through the sequence of numbers $0, 1, 2, \dots, n-1$.

§ 53. Application to the cubic and biquadratic equations

The aim that Tschirnhausen already had in view in this transformation was to determine the substitution coefficients $\alpha_0, \alpha_1, \dots, \alpha_{n-1}$ so that in the transformed equation $\Phi(y) = 0$ so many terms vanish that the equation becomes solvable. This is easily achieved for equations of the third and fourth degrees, although it does not seem simple, and not possible without extensive calculations, to derive by this method the formulas otherwise known.

First take the cubic equation

$$x^3 + ax^2 + bx + c = 0, \tag{1}$$

which we transform by the substitution

$$y = \alpha + \beta x + \gamma x^2. \tag{2}$$

In order to reduce the equation for y,

$$y^3 + Ay^2 + By + C = 0, \tag{3}$$

to a pure cubic equation, we have to determine the ratios $\alpha : \beta : \gamma$ from the two equations

$$A = 0, \qquad B = 0, \tag{4}$$

of which the first is linear and the second quadratic.

Equations (4) are equivalent to

$$\sigma_1 = 0, \qquad \sigma_2 = 0, \tag{5}$$

and hence, by (16) of the preceding paragraph, give

$$\alpha s_0 + \beta s_1 + \gamma s_2 = 0,$$

$$\alpha^2 s_0 + 2\alpha\beta s_1 + 2\alpha\gamma s_2 + \beta^2 s_2 + 2\beta\gamma s_3 + \gamma^2 s_4 = 0,$$

where $s_0 = 3$. If the latter equation is multiplied by s_0 and the square of the former is subtracted, one obtains

$$\beta^2(s_0 s_2 - s_1^2) + 2\beta\gamma(s_0 s_3 - s_1 s_2) + \gamma^2(s_0 s_4 - s_2^2) = 0. \tag{6}$$

The discriminant of this quadratic equation,

$$(s_0 s_3 - s_1 s_2)^2 - (s_0 s_2 - s_1^2)(s_0 s_4 - s_2^2),$$

when expanded gives

$$-s_0(s_0 s_2 s_4 - s_0 s_3^2 - s_1^2 s_4 - s_2^3 + 2 s_1 s_2 s_3),$$

and hence, if D denotes the discriminant of the cubic equation (1) [§ 46, (9)], is equal to $-3D$. The solution of (6) therefore gives

$$\frac{\beta}{\gamma} = \frac{-(s_0 s_3 - s_1 s_2) + \sqrt{-3D}}{s_0 s_2 - s_1^2}. \tag{7}$$

The equation (3) has thereby been reduced to a pure equation, whose solution requires only the extraction of a cube root. The choice of sign for the square root occurring in (7) is left to us.

To use the Tschirnhausen transformation for the biquadratic equation, one may proceed, for instance, by determining $\alpha_0, \alpha_1, \alpha_2, \alpha_3$ in the transformed equation

$$y^4 + b_1 y^3 + b_2 y^2 + b_3 y + b_4 = 0 \tag{8}$$

so that

$$b_1 = 0, \qquad b_3 = 0. \tag{9}$$

Then (8) passes into a quadratic equation for y^2, from whose roots the square roots must still be drawn; thus after the equations (9) have been solved, a square root has still to be drawn twice.

If by means of the first equation (9) one eliminates α_0 from the second, then a homogeneous cubic equation in $\alpha_1, \alpha_2, \alpha_3$ arises. One may therefore choose one of the two ratios $\alpha_1 : \alpha_2 : \alpha_3$ arbitrarily, for example put $\alpha_3 = 0$, and then obtains a cubic equation to determine the other. Thus the solution of the biquadratic equation has been reduced to that of a cubic and to square roots.

§ 54. The Tschirnhausen transformation of the equation of the fifth degree

If one wants to apply the Tschirnhausen transformation to the equation of the fifth degree

$$x^5 + a_1x^4 + a_2x^3 + a_3x^2 + a_4x + a_5 = 0, \tag{1}$$

one must put

$$y = \alpha_0 + \alpha_1 x + \alpha_2 x^2 + \alpha_3 x^3 + \alpha_4 x^4, \tag{2}$$

and obtains the transform

$$y^5 + b_1y^4 + b_2y^3 + b_3y^2 + b_4y + b_5 = 0. \tag{3}$$

The most immediate thought would be to determine the ratios of the five unknowns α so that

$$b_1 = 0, \qquad b_2 = 0, \qquad b_3 = 0, \qquad b_4 = 0 \tag{4}$$

would hold, whereby (3) would be brought back to a pure equation; this was probably also the aim Tschirnhausen had in view. But among the four equations (4), the first is linear, the second quadratic, the third of the third degree, and the fourth of the fourth degree. The final equation derivable from them will therefore be, by § 50, of degree $2 \cdot 3 \cdot 4 = 24$, and hence no benefit for the solution of the equation of the fifth degree can be drawn from it. One therefore seeks first to satisfy only the two equations

$$b_1 = 0, \qquad b_2 = 0, \tag{5}$$

whereby equation (3) passes into a form that, following F. Klein's lectures on the icosahedron, is called a principal equation of the fifth degree.

To satisfy the two equations (5), we have at our disposal the four ratios $\alpha_1 : \alpha_2 : \alpha_3 : \alpha_4$, and we can still use these quantities in many ways to simplify the equation of the fifth degree.

We shall return to this subject more fully in a later section, and confine ourselves here to indicating the aims in general. For clarity we use a geometrical formulation, which, however, is not essential for understanding the algebraic theory as it will be given later.

If, with the help of the linear equation $b_1 = 0$, we express one of the five quantities α, say α_0, in terms of the others, then we may regard the four remaining quantities $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ as homogeneous coordinates of a point in space. The equation $b_2 = 0$ then represents a surface of the second order, $b_3 = 0$ a surface of the third order, $b_4 = 0$ a surface of the fourth order, and $b_5 = 0$ a surface of the fifth order.

If one wanted to set b_2, b_3, b_4 simultaneously equal to zero and thereby reduce the given equation to a pure one, then one would have to determine one of the 24 intersection points of three surfaces of the second, third, and fourth orders; this depends on an equation of the 24th degree.

Instead of proceeding in this way, one seeks to determine a straight line on the surface $b_2 = 0$. On every surface of the second order there is one or two systems of such straight lines, real or imaginary, and one can determine one of these lines by square roots, as will be explained later. The determination of an intersection point of one of these straight lines with the surface $b_3 = 0$ then leads to an equation of the third degree. If, moreover, by determining a common factor of the α one makes the coefficient $b_4 = 1$, the equation for y obtains the form

$$y^5 + y + b = 0, \tag{6}$$

so that y is determined as a function of one variable b.

This equation form is usually named after the English mathematician Jerrard, who made it known in 1834. It had, however, already been published much earlier, in 1786, by E. J. Bring in Lund.[15] We shall therefore call it the Bring-Jerrard form.

[15] Compare F. Klein, Vorlesungen über das Ikosaeder, p. 143.

Just as well, one could make $b_4 = 0$ by an equation of the fourth degree and would obtain a second normal form of the equation of the fifth degree:

$$y^5 + y^2 + b = 0, \tag{7}$$

which has the same justification as the Bring-Jerrard form.

For the solution of the equation of the fifth degree, of course, nothing is gained by these considerations. Their use consists in reducing the equation of the fifth degree to a normal form depending on only one variable coefficient. This can be achieved in infinitely many different ways, and is achieved most simply, namely without solving an equation of the third or fourth degree, if one regards the equation of the fifth degree $b_5 = 0$ itself as a transformation of the given equation of the fifth degree. For once the equation $b_5 = 0$ has been solved, one of the five values of y is equal to 0, and the given equation of the fifth degree is at the same time solved.

§ 55. Introduction of the linear transformation

When, in any functions depending on the m variables $x_1, x_2, \dots, x_m$, we substitute for these variables linear expressions depending on m new variables $x'_1, x'_2, \dots, x'_m$,

$$\begin{aligned} x_1 &= \alpha_1^{(1)} x'_1 + \alpha_1^{(2)} x'_2 + \cdots + \alpha_1^{(m)} x'_m, \\ x_2 &= \alpha_2^{(1)} x'_1 + \alpha_2^{(2)} x'_2 + \cdots + \alpha_2^{(m)} x'_m, \\ &\cdots \\ x_m &= \alpha_m^{(1)} x'_1 + \alpha_m^{(2)} x'_2 + \cdots + \alpha_m^{(m)} x'_m, \end{aligned} \tag{1}$$

we call this a linear substitution, and the result a linear transformation.

The quantity

$$r = \begin{vmatrix} \alpha_1^{(1)} & \alpha_1^{(2)} & \cdots & \alpha_1^{(m)} \\ \alpha_2^{(1)} & \alpha_2^{(2)} & \cdots & \alpha_2^{(m)} \\ \vdots & \vdots & & \vdots \\ \alpha_m^{(1)} & \alpha_m^{(2)} & \cdots & \alpha_m^{(m)} \end{vmatrix} \tag{2}$$

is called the substitution determinant. It must always be different from zero, since otherwise (1) would express a dependence among the variables $x_1, x_2, \dots, x_m$.

Under a linear substitution, every homogeneous function of the variables x passes into a homogeneous function of the same degree in the variables x'.

Thus, for example, every linear function

$$y = \sum_i a_i x_i$$

passes into a similar function

$$y = \sum_i a'_i x'_i,$$

where

$$a'_k = \sum_i a_i \alpha_i^{(k)}. \tag{3}$$

If we consider m such linear functions

$$y_\nu = \sum_i a_{\nu,i} x_i \qquad \nu = 1, 2, \dots, m,$$

we obtain m transformed functions

$$y_\nu = \sum_i a'_{\nu,i} x'_i \qquad \nu = 1, 2, \dots, m.$$

The determinants of the systems y_ν, y'_ν,

$$\Delta = \sum \pm a_{1,1} a_{2,2} \cdots a_{m,m}, \qquad \Delta' = \sum \pm a'_{1,1} a'_{2,2} \cdots a'_{m,m},$$

stand, by the multiplication theorem for determinants, in the relation

$$\Delta' = r\Delta, \tag{4}$$

so that one cannot vanish without the other.

§ 56. Quadratic forms

Of particular importance is the transformation of homogeneous functions of the second degree, or quadratic forms. We denote them, as already in the first section, by

$$\varphi(x_1, x_2, \ldots, x_m) = \sum_{i,k} a_{i,k} x_i x_k, \qquad a_{i,k} = a_{k,i}. \tag{1}$$

By the substitution § 55, (1), φ passes into

$$\Phi(x'_1, x'_2, \ldots, x'_m) = \sum_{i,k} a'_{i,k} x'_i x'_k. \tag{2}$$

Put $x + \xi$ for x, and similarly $x' + \xi'$ for x', so that the ξ' stand in the same dependence on the ξ as the x' do on the x. If, for abbreviation, we set

$$\frac{1}{2}\varphi'(x_i) = u_i, \qquad \frac{1}{2}\Phi'(x'_i) = u'_i,$$

then, by comparison of the terms of the first dimension (§ 16), we obtain

$$\sum \xi_i u_i = \sum \xi'_i u'_i, \tag{3}$$

and hence, as in § 55, (3),

$$u'_k = \sum_i \alpha_i^{(k)} u_i. \tag{4}$$

Now

$$u_k = \sum_i a_{k,i} x_i, \qquad u'_k = \sum_i a'_{k,i} x'_i, \tag{5}$$

and consequently (4) gives us the transformation of m linear functions. The coefficients of the given system are $a'_{k,h}$, and those of the transformed system, obtained by substituting the first system (5) into (4), are

$$b_{k,h} = \sum_i \alpha_i^{(k)} a_{i,h}. \tag{6}$$

The determinant of the first system

$$H' = \sum \pm a'_{1,1} a'_{2,2} \cdots a'_{m,m}$$

is, by § 55, (4),

$$= r \sum \pm b_{1,1} b_{2,2} \cdots b_{m,m},$$

and, by (6), the determinant of the $b_{k,h}$ is

$$= r \sum \pm a_{1,1} a_{2,2} \cdots a_{m,m}.$$

If we therefore put

$$H = \sum \pm a_{1,1} a_{2,2} \cdots a_{m,m}, \tag{7}$$

we have the important theorem

$$H' = r^2 H. \tag{8}$$

The determinant H is called the determinant of the form $\varphi(x)$.

It therefore changes, under the linear transformation of the function φ, only by the factor r^2, and it cannot vanish for the transformed form if it does not vanish for the original one.

The vanishing of the determinant H means that the function $\varphi(x)$ may be regarded as a function of fewer than m variables that are linear functions of the x.

For, first, if by a linear transformation $\varphi(x)$ passes into the function $\Phi(x')$, which is independent of one of the variables, say x'_1, then H' vanishes, since all the elements in one of the rows and columns vanish; consequently, by (8), $H = 0$ as well.

Second, however, H is the determinant of the system of linear equations

$$\varphi'(x_i) = 0 \qquad i = 1, 2, \ldots, m, \tag{9}$$

and, if this determinant vanishes, then (§ 24, III) there is a system of quantities

$$x_1^{(0)}, x_2^{(0)}, \ldots, x_m^{(0)}, \tag{10}$$

whose elements do not all vanish and which, when substituted for the x, satisfy equations (9).

Now if x_i, x'_i are any two systems of quantities, and λ is likewise an arbitrary quantity, the following identities hold:

$$\begin{aligned} \varphi(x + \lambda x') &= \varphi(x) + \lambda \sum x_i \varphi'(x'_i) + \lambda^2 \varphi(x'), \\ \varphi(x') &= \frac{1}{2} \sum x'_i \varphi'(x'_i), \end{aligned} \tag{11}$$

[§ 16, (12) to (15)]. Thus, when for x'_i one substitutes the values $x_i^{(0)}$, one obtains

$$\varphi(x_i + \lambda x_i^{(0)}) = \varphi(x), \tag{12}$$

quite independently of λ. If we now assume that $x_1^{(0)}$ is different from zero and set

$$\lambda = -\frac{x_1}{x_1^{(0)}},$$

and further set

$$\begin{aligned} y_2 &= x_2 x_1^{(0)} - x_1 x_2^{(0)}, \\ y_3 &= x_3 x_1^{(0)} - x_1 x_3^{(0)}, \\ &\cdots \\ y_m &= x_m x_1^{(0)} - x_1 x_m^{(0)}, \end{aligned} \tag{13}$$

then it follows that

$$x_1^{(0)2} \varphi(x) = \varphi(0, y_2, y_3, \ldots, y_m), \tag{14}$$

so that $\varphi(x)$ is expressed through the $m - 1$ variables $y_2, y_3, \ldots, y_m$.

§ 57. Transformation of quadratic forms into a sum of squares

We at once make an important application of the theorem just proved to the discriminant of the function (11), regarded as a function of the second degree in λ, without now assuming that the determinant of the function φ vanishes. This discriminant is

$$\psi(x) = \left[\sum x_i \varphi'(x'_i)\right]^2 - 4\varphi(x)\varphi(x'). \tag{1}$$

If, in this expression, we put $x_i + \lambda x'_i$ in place of x_i, then, after substituting for $\varphi(x + \lambda x')$ the expression § 56, (11) and setting $\sum x'_i \varphi'(x'_i) = 2\varphi(x')$, we get

$$\psi(x + \lambda x') = \psi(x),$$

which shows that $\psi(x)$ depends on only $m - 1$ linear combinations of the m variables. If for $x'_1, x'_2, \ldots, x'_m$ we put any special values for which $\varphi(x')$ does not vanish, and assume that x'_1 is not zero, then

$$x_1'^2 \psi(x_1, x_2, \ldots, x_m) = \psi(0, x_2 x'_1 - x_1 x'_2, \ldots, x_m x'_1 - x_1 x'_m),$$

hence it is a function of the $m - 1$ variables

$$y_2 = x_2 x'_1 - x_1 x'_2, \ldots, y_m = x_m x'_1 - x_1 x'_m.$$

This may also be confirmed by the fact that

$$\psi'(x_1), \psi'(x_2), \ldots, \psi'(x_m)$$

all vanish simultaneously for $x_1, x_2, \ldots, x_m = x'_1, x'_2, \ldots, x'_m$, so that the determinant of the function ψ must vanish.

Formula (1) now contains a very important result, most clearly seen when we give this formula the following form. We set

$$\psi(x) = -4\varphi(x')\chi(y_2, \ldots, y_m), \qquad \sum x_i \varphi'(x'_i) = 2\sqrt{\varphi(x')}\, Y_1, \tag{2}$$

and obtain

$$\varphi(x) = Y_1^2 + \chi(y_2, y_3, \ldots, y_m), \tag{3}$$

so that $\varphi(x)$ is represented as the sum of the square of a linear function Y_1 and a function χ of $m - 1$ variables.

From this follows the important theorem: A quadratic form in m variables can always be represented as a sum of squares of m or fewer linear functions.

This theorem is obviously true for a function of one variable, since such a function is itself a square. From (3), however, the truth of the theorem for a function of m variables follows if its truth is assumed for a function of $m - 1$ variables.

As (2) shows, the representation is possible in infinitely many different ways, because the entirely arbitrary quantities x' still occur in this formula.

A representation by fewer than m squares is possible only when the determinant of the function φ vanishes.

Since in the expression (2) for Y_1 a square root occurs, Y_1 may be real or imaginary even when the coefficients of φ and the x and x' are assumed real. But if, in the case where Y_1 is imaginary, we put iY_1 in place of Y_1, then we can also state our theorem as follows:

A real quadratic form in m variables can be represented as a sum of at most m positive or negative squares of real linear functions of the x.[16]

§ 58. Law of inertia of quadratic forms

For the decomposition of a quadratic form into squares, the following theorem now holds. Sylvester gave it the name of the law of inertia of quadratic forms (Philos. Mag. 1852).

[16] For the general theory of the linear transformation of forms of the second degree, besides the investigations of Jacobi and Hesse (Hesse, "Vorlesungen über analytische Geometrie des Raumes", third edition, edited by Gundelfinger, Leipzig 1876), especially the works of Weierstrass and Kronecker (Monthly Reports of the Berlin Academy, 1868, 1874) are important. A detailed historical account of the development of the question is given in the "Bericht über den gegenwärtigen Stand der Invariantentheorie" by Franz Meyer (first annual report of the Deutsche Mathematiker-Vereinigung, Berlin 1892).

In whatever way one decomposes a real quadratic form $\varphi(x)$ into a sum of positive and negative squares of linear functions, the number of positive and negative squares, and therefore also their total number, is always the same, provided that there is no linear dependence among these linear functions.

The proof of this important theorem is very simple.

Let

$$\begin{aligned} \varphi(x) &= Y_1^2 + Y_2^2 + \cdots + Y_\nu^2 - Y_1'^2 - Y_2'^2 - \cdots - Y_{\nu'}'^2 \\ &= Z_1^2 + Z_2^2 + \cdots + Z_\mu^2 - Z_1'^2 - Z_2'^2 - \cdots - Z_{\mu'}'^2 \end{aligned} \tag{1}$$

be two decompositions of $\varphi(x)$ into squares. Thus $Y_1, Y_2, \ldots, Y_\nu, Y_1', Y_2', \ldots, Y_{\nu'}'$ are homogeneous linear functions of x among which no linear relation with constant coefficients exists, so that $\nu + \nu' \leq m$. The same assumptions are made about the functions $Z_1, Z_2, \ldots, Z_\mu, Z_1', Z_2', \ldots, Z_{\mu'}'$, so that $\mu + \mu' \leq m$ as well.

Suppose that

$$\nu < \mu.$$

Then we can subject the variables x to the linear equations

$$\begin{aligned} Y_1 = 0, \quad & Y_2 = 0, \ldots, Y_\nu = 0, \\ Z_1' = 0, \quad & Z_2' = 0, \ldots, Z_{\mu'}' = 0, \end{aligned} \tag{2}$$

whose number $\nu + \mu'$ is less than m.

If all the functions $Z_1, Z_2, \ldots, Z_\mu$ were linearly dependent on the functions $Y_1, Y_2, \ldots, Y_\nu, Z_1', Z_2', \ldots, Z_{\mu'}'$, then from these μ equations one could, by eliminating the ν variables $Y_1, Y_2, \ldots, Y_\nu$, derive a linear equation among $Z_1, Z_2, \ldots, Z_\mu, Z_1', \ldots, Z_{\mu'}'$, which by hypothesis is not to exist. Thus the vanishing of all $Z_1, Z_2, \ldots, Z_\mu$ is not a necessary consequence of equations (2), and we may add to equations (2) a non-homogeneous equation, for instance

$$Z_1 = 1.$$

Then, for the m unknowns x, we have a system of m or fewer equations that are independent of one another and hence do not contradict one another.

But then the second representation (1) gives a positive value for $\varphi(x)$, while the first gives a negative or vanishing value, a contradiction. Hence

$$\nu \geq \mu,$$

and, since one similarly concludes that

$$\mu \geq \nu,$$

only $\nu = \mu$ remains. In the same way one may prove that $\nu' = \mu'$, and this proves the theorem completely.

§ 59. Transformation of forms of the nth degree

When a form $F(x)$ of the nth degree in m variables is transformed by a linear substitution [§ 55, (1)] into $\Phi(x')$, the linear function of the variables $\xi_1, \xi_2, \ldots, \xi_m$

$$\sum_i \xi_i F'(x_i) \tag{1}$$

is simultaneously transformed, by the same substitution applied to the ξ_i, into

$$\sum_i \xi_i' \Phi'(x_i'), \tag{2}$$

because both expressions represent the coefficient of the first power of t in the expansion of

$$F(x_i + t\xi_i) = \Phi(x'_i + t\xi'_i)$$

in powers of t (§ 16). Likewise, the quadratic function

$$\sum_{i,k} \xi_i \xi_k F''(x_i x_k) \tag{3}$$

is transformed into

$$\sum_{i,k} \xi'_i \xi'_k \Phi''(x'_i x'_k). \tag{4}$$

If we apply the first result to a system of m functions $F_1(x), F_2(x), \dots, F_m(x)$ of arbitrary degrees and denote by Δ the determinant

$$\Delta = \begin{vmatrix} F'_1(x_1) & F'_1(x_2) & \cdots & F'_1(x_m) \\ F'_2(x_1) & F'_2(x_2) & \cdots & F'_2(x_m) \\ \vdots & \vdots & & \vdots \\ F'_m(x_1) & F'_m(x_2) & \cdots & F'_m(x_m) \end{vmatrix},$$

while Δ' denotes the corresponding determinant formed from the functions $\Phi_1, \Phi_2, \dots, \Phi_m$, then by § 55, (4), again denoting the substitution determinant by r, we obtain

$$\Delta' = r\Delta. \tag{5}$$

The function Δ is called the functional determinant of the m functions $F_1, F_2, \dots, F_m$.

Likewise, from the transformation (3), (4), if H denotes the determinant

$$H = \begin{vmatrix} F''(x_1, x_1) & F''(x_1, x_2) & \cdots & F''(x_1, x_m) \\ F''(x_2, x_1) & F''(x_2, x_2) & \cdots & F''(x_2, x_m) \\ \vdots & \vdots & & \vdots \\ F''(x_m, x_1) & F''(x_m, x_2) & \cdots & F''(x_m, x_m) \end{vmatrix}$$

and H' the corresponding determinant for the transformed function Φ, we obtain by § 56, (8)

$$H' = r^2 H. \tag{6}$$

H is called the Hessian determinant[17] of the function F. In both formulas (5) and (6), r denotes the substitution determinant.

§ 60. Invariants and covariants

In the linear transformation of homogeneous functions, certain functions of the coefficients always occur which, when the coefficients of the given form are replaced by the coefficients of the transformed form, change in no other way than by acquiring a factor that is a power of the substitution determinant. Such functions also occur under the simultaneous transformation of a system of several forms; they depend on the coefficients of all forms of the system. Examples are, for a system of n linear forms, the determinant of the system, and, for a quadratic form, the determinant of that form.

These functions are called invariants of the form, or, when they depend on several forms, simultaneous invariants of the system.

[17] Named after Hesse, who frequently used this determinant in geometrical investigations, especially in the theory of curves of the third order and in elimination problems, and recognized its significance.

Thus, if $I(a)$ is a function of the coefficients a of a form $F(x)$, and if a' are the coefficients of the transformed form $\Phi(x')$, then $I(a)$ is called an invariant if the identical relation

$$I(a') = r^\lambda I(a) \tag{1}$$

holds. We shall call the exponent λ the weight of the invariant. If this exponent is zero, then I is called an absolute invariant.

One usually considers only integral rational invariants, that is, such functions I as depend rationally on the coefficients a and are moreover integral homogeneous functions of them. Simultaneous invariants are to be homogeneous with respect to the coefficients of each of the forms on which they depend.

Besides invariants, there are also functions that, in addition to the coefficients, contain the variables themselves and otherwise have the same properties as the invariants (1), namely the "invariant property." Thus, if $C(x, a)$ is such a function, then

$$C(x', a') = r^\lambda C(x, a). \tag{2}$$

Such functions are called covariants of the form $F(x)$. The functional determinant is a simultaneous covariant of a system of m functions, and the Hessian determinant of a function of degree higher than the second is a covariant of a single form. Among covariants, too, one chiefly considers integral rational and homogeneous functions of x.

If m is the number of variables x, n the degree of the function $F(x)$, and ν and μ the degrees of $C(x, a)$ with respect to x and to a, then the numbers λ, μ, ν, m, n satisfy the relation

$$n\mu = m\lambda + \nu. \tag{3}$$

This is proved by comparing the degree of the right and left sides of (2) with respect to the substitution coefficients. Namely, the a' are of degree n in the substitution coefficients, r is of degree m, and x is of degree one; this gives (3). The corresponding formula for invariants is obtained by putting $\nu = 0$, just as invariants may in general be regarded as a special case of covariants.

We shall briefly discuss a general rule for the formation of invariants and covariants, which we have already used in the special cases of § 59.

If, according to § 16, (4), we arrange the function

$$F(x_1 + t\xi_1, x_2 + t\xi_2, \ldots, x_m + t\xi_m)$$

by powers of t, then for the coefficient of t^ν we obtain

$$F_\nu(x, \xi) = \frac{1}{\Pi(\nu)} \sum_{\alpha_1+\alpha_2+\cdots+\alpha_m=\nu} \frac{\Pi(\nu)}{\Pi(\alpha_1)\cdots\Pi(\alpha_m)} \xi_1^{\alpha_1}\xi_2^{\alpha_2}\cdots\xi_m^{\alpha_m} \mathrm{D}_{\alpha_1,\alpha_2,\ldots,\alpha_m}(F). \tag{4}$$

If we apply the linear transformation (1), § 55, simultaneously to the variables ξ and x, then $F_\nu(x, \xi)$ passes into $\Phi_\nu(x', \xi')$, which is obtained from F_ν by replacing at the same time all letters x, ξ, a by x', ξ', a' and F by Φ. Hence:

I. If $F_\nu(x, \xi)$ is regarded as a form of the νth degree in ξ and one forms an invariant of this form, whose coefficients therefore still depend on the x, one obtains a covariant of the form F.

Since the variables x and ξ undergo the same transformation, it follows similarly that:

II. If one forms a covariant of the function $F_\nu(x, \xi)$, regarded as a function of ξ, and then replaces the ξ in it by the x, one obtains a covariant of the original function $F(x)$.

The same theorems also hold for simultaneous invariants and covariants.

For the functions $F_\nu(x, \xi)$ one has the theorem

$$F_\nu(x, \xi) = F_{n-\nu}(\xi, x), \tag{5}$$

which follows by comparing corresponding powers of t on both sides of the identity

$$F(x_1 + t\xi_1, \ldots, x_m + t\xi_m) = t^n F(t^{-1}x_1 + \xi_1, \ldots, t^{-1}x_m + \xi_m).$$

The functions

$$P_\nu(x, \xi) = \frac{\Pi(\nu)\Pi(n-\nu)}{\Pi(n)} F_\nu(x, \xi)$$

are called the polars of the function $F(x)$; more precisely, $P_\nu(x, \xi)$ is called the νth polar. They too satisfy the theorem

$$P_\nu(x, \xi) = P_{n-\nu}(\xi, x).$$

The constant factor $\Pi(\nu)\Pi(n-\nu) : \Pi(n)$ is added so that, when the form $F(x)$ is written with polynomial coefficients, the functions P_ν have coefficients as simple as possible.

For the binary form $f(x, y)$, to which in what follows we shall mainly apply these definitions, the following expressions for the polars result:

$$P_1(x, \xi) = \frac{1}{n}\{\xi f'(x) + \eta f'(y)\},$$

$$P_2(x, \xi) = \frac{1}{n(n-1)}\{\xi^2 f''(x, x) + 2\xi\eta f''(x, y) + \eta^2 f''(y, y)\},$$

and in general

$$\begin{aligned} &n(n-1)\cdots(n-\nu+1)P_\nu(x, \xi) \\ &\quad = \xi^\nu \frac{\partial^\nu f}{\partial x^\nu} + \nu\xi^{\nu-1}\eta \frac{\partial^\nu f}{\partial x^{\nu-1}\partial y} + \cdots + \eta^\nu \frac{\partial^\nu f}{\partial y^\nu}. \end{aligned}$$

§ 61. Linear transformation of binary forms

An integral rational function of degree n in x,

$$f(x) = a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \cdots + a_{n-1}x + a_n, \tag{1}$$

is converted into a binary form when x is replaced by $x : y$ and the result is multiplied by y^n:

$$f(x, y) = a_0 x^n + a_1 x^{n-1} y + a_2 x^{n-2} y^2 + \cdots + a_n y^n. \tag{2}$$

If the roots of $f(x)$ are known, then $f(x)$ can be decomposed into n linear factors:

$$f(x) = a_0(x - \alpha_1)(x - \alpha_2)\cdots(x - \alpha_n), \tag{3}$$

and from this it follows that

$$f(x, y) = a_0(x - \alpha_1 y)(x - \alpha_2 y)\cdots(x - \alpha_n y). \tag{4}$$

But if $\alpha_1, \alpha_2, \ldots, \alpha_n$ are themselves replaced by ratios $\alpha_1 : \beta_1, \alpha_2 : \beta_2, \ldots, \alpha_n : \beta_n$, one obtains

$$f(x, y) = A(x\beta_1 - y\alpha_1)(x\beta_2 - y\alpha_2)\cdots(x\beta_n - y\alpha_n), \tag{5}$$

provided that

$$a_0 = A\beta_1\beta_2\cdots\beta_n.$$

For the factors of $f(x, y)$ that will now occur more frequently, we introduce the abbreviated notation

$$x\beta_1 - y\alpha_1 = (x\beta_1). \tag{6}$$

Similarly, we shall put

$$x_1 y_2 - x_2 y_1 = (x_1 y_2), \tag{7}$$

where x_1, y_1 and x_2, y_2 are two arbitrary pairs of variables. If we use the inhomogeneous representation, we shall write still more briefly

$$x - \alpha_1 = (0,1), \quad x - \alpha_2 = (0,2), \ldots, x - \alpha_n = (0,n), \\ \alpha_1 - \alpha_2 = (1,2), \ldots, \alpha_i - \alpha_k = (i,k). \tag{8}$$

If, in the form (2), we make the linear substitution

$$x = \alpha x' + \beta y', \qquad y = \gamma x' + \delta y' \tag{9}$$

with non-zero determinant

$$r = \alpha\delta - \beta\gamma, \tag{10}$$

then the form $f(x,y)$ passes into a form $F(x', y')$ of degree n, and

$$F(x', y') = a_0' x'^n + a_1' x'^{n-1} y' + a_2' x'^{n-2} y'^2 + \cdots + a_n' y'^n. \tag{11}$$

The coefficients $a_0', a_1', \ldots, a_n'$ of the transformed form are obtained as linear and homogeneous expressions in the coefficients $a_0, a_1, \ldots, a_n$, and as homogeneous expressions of degree n in the substitution coefficients $\alpha, \beta, \gamma, \delta$. Thus, by setting $x' = 1, y' = 0$ or $x' = 0, y' = 1$, one obtains

$$a_0' = f(\alpha, \gamma), \qquad a_n' = f(\beta, \delta). \tag{12}$$

The remaining coefficients a' can be expressed through the derivatives of the function $f(x,y)$, for example

$$(n-1)a_1' = \beta f'(\alpha) + \delta f'(\gamma),$$

which we shall not carry out further.

If we apply the transformation (9) simultaneously to the two pairs of variables x, y and α_k, β_k, then we obtain the transformed linear factors of $f(x,y)$, namely

$$(x'\beta_k') = r(x\beta_k), \tag{13}$$

as follows either from the multiplication rule for determinants or by direct calculation. These factors $(x\beta_k)$ are therefore covariants of $f(x,y)$, though irrational ones, since they do not depend rationally on the coefficients of $f(x)$. Similarly one obtains

$$(\alpha_h'\beta_k') = r(\alpha_h\beta_k), \tag{14}$$

according to which the determinants $(\alpha_h\beta_k)$ are to be regarded as irrational invariants. We shall now explain how rational invariants and covariants may be formed from them.

For this purpose it is best to consider the linear transformation in the non-homogeneous form

$$x = \frac{\alpha x' + \beta}{\gamma x' + \delta} \tag{15}$$

and to apply it to the function $f(x)$ in (1). Then, if

$$F(x') = a_0' x'^n + a_1' x'^{n-1} + a_2' x'^{n-2} + \cdots + a_n'$$

is put, one obtains

$$F(x') = (\gamma x' + \delta)^n f(x). \tag{16}$$

We also write this equation as

$$F(x') = (\alpha x' + \beta)^n \left(a_0 + a_1 \frac{1}{x} + a_2 \frac{1}{x^2} + \cdots \right),$$

from which, if $x' = -\delta : \gamma$, so that $x = \infty$, one obtains

$$r^n a_0 = (-\gamma)^n F\left(-\frac{\delta}{\gamma}\right).$$

Or, if $\alpha'_1, \alpha'_2, \ldots, \alpha'_n$ depend on $\alpha_1, \alpha_2, \ldots, \alpha_n$ in the same way as x' depends on x, so that

$$F(x') = a'_0(x' - \alpha'_1)(x' - \alpha'_2)\cdots(x' - \alpha'_n)$$

is put, it follows that

$$r^n a_0 = a'_0(\gamma\alpha'_1 + \delta)(\gamma\alpha'_2 + \delta)\cdots(\gamma\alpha'_n + \delta). \tag{17}$$

Now, by the transformation (15),

$$\begin{aligned} x - \alpha_i &= \frac{r(x' - \alpha'_i)}{(\gamma x' + \delta)(\gamma\alpha'_i + \delta)}, \\ \alpha_i - \alpha_k &= \frac{r(\alpha'_i - \alpha'_k)}{(\gamma\alpha'_i + \delta)(\gamma\alpha'_k + \delta)}. \end{aligned} \tag{18}$$

Now form a product P from any of the factors

$$(0,1), (0,2), \ldots, (0,n),$$

$$(1,2), (1,3), \ldots, (n-1,n),$$

but in such a way that each of the indices $1, 2, \ldots, n$ occurs altogether equally often, say μ times. The index 0, that is, the variable x, is to occur ν times, and the total number of factors is to be q. Since each factor contains two indices, one has

$$\nu + n\mu = 2q. \tag{19}$$

Then from (17) and (18), if P' is obtained from P by replacing x, α_i by x', α'_i, one obtains

$$a_0'^{\mu} P' = a_0^{\mu} r^{n\mu - q}(\gamma x' + \delta)^{\nu} P. \tag{20}$$

We now form

$$C(x, y, a) = a_0^{\mu} y^{\nu} \sum P\left(\frac{x}{y}\right), \tag{21}$$

where the sum is taken over all values obtained from P by permuting the indices $1, 2, 3, \ldots, n$ with one another in all possible ways. This sum is then a symmetric function of the roots of (1), and may therefore be expressed as an integral rational function of the ratios $a_1 : a_0, a_2 : a_0, \ldots, a_n : a_0$, with no term exceeding the μth order. Thus $C(x, y, a)$ is an integral homogeneous function of $a_0, a_1, \ldots, a_n$ of order μ, and an integral homogeneous function of x, y of order ν. Since, by (19), (20), it satisfies the condition

$$C(x', y', a') = r^{\lambda} C(x, y, a), \tag{22}$$

$$\lambda = n\mu - q, \qquad 2\lambda = n\mu - \nu, \tag{23}$$

it is a covariant, and in the special case where $\nu = 0$ an invariant.

The weight λ is, as the second expression shows, always positive or zero, since $n\mu$ is at least equal to ν. The extreme value $n\mu = \nu$, or $\lambda = 0$, occurs only when every one of the indices $1, 2, 3, \ldots, n$ occurs in P only in connection with 0, that is, when P is a power of $f(x)$ itself.

§ 62. Binary cubic forms

A binary quadratic form gives no invariant constructions other than the discriminant. We shall therefore not concern ourselves with these, and pass at once to the consideration of the cubic forms

$$f(x,y) = a_0x^3 + a_1x^2y + a_2xy^2 + a_3y^3. \tag{1}$$

Here we form, as the first quadratic covariant, the Hessian determinant. We shall always follow the rule here that the forms are written with integral numerical coefficients having no common factor. Thus in the determinant formed from the second derivatives of (1) we must discard the factor 4, and obtain as first covariant

$$H = \begin{vmatrix} 3a_0x + a_1y & a_1x + a_2y \\ a_1x + a_2y & a_2x + 3a_3y \end{vmatrix},$$

or

$$H = A_0x^2 + A_1xy + A_2y^2, \tag{2}$$

where, for abbreviation,

$$\begin{aligned} A_0 &= 3a_0a_2 - a_1^2, \\ A_1 &= 9a_0a_3 - a_1a_2, \\ A_2 &= 3a_1a_3 - a_2^2. \end{aligned} \tag{3}$$

The discriminant of this quadratic form is an invariant of the given cubic form. It contains, however, the factor 3 in all numerical coefficients, and therefore we put

$$3D = 4A_0A_2 - A_1^2. \tag{4}$$

Then D agrees with the discriminant of the cubic form [§ 46, (10)]:

$$D = a_1^2a_2^2 + 18a_0a_1a_2a_3 - 4a_0a_2^3 - 4a_1^3a_3 - 27a_0^2a_3^2. \tag{5}$$

We obtain a further cubic covariant by forming the functional determinant of $f(x,y)$ and $H(x,y)$:

$$Q(x,y) = f'(x)H'(y) - f'(y)H'(x), \tag{6}$$

or

$$Q = \begin{vmatrix} 3a_0x^2 + 2a_1xy + a_2y^2 & 2A_0x + A_1y \\ a_1x^2 + 2a_2xy + 3a_3y^2 & A_1x + 2A_2y \end{vmatrix}, \tag{7}$$

from which no numerical factor can be removed. We set down here, in order, the computed coefficients of x^3, x^2y, xy^2, y^3, whose formation presents no difficulty:

$$\begin{aligned} &27a_0^2a_3 - 9a_0a_1a_2 + 2a_1^3, \\ &27a_0a_1a_3 - 18a_0a_2^2 + 3a_1^2a_2, \\ &-27a_0a_2a_3 + 18a_1^2a_3 - 3a_1a_2^2, \\ &-27a_0a_3^2 + 9a_1a_2a_3 - 2a_2^3. \end{aligned}$$

The behavior of H, D, Q under a linear transformation follows from § 60, (2), (3). If H', D', Q' are the corresponding constructions for a transformed form, then

$$H' = r^2H, \qquad D' = r^6D, \qquad Q' = r^3Q. \tag{8}$$

We can use the covariants to transform the cubic form into a normal form which at the same time gives the solution of the cubic equation.

We choose the normal form

$$f(x,y) = F(\xi,\eta) = \xi^3 + \eta^3, \tag{9}$$

where ξ, η are linear functions of x, y; as will appear at once, this form can always be produced when D does not vanish, that is, when the equation $f = 0$ does not have two equal roots. The solutions of $f = 0$ then follow from the linear equations

$$\xi + \eta = 0, \qquad \xi + \varepsilon\eta = 0, \qquad \xi + \varepsilon^2\eta = 0,$$

where ε is a third root of unity.

To form the forms H', D', Q' for the normal form (9), we have to put $a'_0 = a'_3 = 1$, $a'_1 = a'_2 = 0$, and we obtain

$$H' = 9\xi\eta, \qquad D' = -27, \qquad Q' = 27(\xi^3 - \eta^3).$$

If the normal form (9) is derived from the general form $f(x, y)$ by a linear transformation, then by (8) one obtains

$$\begin{aligned} 9\xi\eta &= r^2 H, \\ -27 &= r^6 D, \\ 27(\xi^3 - \eta^3) &= r^3 Q, \\ \xi^3 + \eta^3 &= f. \end{aligned} \tag{10}$$

It follows that

$$r^6 = -\frac{27}{D}, \qquad 2\xi^3 = \frac{r^3 Q}{27} + f, \qquad 2\eta^3 = -\frac{r^3 Q}{27} + f,$$

and hence, after substituting for r^3 the value obtained from the first equation,

$$\begin{aligned} 6\xi^3\sqrt{-3D} &= Q + 3\sqrt{-3D}\, f, \\ 6\eta^3\sqrt{-3D} &= -Q + 3\sqrt{-3D}\, f. \end{aligned} \tag{11}$$

Thus ξ and η are determined by two cube roots, of which, by the first equation (10), one is determined by the other.

This also contains the proof that, when D is different from zero, the normal form can be produced by linear transformation. Under this hypothesis H splits into two distinct linear factors; and if, up to constant factors, these are chosen as the new variables ξ, η of a linear transformation, then $A'_0 = 0$, $A'_2 = 0$. From the identical relations

$$3a'_3A'_0 + a'_1A'_2 = a'_2A'_1, \qquad a'_2A'_0 + 3a'_0A'_2 = a'_1A'_1,$$

it follows, unless $A'_1 = 0$ also, which is excluded by the non-vanishing of D, that $a'_1 = 0$, $a'_2 = 0$; that is, the transformed form of f contains only the cubes of ξ and η.

If one raises the first equation (10) to the third power and eliminates ξ^3, η^3, r^6 by means of (11) and of the second equation (10), one obtains the following identical relation among the covariants:

$$4H^3 + Q^2 + 27Df^2 = 0. \tag{12}$$

In order to carry out the transformation into the normal form, that is, actually to find the functions ξ, η, one decomposes the function H into its linear factors:

$$4A_0H = (2A_0x + A_1y + \sqrt{-3D}\, y)(2A_0x + A_1y - \sqrt{-3D}\, y).$$

Then ξ, η differ from these two factors of H only by one constant factor each. Thus, denoting two constant factors by h, k, we put

$$\begin{aligned} 2\xi &= h(2A_0x + A_1y - \sqrt{-3D}\, y), \\ 2\eta &= k(2A_0x + A_1y + \sqrt{-3D}\, y), \end{aligned} \tag{13}$$

from which, by multiplication and with reference to the first two equations (10), it follows that

$$3hkA_0\sqrt[3]{-D} = 1, \tag{14}$$

and the equations (11), by comparing the coefficients of x^3, give

$$\begin{aligned} 6h^3A_0^3\sqrt{-3D} &= q_0 + 3\sqrt{-3D}\,a_0, \\ 6k^3A_0^3\sqrt{-3D} &= -q_0 + 3\sqrt{-3D}\,a_0, \end{aligned} \tag{15}$$

where q_0 is the coefficient of x^3 in Q, namely

$$q_0 = 27a_0^2a_3 - 9a_0a_1a_2 + 2a_1^3. \tag{16}$$

That the signs of $\sqrt{-3D}$ are to be taken as in formulas (13), with regard to the signs in (11), is seen most simply by carrying out the calculation for the special case $a_0 = 0$, $a_2 = 0$.

§ 63. The complete system of forms of the binary cubic form

We shall still show that the invariants and covariants of the cubic form are exhausted by the functions D, f, H, Q, that is, that all invariants and covariants of a cubic form can be expressed rationally by these four forms.

Let

$$C = C(x, y, a)$$

be a covariant of degree ν in the variables and of degree μ in the coefficients, and suppose that it has only integral numerical coefficients.

Then, for any transformed form,

$$C' = C(\xi, \eta, a') = r^\lambda C(x, y, a), \tag{1}$$

where

$$\lambda = \frac{3\mu - \nu}{2}. \tag{2}$$

We now understand by ξ, η the variables of the normal form considered above. If ε denotes a third root of unity, then

$$\xi = \varepsilon\xi', \qquad \eta = \varepsilon^2\eta'$$

is a linear substitution with determinant 1, by which the normal form § 62, (9) is carried into itself. Hence C' cannot change when ξ, η are replaced by $\varepsilon\xi, \varepsilon^2\eta$; therefore C' can depend rationally only on $\xi\eta, \xi^3, \eta^3$.

The interchange of ξ with η corresponds to a substitution with determinant -1. Therefore C' remains unchanged or changes its sign according as λ is even or odd. Consequently C' consists of a sum of terms of the form

$$M(\xi\eta)^\alpha(\xi^{3\beta} \pm \eta^{3\beta}),$$

where M is an integer factor, and the upper or lower sign applies according as λ is even or odd. In the latter case this expression is divisible by $\xi^3 - \eta^3$, and the quotient can be expressed rationally and integrally by $\xi^3 + \eta^3$ and $\xi\eta$, since a symmetric function of two quantities can be represented in this way by their sum and product. Thus, putting $\rho = 0$ or $= 1$ according as λ is even or odd, we obtain for C' an expression of the form

$$C'(\xi, \eta) = (\xi^3 - \eta^3)^\rho \sum M(\xi\eta)^\alpha(\xi^3 + \eta^3)^\beta,$$

where the M are integral coefficients and α, β run through a series of positive integers satisfying

$$3\rho + 3\beta + 2\alpha = \nu. \tag{3}$$

From (1), after multiplying by a sufficiently high power of 3 and expressing $\xi\eta, \xi^3 + \eta^3, \xi^3 - \eta^3$ by H, f, Q and r by D^{-1} according to § 62, (10), it follows that

$$3^\sigma C = Q^\rho \sum M D^\gamma H^\alpha f^\beta, \tag{4}$$

where

$$6\gamma = \lambda - 3\rho - 2\alpha, \tag{5}$$

and where the M are again integral factors.

To every value of γ there belongs, by (5), a definite value of α, and by (3) a definite value of β; likewise, through a value of α or of β the two other numbers are determined each time. Thus in (4) every exponent of D, H or f occurs only once.

That γ is an integer follows easily from (2), (3), (5). Since by definition $\lambda - 3\rho$ is an even number, 3γ is first of all an integer; furthermore, by (3),

$$\lambda - 2\alpha = \frac{3}{2}(\mu - \nu + 2\beta + 2\rho),$$

and hence 6γ, and therefore 3γ, is divisible by 3. It follows that γ is also an integer.

That γ must also be non-negative is seen as follows. Suppose that negative powers of D occur on the right side of (4). Multiply the equation on both sides by so high a power of D that no negative powers of D appear on the right, but also so that the right side does not acquire the factor D. If in the equation so obtained we put

$$a_0 = 0, \qquad a_1 = 1, \qquad a_2 = 0, \qquad a_3 = 0, \tag{6}$$

then

$$D = 0, \qquad H = -x^2, \qquad f = x^2 y, \qquad Q = 2x^3. \tag{7}$$

Thus the left side would vanish while the right side would not vanish, which is a contradiction.

Finally we can also prove that on the left side of (4) no power of 3 can occur which is not contained in all factors M and can therefore be cancelled.

In § 2 we proved the theorem that the product of two primitive integral rational functions of arbitrary variables is again a primitive function. By a primitive function there is meant one whose coefficients are integers without a common divisor. Here it remains to show that a sum of the form

$$Q^\rho \sum M D^\gamma H^\alpha f^\beta,$$

where the M are integers without a common divisor, cannot become imprimitive, and in particular cannot acquire the factor 3, when the expressions in the a, x, y are substituted for Q, D, H, f. By the theorem just mentioned, since Q is a primitive function, it is enough to show this for the form

$$\sum M D^\gamma H^\alpha f^\beta. \tag{8}$$

Here we may further omit all terms in which M already has the factor 3, and finally, again by the theorem just mentioned, we may suppose that the expression is not divisible by D, hence that it contains a term with $\gamma = 0$. Assume then that, under these hypotheses, the developed expression (8) has the divisor 3, and substitute the special values (6), (7), that is, put $D = 0, H = -x^2, f = x^2 y$. The expression then reduces to the single term in which $\gamma = 0$, $2\alpha = \lambda$, namely

$$(-1)^\alpha M 2^{\alpha+\rho} y^\beta,$$

and it would follow that this M must have the divisor 3, contrary to the assumption. Thus we have proved:

Every integral covariant of the cubic form can be represented integrally and rationally, with integral coefficients, by one of the two expressions

$$\sum M D^\gamma H^\alpha f^\beta, \qquad Q \sum M D^\gamma H^\alpha f^\beta.$$

The invariants are included among the covariants as a special case, and are all powers of D.

§ 64. Biquadratic forms

We now pass to the consideration of the biquadratic form

$$f(x, y) = a_0x^4 + a_1x^3y + a_2x^2y^2 + a_3xy^3 + a_4y^4. \tag{1}$$

We first have the Hessian determinant as a covariant of the fourth order, which, in order to free it from a numerical factor, we divide by 3:

$$H = \frac{1}{3}\begin{vmatrix} 12a_0x^2 + 6a_1xy + 2a_2y^2 & 3a_1x^2 + 4a_2xy + 3a_3y^2 \\ 3a_1x^2 + 4a_2xy + 3a_3y^2 & 2a_2x^2 + 6a_3xy + 12a_4y^2 \end{vmatrix}, \tag{2}$$

or

$$H = A_0x^4 + A_1x^3y + A_2x^2y^2 + A_3xy^3 + A_4y^4,$$

where

$$\begin{aligned} A_0 &= 8a_0a_2 - 3a_1^2, & A_4 &= 8a_2a_4 - 3a_3^2, \\ A_1 &= 24a_0a_3 - 4a_1a_2, & A_3 &= 24a_1a_4 - 4a_2a_3, \\ A_2 &= 48a_0a_4 + 6a_1a_3 - 4a_2^2, \end{aligned}$$

and a covariant of the sixth degree

$$T = \frac{1}{12}\begin{vmatrix} f'(x) & f'(y) \\ H'(x) & H'(y) \end{vmatrix}. \tag{3}$$

The invariants of the biquadratic form are most simply formed from the root differences, according to § 61.

We thus obtain an invariant of the second and one of the third order in the coefficients:

$$A = \frac{1}{2}a_0^2[(12)^2(34)^2 + (13)^2(24)^2 + (14)^2(23)^2], \tag{4}$$

$$B = a_0^3 \sum (12)^2(34)^2(13)(42). \tag{5}$$

These expressions may be written more clearly if one puts

$$U = (12)(34), \qquad V = (13)(42), \qquad W = (14)(23), \tag{6}$$

namely

$$\begin{aligned} A &= \frac{1}{2}a_0^2(U^2 + V^2 + W^2), \\ B &= a_0^3\{U^2(V - W) + V^2(W - U) + W^2(U - V)\} \\ &= a_0^3(W - V)(U - W)(V - U). \end{aligned} \tag{7}$$

The necessary formulas for computing these quantities were already developed in § 47. From the formulas (10) there one obtains

$$U = v^2 - w^2, \qquad V = w^2 - u^2, \qquad W = u^2 - v^2,$$

where u^2, v^2, w^2 are the roots of the cubic resolvent

$$z^3 + 2az^2 + (a^2 - 4c)z - b^2 = 0.$$

It follows that

$$\begin{aligned} A &= a_0^2\{(u^2 + v^2 + w^2)^2 - 3(v^2w^2 + w^2u^2 + u^2v^2)\} \\ &= -72ac + 27b^2, \end{aligned}$$

$$\begin{aligned} B &= a_0^3(2u^2 - v^2 - w^2)(2v^2 - w^2 - u^2)(2w^2 - u^2 - v^2) \\ &= a_0^3(3u^2 + 2a)(3v^2 + 2a)(3w^2 + 2a) \\ &= a_0^3(2a^3 - 72ac + 27b^2), \end{aligned}$$

so that A, B have the same meaning as in § 47, and, expressed by the coefficients a_0, a_1, a_2, a_3, a_4, are represented as follows:

$$A = a_2^2 - 3a_1a_3 + 12a_0a_4, \tag{8}$$

$$B = 27a_0a_3^2 + 27a_1^2a_4 + 2a_2^3 - 72a_0a_2a_4 - 9a_1a_2a_3. \tag{9}$$

We had also already formed there the discriminant of the biquadratic equation as a third invariant, but one depending on A and B,

$$D = a_0^6U^2V^2W^2, \tag{10}$$

and found

$$27D = 4A^3 - B^2. \tag{11}$$

From the formula of § 67, $2\lambda = n\mu - \nu$, we obtain for the invariant constructions found so far, which we denote by accents for a transformed form, the following relations:

$$H' = r^2H, \qquad T' = r^3T, \tag{12a}$$

$$A' = r^4A, \qquad B' = r^6B, \qquad D' = r^{12}D. \tag{12b}$$

§ 65. Solution of the biquadratic equation

We want to produce a normal form by a linear substitution with determinant 1. For its derivation we assume the given biquadratic form to be decomposed into linear factors:

$$f(x, y) = a_0(x - \alpha y)(x - \beta y)(x - \gamma y)(x - \delta y). \tag{1}$$

We put

$$\begin{aligned} x\sqrt{\alpha - \beta} &= \beta\xi - \alpha\eta, \\ y\sqrt{\alpha - \beta} &= \xi - \eta, \qquad r = 1, \end{aligned} \tag{2}$$

so that

$$\begin{aligned} x - \alpha y &= -\sqrt{\alpha - \beta}\,\xi, \\ x - \beta y &= -\sqrt{\alpha - \beta}\,\eta, \\ (x - \gamma y)\sqrt{\alpha - \beta} &= (\beta - \gamma)\xi - (\alpha - \gamma)\eta, \\ (x - \delta y)\sqrt{\alpha - \beta} &= (\beta - \delta)\xi - (\alpha - \delta)\eta. \end{aligned} \tag{3}$$

Thus $f(x, y)$ goes over into

$$F(\xi, \eta) = a_1'\xi^3\eta + a_2'\xi^2\eta^2 + a_3'\xi\eta^3. \tag{4}$$

By (3), the coefficients a_1', a_2', a_3' are easily expressed through $\alpha, \beta, \gamma, \delta$:

$$\begin{aligned} a_1' &= a_0(\beta - \gamma)(\beta - \delta), \\ a_3' &= a_0(\alpha - \gamma)(\alpha - \delta), \\ a_2' &= -a_0\{(\alpha - \gamma)(\beta - \delta) + (\alpha - \delta)(\beta - \gamma)\}. \end{aligned} \tag{5}$$

By interchanging the roots we can produce the normal form (4) in six different ways, but these yield only three distinct values of a_2'. These three values are, if as above

$$\begin{aligned} U &= (\alpha - \beta)(\gamma - \delta), \\ V &= (\alpha - \gamma)(\delta - \beta), \\ W &= (\alpha - \delta)(\beta - \gamma), \\ U + V + W &= 0, \end{aligned} \tag{6}$$

then

$$a_2' = a_0(V - W), \qquad a_2'' = a_0(W - U), \qquad a_2''' = a_0(U - V). \tag{7}$$

If a_2', a_2'', a_2''' are known, then U, V, W can also be determined by

$$\begin{aligned} 3a_0U &= a_2''' - a_2'', \\ 3a_0V &= a_2' - a_2''', \\ 3a_0W &= a_2'' - a_2'. \end{aligned} \tag{8}$$

It follows that, if of the three quantities a_2', a_2'', a_2''' two are equal, one of the quantities U, V, W vanishes, so that two of the roots of the biquadratic equation are equal, and consequently its discriminant vanishes.

Let us also put

$$\begin{aligned} \alpha\beta + \gamma\delta &= u, \\ \alpha\gamma + \delta\beta &= v, \\ \alpha\delta + \beta\gamma &= w. \end{aligned} \tag{9}$$

Then

$$a_0(u + v + w) = a_2$$

[the sum of the products of the roots of $f(x, y)$ taken two at a time], and

$$U = v - w, \qquad V = w - u, \qquad W = u - v.$$

Thus

$$\begin{aligned} 3a_0u &= a_2 - a_2', \\ 3a_0v &= a_2 - a_2'', \\ 3a_0w &= a_2 - a_2'''. \end{aligned} \tag{10}$$

Consequently, together with the quantities a_2', a_2'', a_2''', the quantities u, v, w are also known. But if one knows u, v, w, then the solution of the biquadratic equation has been reduced to square roots. For

$$\alpha\beta + \gamma\delta = u, \qquad a_0\alpha\beta\gamma\delta = a_4,$$

and therefore

$$2\alpha\beta = u + \sqrt{\frac{a_0u^2 - 4a_4}{a_0}}, \qquad 2\gamma\delta = u - \sqrt{\frac{a_0u^2 - 4a_4}{a_0}}.$$

Since the products

$$\alpha\gamma, \qquad \alpha\delta, \qquad \beta\gamma, \qquad \beta\delta$$

can likewise be determined, α^2, for example, can be computed from

$$\frac{\alpha\beta \cdot \alpha\gamma}{\beta\gamma}.$$

It remains only to form the cubic equation whose roots are a_2', a_2'', a_2'''. This is obtained at once from the invariants.

If the invariants A', B' are formed from the transformed form (4), then by § 64, (8), (9), (12), since $r = 1$, one obtains

$$\begin{aligned} A &= a_2'^2 - 3a_1'a_3', \\ B &= 2a_2'^3 - 9a_1'a_2'a_3', \\ D &= a_1'^2a_2'^2a_3'^2 - 4a_1'^3a_3'^3. \end{aligned} \tag{11}$$

Eliminating $a_1'a_3'$ gives

$$a_2'^3 - 3a_2'A + B = 0.$$

Hence a_2', a_2'', a_2''' are the roots of the cubic equation in z

$$z^3 - 3Az + B = 0. \tag{12}$$

This is probably the simplest form that can be given to the cubic resolvent of the biquadratic equation.

§ 66. The covariants

If we form the covariant H, according to § 64, (2), for the transformed form

$$f(x,y) = F(\xi,\eta) = a_1'\xi^3\eta + a_2'\xi^2\eta^2 + a_3'\xi\eta^3, \tag{1}$$

we obtain

$$-H = 3a_1'^2\xi^4 + 4a_1'a_2'\xi^3\eta - (6a_1'a_3' - 4a_2'^2)\xi^2\eta^2 + 4a_2'a_3'\xi\eta^3 + 3a_3'^2\eta^4, \tag{2}$$

and from this it follows that

$$H + 4a_2'f = -3(a_1'\xi^2 - a_3'\eta^2)^2. \tag{3}$$

Thus, apart from the factor -3, this combination is the square of a quadratic form; the same holds when a_2' is replaced by a_2'', a_2'''. For abbreviation, we therefore put

$$\begin{aligned} H + 4a_2'f &= -3\psi_1^2, \\ H + 4a_2''f &= -3\psi_2^2, \\ H + 4a_2'''f &= -3\psi_3^2. \end{aligned} \tag{4}$$

Of the three functions ψ_1, ψ_2, ψ_3, no two can have a common divisor if the discriminant D is assumed different from zero. For then the three quantities a_2', a_2'', a_2''' are also distinct. Thus, if ψ_1 and ψ_2 vanish, then H and f must vanish; that is, a common divisor of ψ_1 and ψ_2 would have to be a common divisor of H and f. But ξ was an arbitrary linear divisor of f. By (2), this can be a divisor of H only when $a_3' = 0$, hence again only when the discriminant D vanishes [§ 65, (11)].

Now the functional determinant T of f and H is identical with the functional determinant of f and $H + \lambda f$, whatever λ may be; and hence, putting $\lambda = 4a_2'$, one obtains

$$T = -\frac{1}{3}\psi_1\left[F'(\xi)\psi_1'(\eta) - F'(\eta)\psi_1'(\xi)\right].$$

Thus T is divisible by ψ_1, and by the same reasoning by ψ_2 and ψ_3; consequently T is, up to a factor independent of ξ, η, identical with the product $\psi_1\psi_2\psi_3$.

If we now form the product of the three equations (4), then, with reference to the cubic equation § 65, (12), satisfied by a_2', a_2'', a_2''', it follows that

$$H^3 - 48AHf^2 - 64Bf^3 = cT^2,$$

where c is a constant still to be determined. By § 64, (12), c remains unchanged under a linear transformation, and therefore its value is found by equating, in the normal form (1), (2), the terms with the highest power of ξ on both sides: $c = -27$.

We thus have the following identical relation among the invariants and covariants:

$$H^3 - 48AHf^2 - 64Bf^3 = -27T^2. \tag{5}$$

The functions ψ_1, ψ_2, ψ_3 are likewise covariants, although with coefficients not rational in the coefficients of f. We can express them easily in terms of the roots $\alpha, \beta, \gamma, \delta$ of $f = 0$. Putting $y = 1$ for simplicity, it follows from § 65, (3) and (5) that

$$\psi_1 = a_1'\xi^2 - a_3'\eta^2 = a_0\frac{(\beta-\gamma)(\beta-\delta)(x-\alpha)^2 - (\alpha-\gamma)(\alpha-\delta)(x-\beta)^2}{\alpha-\beta}.$$

Here, however, $\alpha - \beta$ cancels from numerator and denominator, and ψ_1 can be represented in two forms; to obtain ψ_2 and ψ_3, one then only has to cyclically permute β, γ, δ.

One finds

$$\begin{aligned}
\frac{\psi_1}{a_0} &= (\gamma-\beta)(x-\alpha)(x-\delta) - (\alpha-\delta)(x-\beta)(x-\gamma) \\
&= (\delta-\beta)(x-\alpha)(x-\gamma) - (\alpha-\gamma)(x-\beta)(x-\delta), \\
\frac{\psi_2}{a_0} &= (\delta-\gamma)(x-\alpha)(x-\beta) - (\alpha-\beta)(x-\gamma)(x-\delta) \\
&= (\beta-\gamma)(x-\alpha)(x-\delta) - (\alpha-\delta)(x-\gamma)(x-\beta), \\
\frac{\psi_3}{a_0} &= (\beta-\delta)(x-\alpha)(x-\gamma) - (\alpha-\gamma)(x-\delta)(x-\beta) \\
&= (\gamma-\delta)(x-\alpha)(x-\beta) - (\alpha-\beta)(x-\delta)(x-\gamma).
\end{aligned} \tag{6}$$

The quantities $\psi_1^2, \psi_2^2, \psi_3^2$ may be regarded as the roots of a cubic equation whose coefficients are covariants. From (4) this equation is readily obtained in the form

$$\left(z + \frac{H + 4a_2' f}{3}\right)\left(z + \frac{H + 4a_2'' f}{3}\right)\left(z + \frac{H + 4a_2''' f}{3}\right) = 0.$$

By § 65, (12), and using expression (5) for T^2, this equation gives

$$z^3 + Hz^2 + \frac{1}{3}(H^2 - 16Af^2)z - T^2 = 0. \tag{7}$$

It is still of interest to form the discriminant of this equation. This is obtained most simply from the expression

$$\Delta = (\psi_1^2 - \psi_2^2)^2(\psi_1^2 - \psi_3^2)^2(\psi_2^2 - \psi_3^2)^2,$$

when one substitutes the expressions (4) and then replaces $(a_2' - a_2'')^2(a_2' - a_2''')^2(a_2'' - a_2''')^2$ by the discriminant of the equation § 65, (12), where D then denotes the discriminant of f. One obtains

$$\Delta = 2^{12} D f^6. \tag{8}$$

§ 67. The complete invariant system of the binary biquadratic form

We shall still prove that the constructions A, B exhaust the system of independent invariants of the binary biquadratic form. While one is usually content to show that every invariant is an integral rational function of A, B, we shall, as we have already done in § 63 for the covariants of the cubic form, also touch upon the question of the numerical coefficients; here a new phenomenon appears.

For example, by § 64, (11), the invariant D is indeed rationally expressible through A, B. But the numerical coefficients are not integers; rather, they have denominator 27, although the coefficients in the developed function D are all integers.

We therefore now consider as integral invariants those integral rational homogeneous functions of the μth degree in the five variables a_0, a_1, a_2, a_3, a_4

$$I(a_0, a_1, a_2, a_3, a_4) = I(a), \tag{1}$$

whose coefficients are integers, and which have the invariant property

$$I' = r^{2\mu} I, \tag{2}$$

where I' or $I(a')$ is the same function of the coefficients of a transformed function.

Such invariants are A, B, D, between which the relation

$$27D = 4A^3 - B^2 \tag{3}$$

holds; and our goal is to prove that all integral invariants are integral rational functions of these three, with integer coefficients.

If in the normal form (4) of § 65 we form the function I', then from (2), since $r = 1$, we first obtain

$$I(a) = I'(0, a_1', a_2', a_3', 0).$$

This function I can depend only on a_2' and on the product $a_1'a_3'$, for the substitution

$$\xi = \lambda\xi', \qquad \eta = \frac{1}{\lambda}\eta'$$

with determinant 1 leaves the normal form § 65, (4), and in it a_2', unchanged, while a_1', a_3' go over into $\lambda^2 a_1'$, $a_3' : \lambda^2$; here λ is arbitrary.

Thus, putting

$$a_2' = z,$$

we have

$$I = \varphi(a_1'a_3', z), \tag{4}$$

where φ is an integral rational function with integer coefficients of the two arguments $a_1'a_3'$ and z.

But by § 65, (11),

$$3a_1'a_3' = z^2 - A.$$

Therefore, if we multiply (4) by a suitable power of 3, say 3^ν, we obtain

$$3^\nu I = \psi(A, z), \tag{5}$$

where ψ is again an integral function of A and z. But by § 65, (12),

$$z^3 = 3Az - B,$$

and hence all higher powers of z can be expressed by the first and second powers. Thus we obtain

$$3^\nu I = \chi(A, B, z), \tag{6}$$

where χ is of degree at most two with respect to z.

The left side of (6), however, remains unchanged when z is replaced by a_2', a_2'', a_2''', that is, by three different values. Consequently the function χ cannot contain the variable z at all (§ 30, II).

Thus

$$3^\nu I = \chi(A, B), \tag{7}$$

where χ is an integral, rational function with integer coefficients.

If in (7) we put, according to (3),

$$B^2 = 4A^3 - 27D,$$

it follows that (7) has one of the two following forms:

$$3^\nu I = \Phi(A, D), \qquad B\Phi(A, D), \tag{8}$$

according as the degree of I is even or odd.

If I is a primitive function of the a, that is, one whose numerical coefficients have no common divisor, then the coefficients of the functions Φ can in any case have no common divisor other than a power of 3.

What remains to be proved is that they are all divisible by 3^ν, or, what is the same, that if $\Phi(A, D)$ is assumed primitive, then $\nu = 0$.

The question is therefore this: can functions with divisor 3 arise from the primitive function $\Phi(A, D)$ or $B\Phi(A, D)$ when A, B, D are replaced by their expressions in the a? Since B is primitive, by § 2 the function $B\Phi(A, D)$ expressed in the a can have the divisor 3 only if $\Phi(A, D)$ has it.

Thus the question is reduced to this: can the primitive function $\Phi(A, D)$ acquire the divisor 3 when A, D are replaced by their expressions?

That this question must be answered in the negative can be seen easily. First imagine that in $\Phi(A, D)$ all terms whose coefficients are divisible by 3 have been removed; for plainly the remaining terms must still have the same property, namely, to acquire the divisor 3 by substitution of the expressions for A, D.

Second, we may assume that $\Phi(A, D)$ does not have the factor D, for by omitting this factor, according to the theorem already mentioned (§ 2), the property in question would not be destroyed.

Then, however, a number divisible by 3 would have to arise from $\Phi(A, D)$ when all a_i, except a_2, are set equal to zero and $a_2 = 1$ is put. This gives $D = 0$, $A = 1$, and therefore $\Phi(1, 0)$ would have to be a number divisible by 3. But this is the coefficient of the highest power of A in $\Phi(A, D)$, which by assumption is not divisible by 3. Thus we have proved:

Every integral invariant of the biquadratic form can be represented rationally and integrally by A, B, D, and indeed in one of the two forms

$$\Phi(A, D), \qquad B\Phi(A, D).$$

Conversely, it is clear by itself that every such expression, if homogeneous in the coefficients a, represents an integral invariant[18].

Sixth Section.

Tschirnhausen Transformation.

§ 68. Hermite's form of the Tschirnhausen transformation

In the fourth section we have already become acquainted with the basic idea of the Tschirnhausen transformation.

The problem was to transform an algebraic equation of degree n,

$$f(x) = a_0 x^n + a_1 x^{n-1} + \cdots + a_{n-1} x + a_n = 0, \tag{1}$$

by a substitution of degree $n - 1$,

$$y = \alpha_0 + \alpha_1 x + \alpha_2 x^2 + \cdots + \alpha_{n-1} x^{n-1}, \tag{2}$$

so as to obtain, in the arbitrary coefficients α, means for simplifying the equation.

Hermite greatly simplified this problem and brought it into connection with invariant theory by giving the substitution (2) a special form[19].

In § 4 we became acquainted with a sequence of functions $f_0, f_1, f_2, \ldots, f_{n-1}$, by means of which the powers of x up to the $(n-1)$st can be expressed rationally; these are the functions which Hermite uses to represent the substitution (2).

Here x denotes a root of the equation (1), while t is to be understood as an indeterminate variable. Then by § 4,

$$\frac{f(t)}{t - x} = t^{n-1} f_0(x) + t^{n-2} f_1(x) + \cdots + t f_{n-2}(x) + f_{n-1}(x) \tag{3}$$

[18] The theory of invariants is presented in detail in the works: Clebsch, *Theory of Binary Algebraic Forms.* Leipzig 1872. Faa di Bruno, *Introduction to the Theory of Binary Forms.* German translation by Walter. Leipzig 1881. P. Gordan, *Lectures on Invariant Theory*, edited by Kerschensteiner, Leipzig 1885. See also Franz Meyer, *Report on the Present State of Invariant Theory*, in the Annual Report of the German Mathematical Society 1890/91 (Berlin 1892).

[19] Hermite, "Sur quelques théorèmes d'algèbre et la résolution de l'équation du quatrième degré," separately issued from the Comptes rendus of the Paris Academy. Paris 1859.

and

$$\begin{aligned}
f_0(x) &= a_0, \\
f_1(x) &= a_0 x + a_1, \\
f_2(x) &= a_0 x^2 + a_1 x + a_2, \\
&\cdots \\
f_{n-1}(x) &= a_0 x^{n-1} + a_1 x^{n-2} + a_2 x^{n-3} + \cdots + a_{n-1}.
\end{aligned} \tag{4}$$

We now take the substitution (2) in the form

$$y = t_{n-1} f_0(x) + t_{n-2} f_1(x) + \cdots + t_1 f_{n-2}(x) + t_0 f_{n-1}(x), \tag{5}$$

where $t_{n-1}, t_{n-2}, \ldots, t_1, t_0$ are the indeterminate quantities which have taken the place of the α, and are not to be confused with the powers of t. But y is obtained from the expression (3) when t^k is replaced by t_k.

As in § 42, let the sign S placed before a function mean that the sum is to be taken over all roots x of equation (1). Then

$$\begin{aligned}
S[f_0(x)] &= n a_0, \\
S[f_1(x)] &= (n-1) a_1, \\
S[f_2(x)] &= (n-2) a_2, \\
&\cdots \\
S[f_{n-1}(x)] &= a_{n-1},
\end{aligned} \tag{6}$$

and hence

$$S(y) = n a_0 t_{n-1} + (n-1) a_1 t_{n-2} + \cdots + 2 a_{n-2} t_1 + a_{n-1} t_0, \tag{7}$$

an expression which is obtained from $f'(t)$ by replacing t^k by t_k.

If with the help of (7) we eliminate the variable t_{n-1} from (5), we get

$$y - \frac{1}{n} S(y) = t_{n-2} F_0(x) + t_{n-3} F_1(x) + \cdots + t_0 F_{n-2}(x), \tag{8}$$

where

$$\begin{aligned}
F_0 &= f_1 - \frac{n-1}{n} a_1 = a_0 x + \frac{a_1}{n}, \\
F_1 &= f_2 - \frac{n-2}{n} a_2 = a_0 x^2 + a_1 x + \frac{2a_2}{n}, \\
&\cdots \\
F_{n-2} &= f_{n-1} - \frac{1}{n} a_{n-1} = a_0 x^{n-1} + a_1 x^{n-2} + \cdots + \frac{n-1}{n} a_{n-1}.
\end{aligned} \tag{9}$$

Thus

$$S[F_0(x)] = 0, \quad S[F_1(x)] = 0, \ldots, \quad S[F_{n-2}(x)] = 0. \tag{10}$$

Putting, according to (3),

$$\begin{aligned}
F(t, x) &= \frac{f(t)}{t - x} - \frac{1}{n} f'(t) \\
&= t^{n-2} F_0(x) + t^{n-3} F_1(x) + \cdots + t F_{n-3}(x) + F_{n-2}(x),
\end{aligned} \tag{11}$$

the right side of (8) is obtained from $F(t, x)$ by replacing t^k by t_k.

If from the outset we take y in the form

$$y = t_{n-2} F_0(x) + t_{n-3} F_1(x) + \cdots + t_0 F_{n-2}(x), \tag{12}$$

then the equation

$$S(y) = 0 \tag{13}$$

is identically satisfied. What use this form of the substitution affords will be shown by the next considerations.